

HOW TO CREATE CLUSTER USING HYPER-V -- Authors: Karthick P.K & Pradeepa

<https://sqlserverscribbles.com/2013/06/18/how-to-create-cluster-using-hyper-v/>

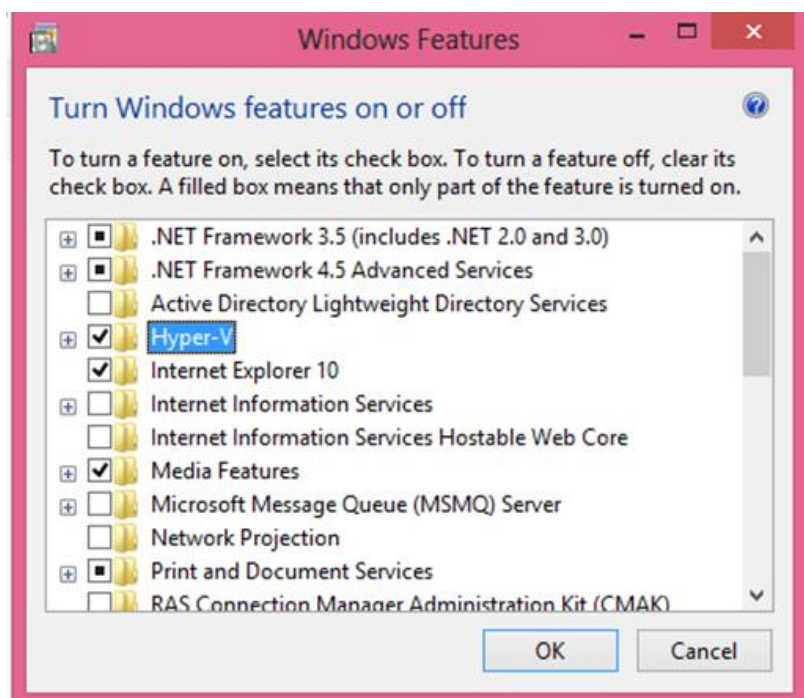
Hyper-V can be used to create virtualized computers to improve the hardware efficiency.

In this blog post I will try to cover how to create windows failover cluster using Hyper-V without using multiple physical computers. Cluster we are building can be used to install Clustered SQL Server and similar cluster aware applications.

1. We will create a Domain controller
2. We will create two member servers which will act as two nodes of the cluster.

We will start the installation by Turning on window feature Hyper-V.

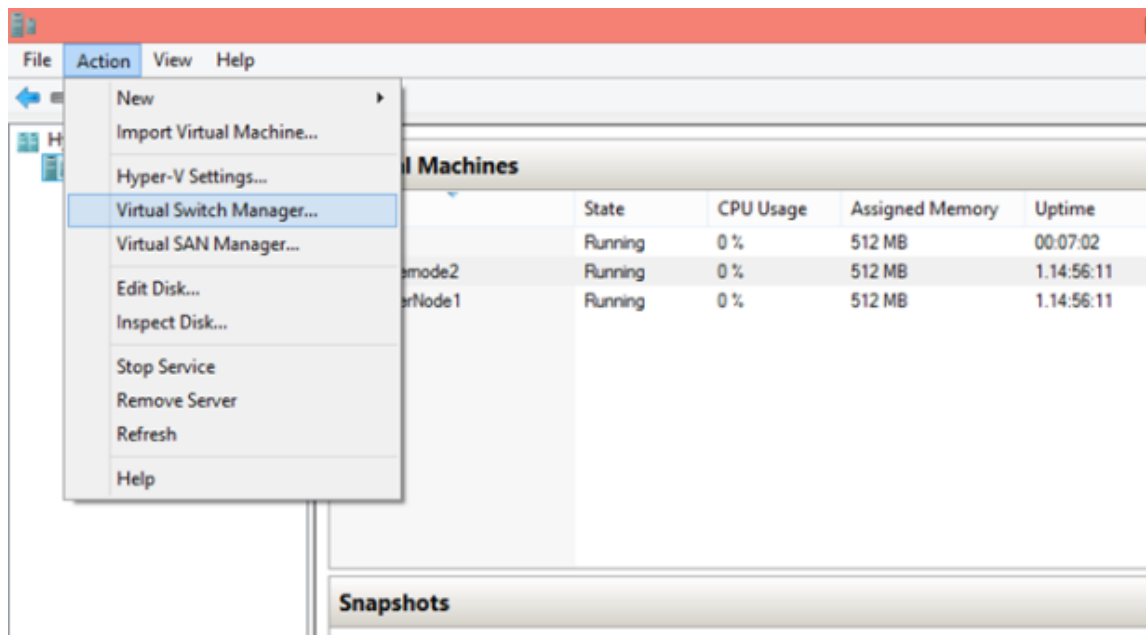
To enable Hyper-V Open control panel—>Programs —>Get Programs->Turn windows features on or off —>Enable Hyper-V as shown in picture below



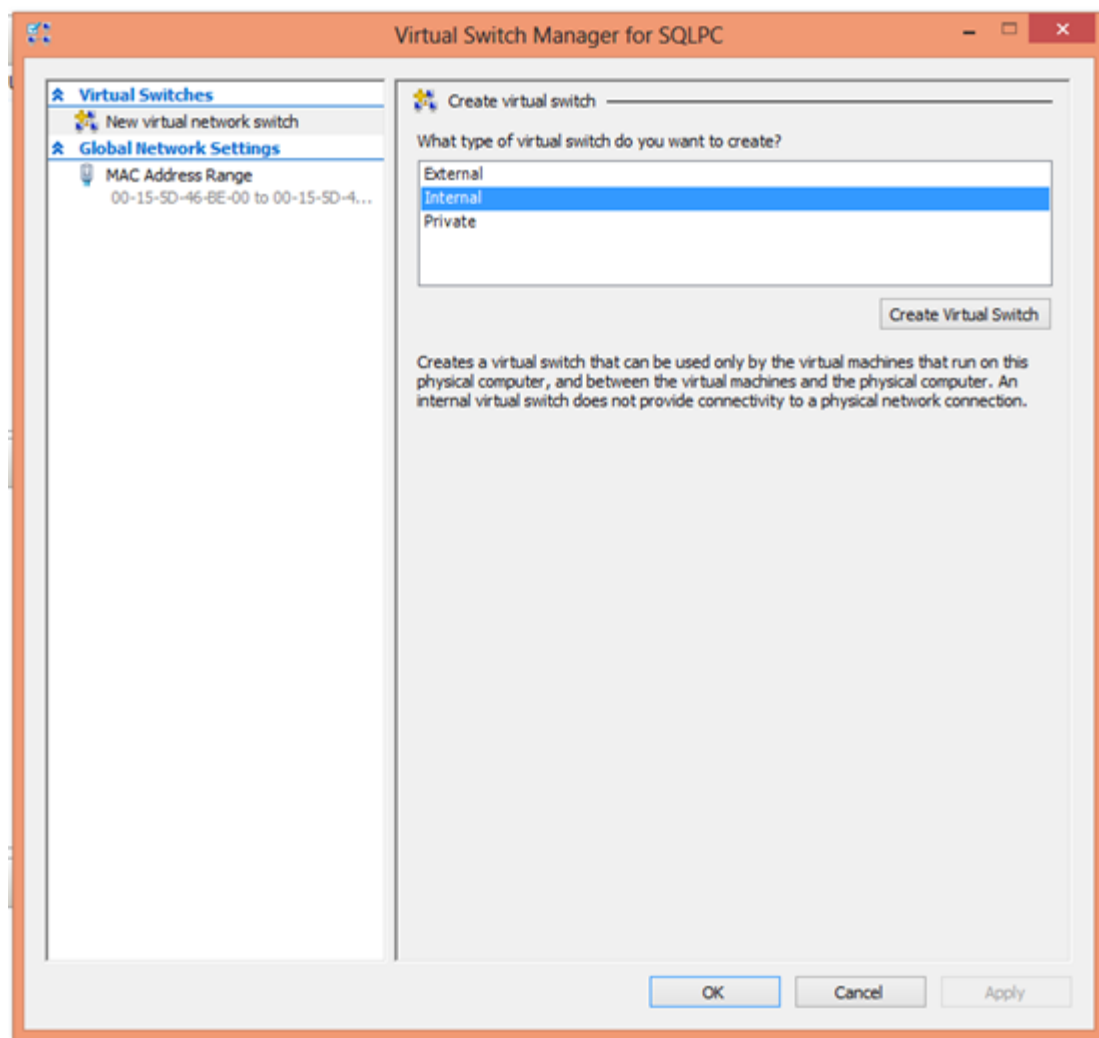
Once the Hyper-v feature is enabled open the hyper v manager and follow the below steps one by one.

REPORT THIS AD

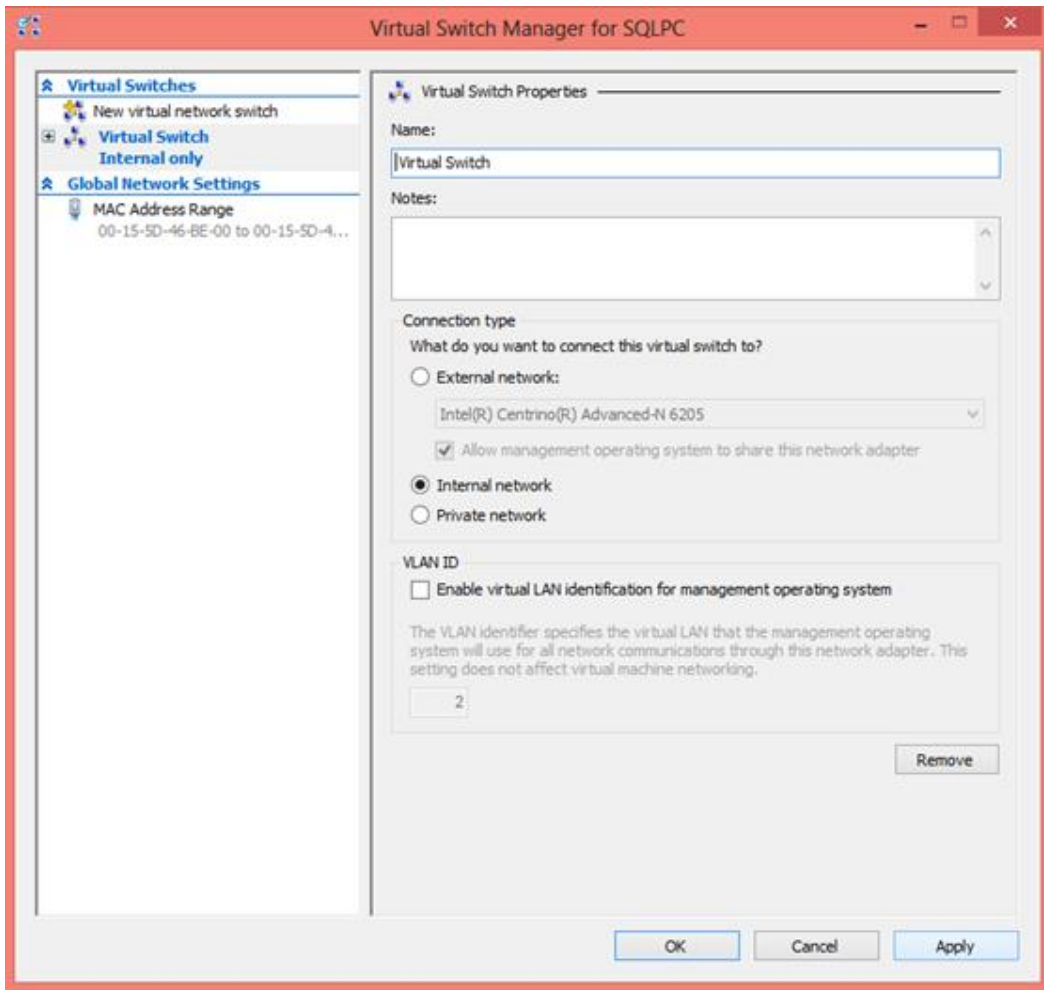
Before creating virtual machine create virtual switch to configure network access between the computers which will create and name it.



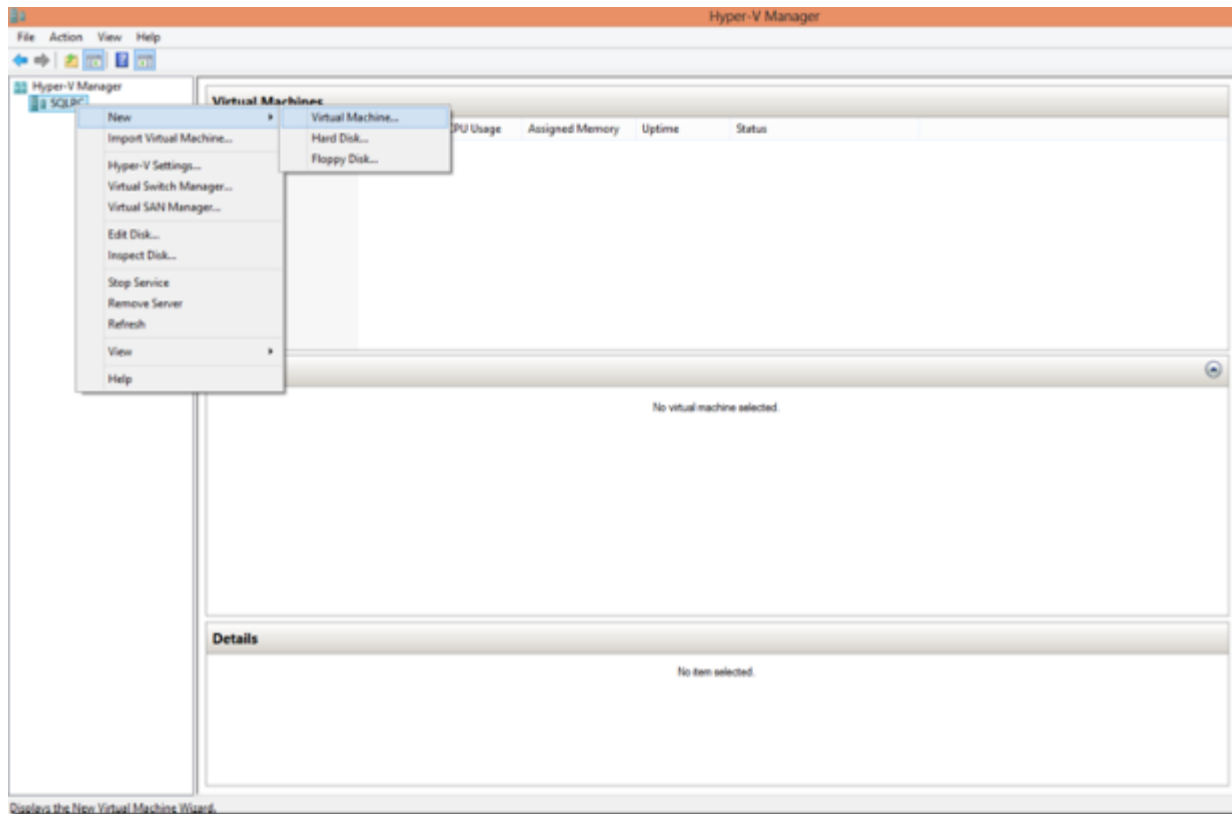
Select the type of virtual switch for your virtual machines.



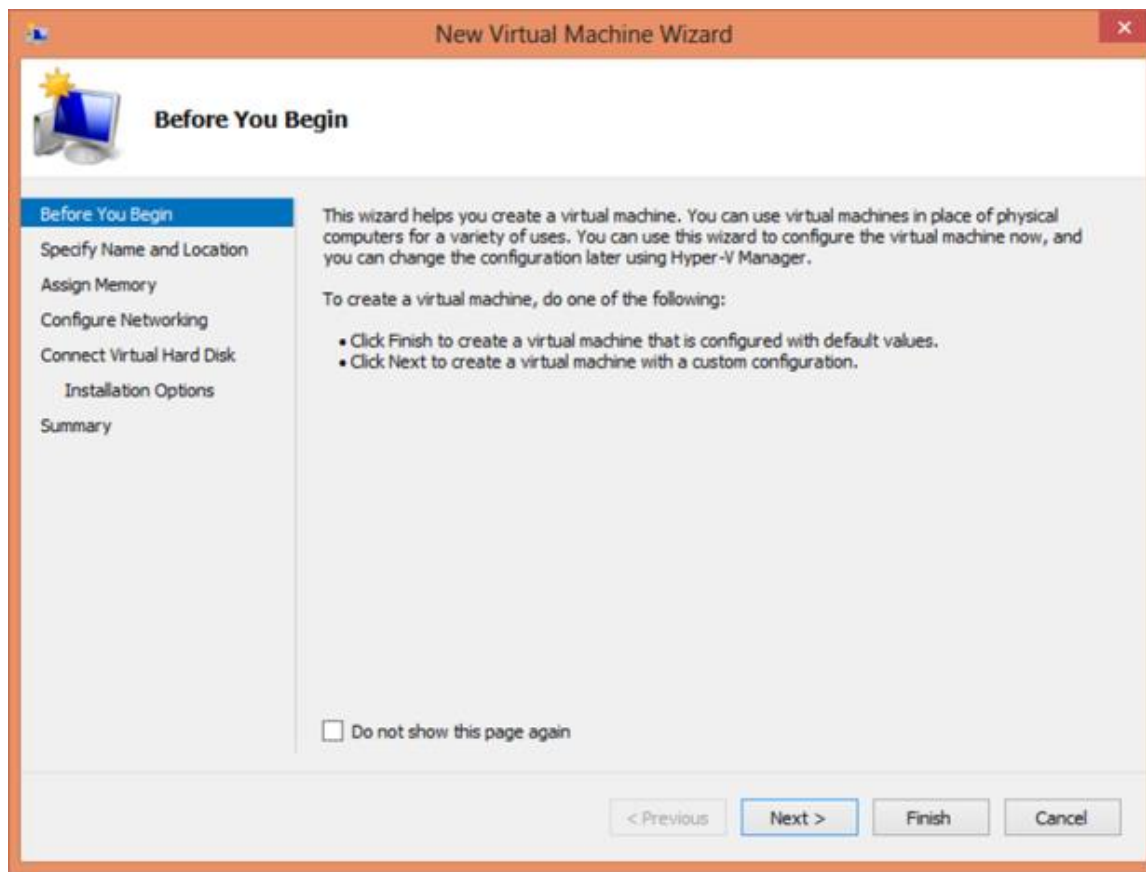
Give the name for your Virtual switch and click ok.



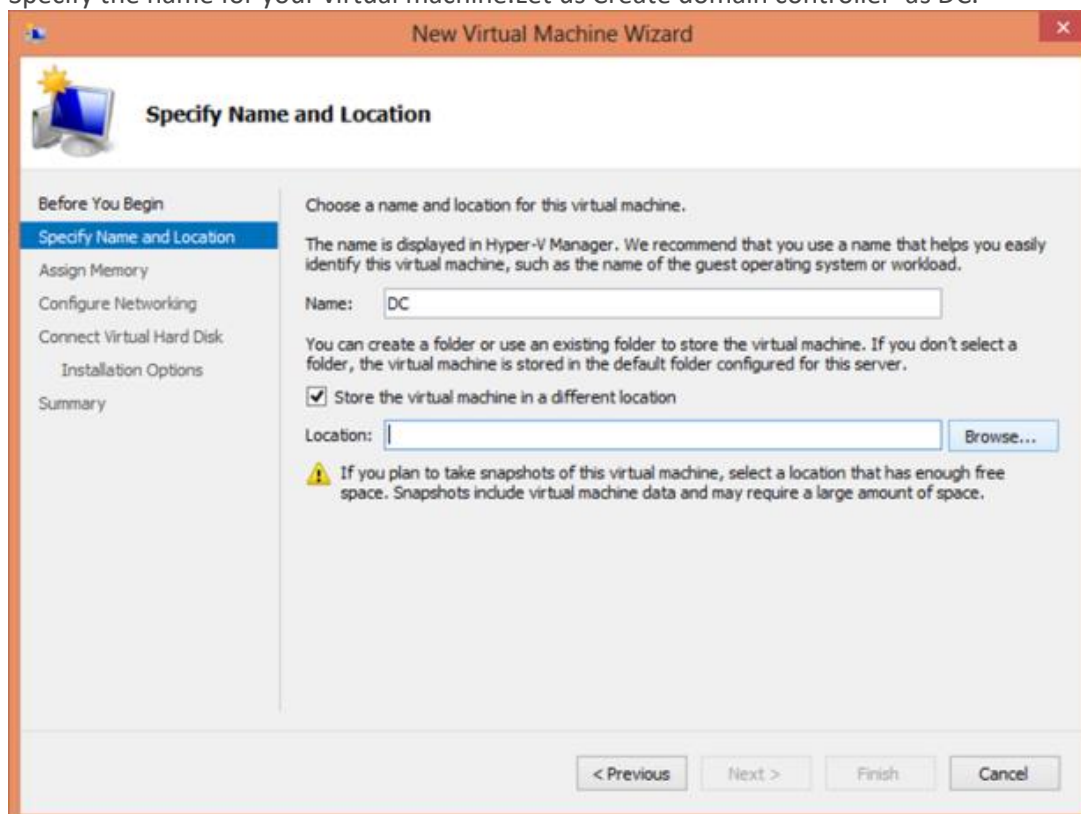
Now create New virtual machine by Right clicking SQLPC → New → Virtual machine.



2.New virtual machine wizard will appear.Click Next to continue.



Specify the name for your virtual machine.Let us Create domain controller as DC.



Create a new folder or use the existing folder to store the virtual machine.Specify that location in location tab.Click Next

New Virtual Machine Wizard

Specify Name and Location

Before You Begin

- Specify Name and Location
- Assign Memory
- Configure Networking
- Connect Virtual Hard Disk
- Installation Options
- Summary

Choose a name and location for this virtual machine.

The name is displayed in Hyper-V Manager. We recommend that you use a name that helps you easily identify this virtual machine, such as the name of the guest operating system or workload.

Name:

You can create a folder or use an existing folder to store the virtual machine. If you don't select a folder, the virtual machine is stored in the default folder configured for this server.

☒ Store the virtual machine in a different location

Location:

 If you plan to take snapshots of this virtual machine, select a location that has enough free space. Snapshots include virtual machine data and may require a large amount of space.

< Previous Next > Finish Cancel

Specify the amount of memory you want to allocate for this machine. Then, click Next.

New Virtual Machine Wizard

Assign Memory

Before You Begin

- Specify Name and Location
- Assign Memory
- Configure Networking
- Connect Virtual Hard Disk
- Installation Options
- Summary

Specify the amount of memory to allocate to this virtual machine. You can specify an amount from 8 MB through 5735 MB. To improve performance, specify more than the minimum amount recommended for the operating system.

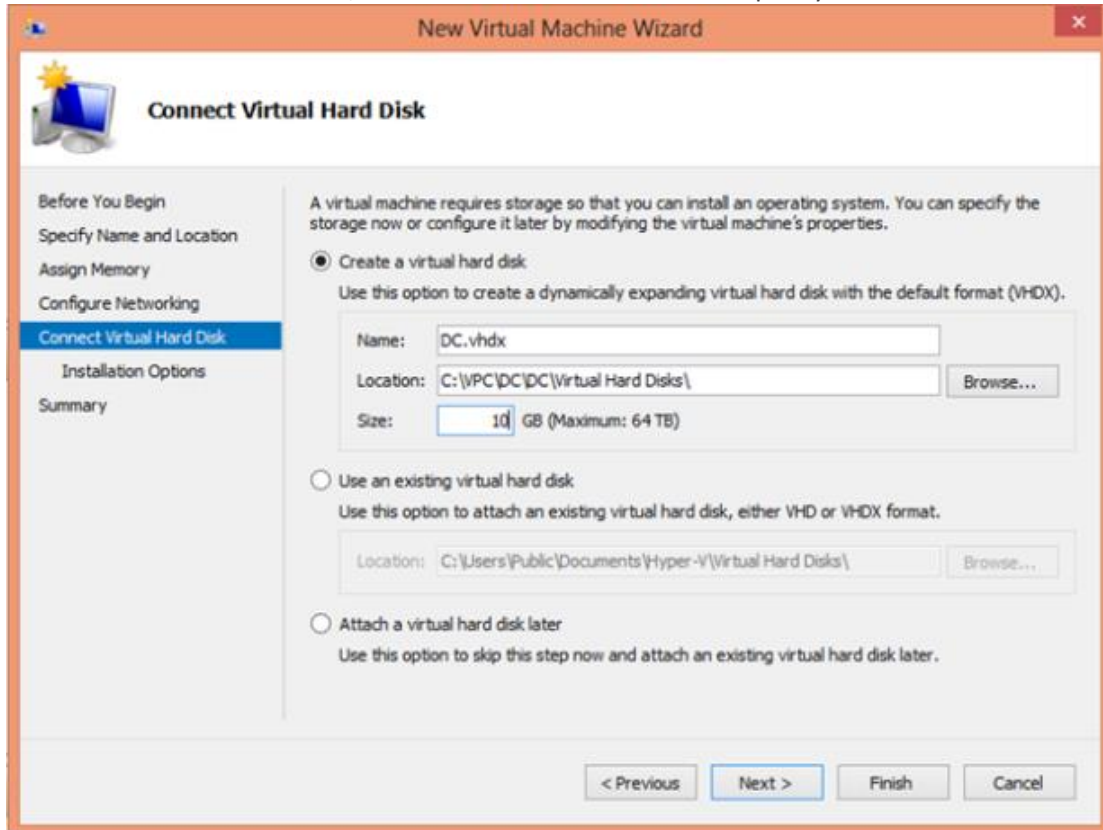
Startup memory: MB

☐ Use Dynamic Memory for this virtual machine.

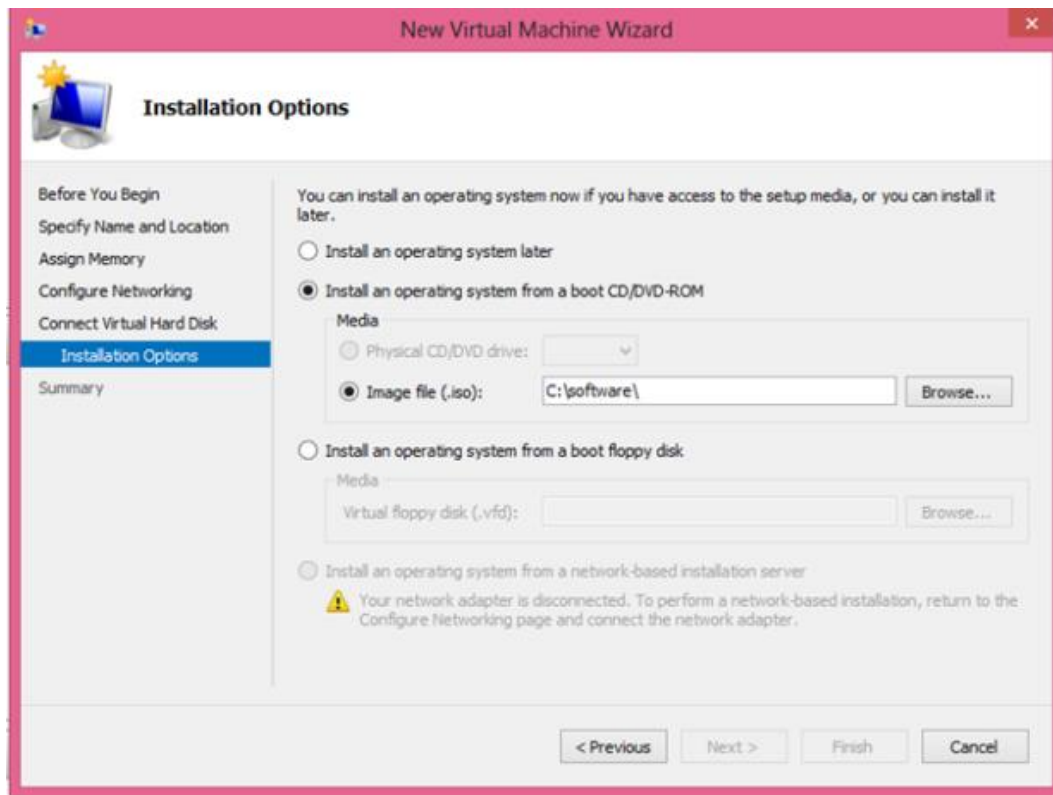
 When you decide how much memory to assign to a virtual machine, consider how you intend to use the virtual machine and the operating system that it will run.

< Previous Next > Finish Cancel

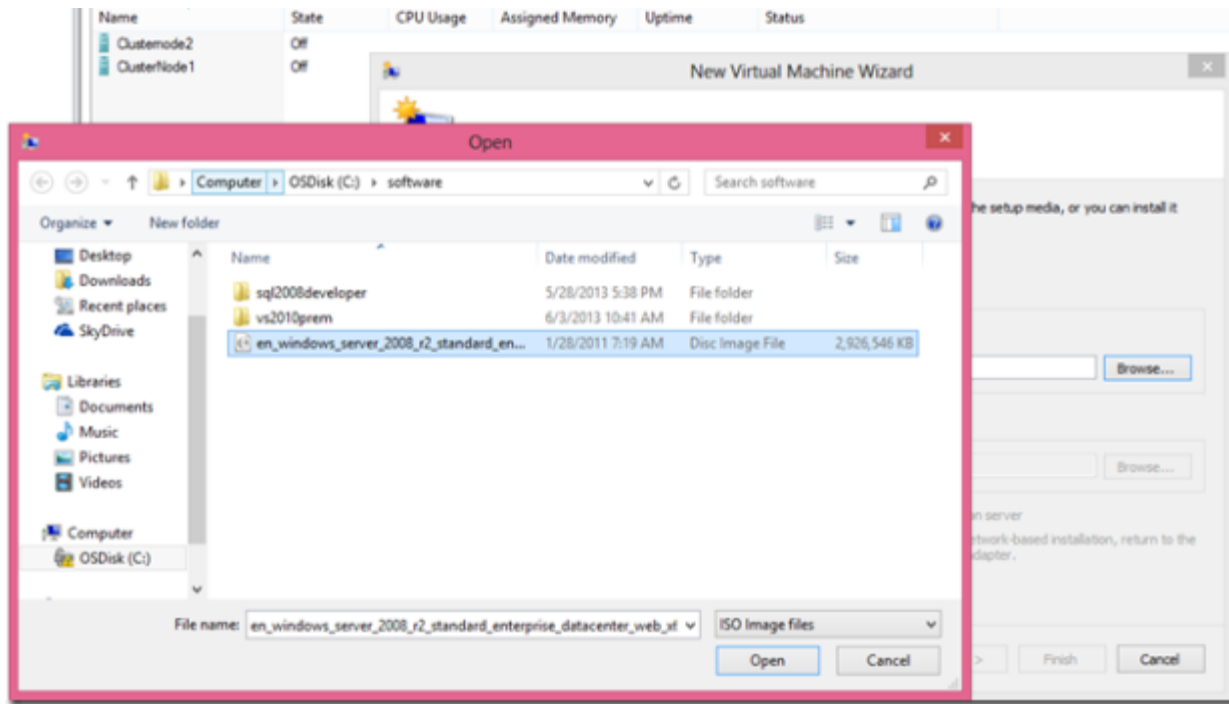
In order to do OS installation, connect to virtual hard disk and specify the size for it. Click Next



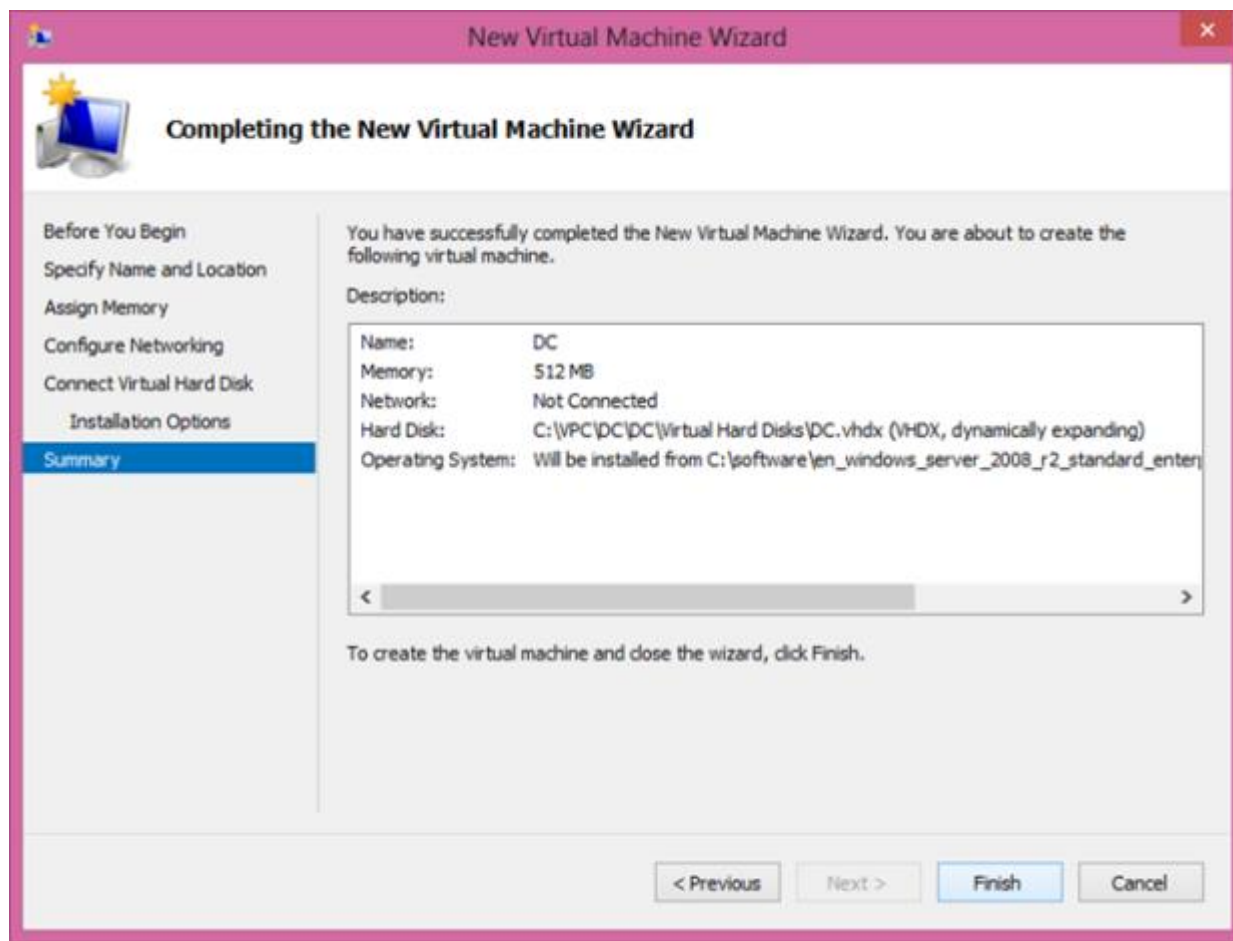
Then install an Operating system from CD or as image file which you have in your system.



Browse and select the file for installation. Click open.



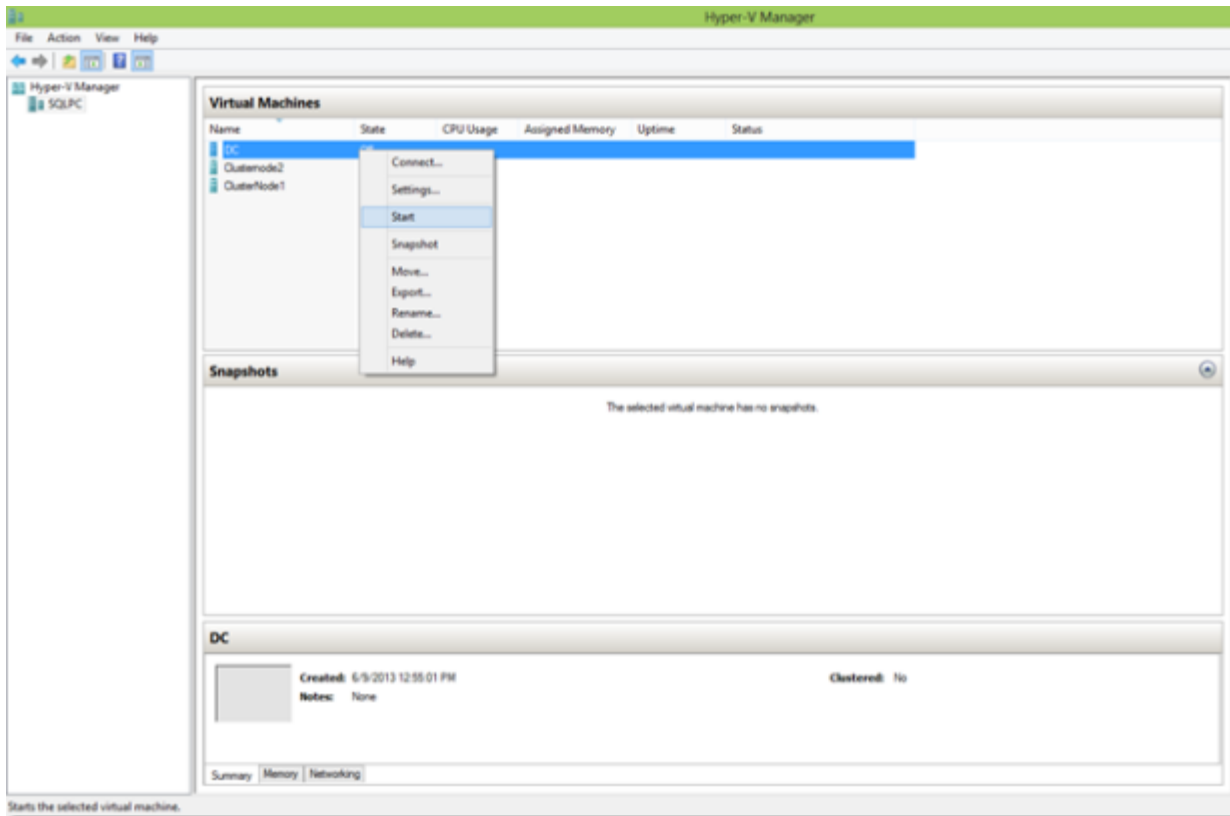
Now,click finish and complete the virtual machine wizard.



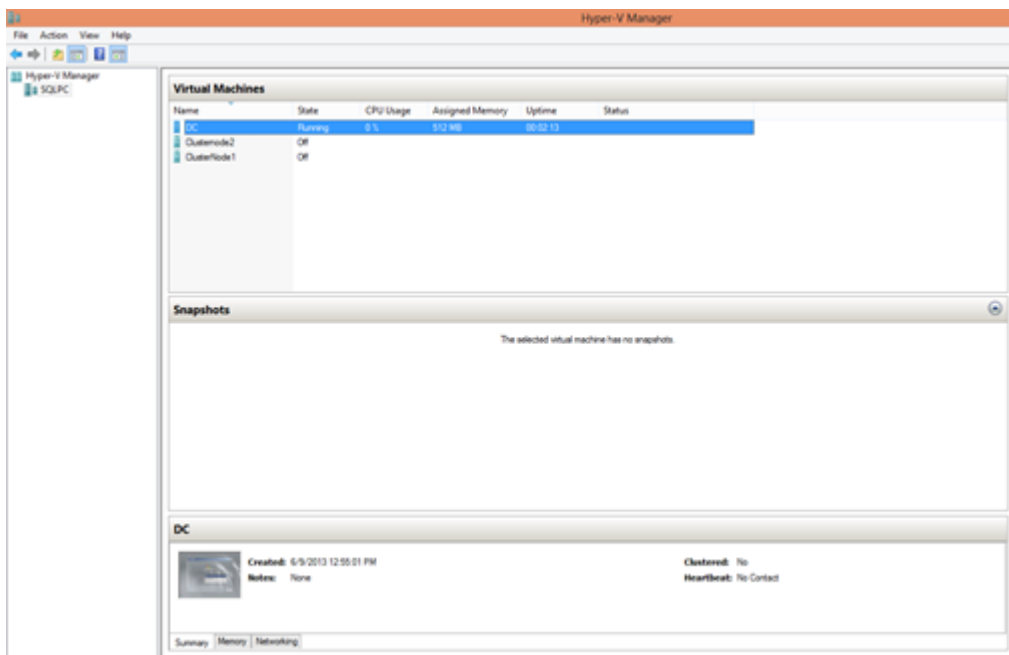
Similarly,Follow the above steps and create two cluster nodes as clusternode1 and clusternode2 .

Now Hyper –V manager will have three virtual machines which we created one as DC and others as clusternodes.

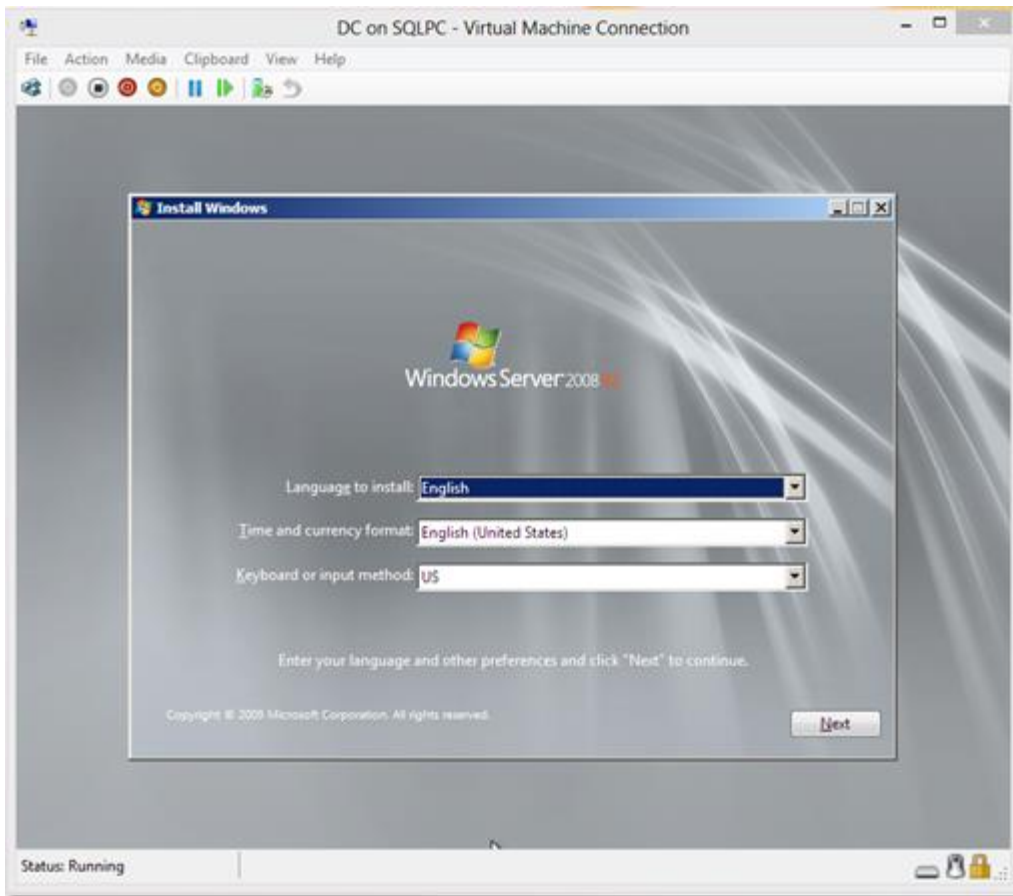
Right click DC->Start->connect.(similarly do for all)



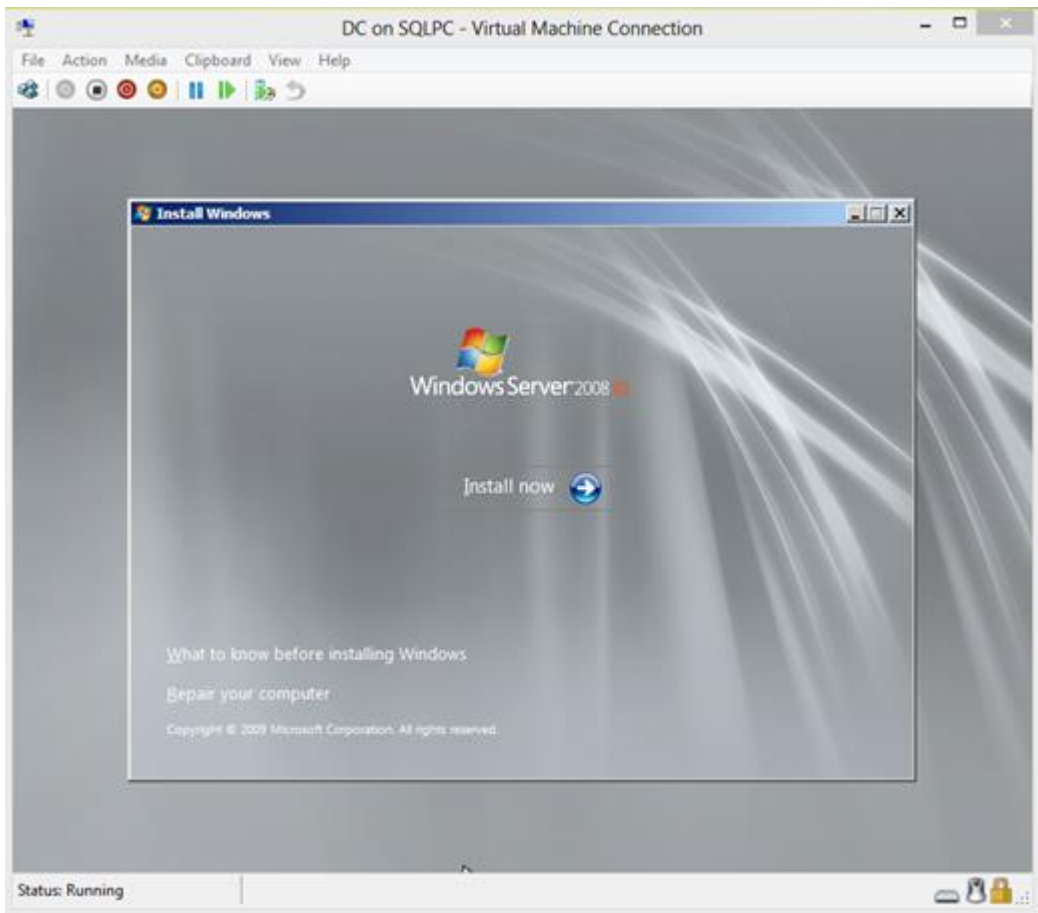
Cluster nodes will start running.



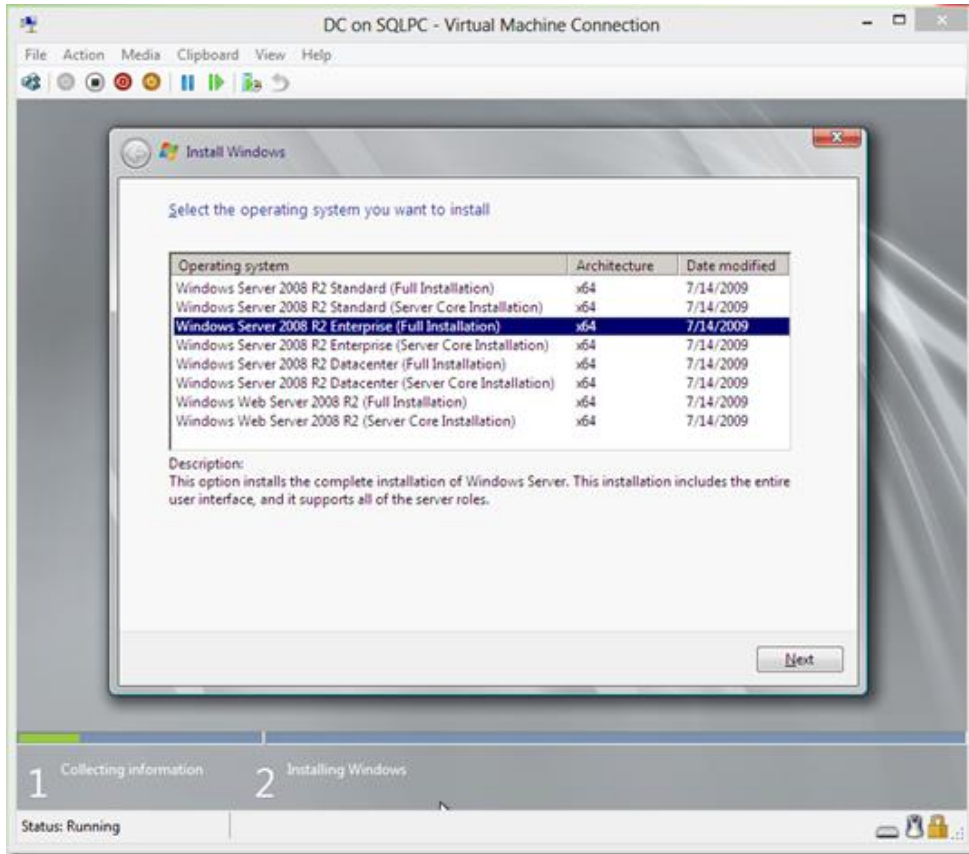
DC virtual machine connection window will appear,click next.



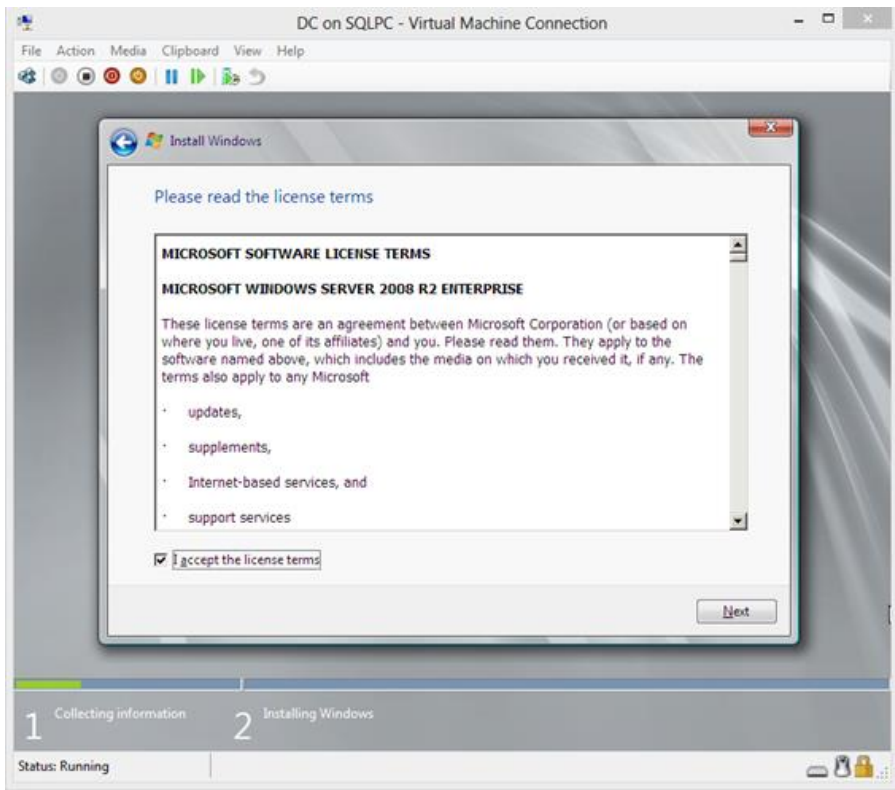
Installation window will appear,click install now to start installation.



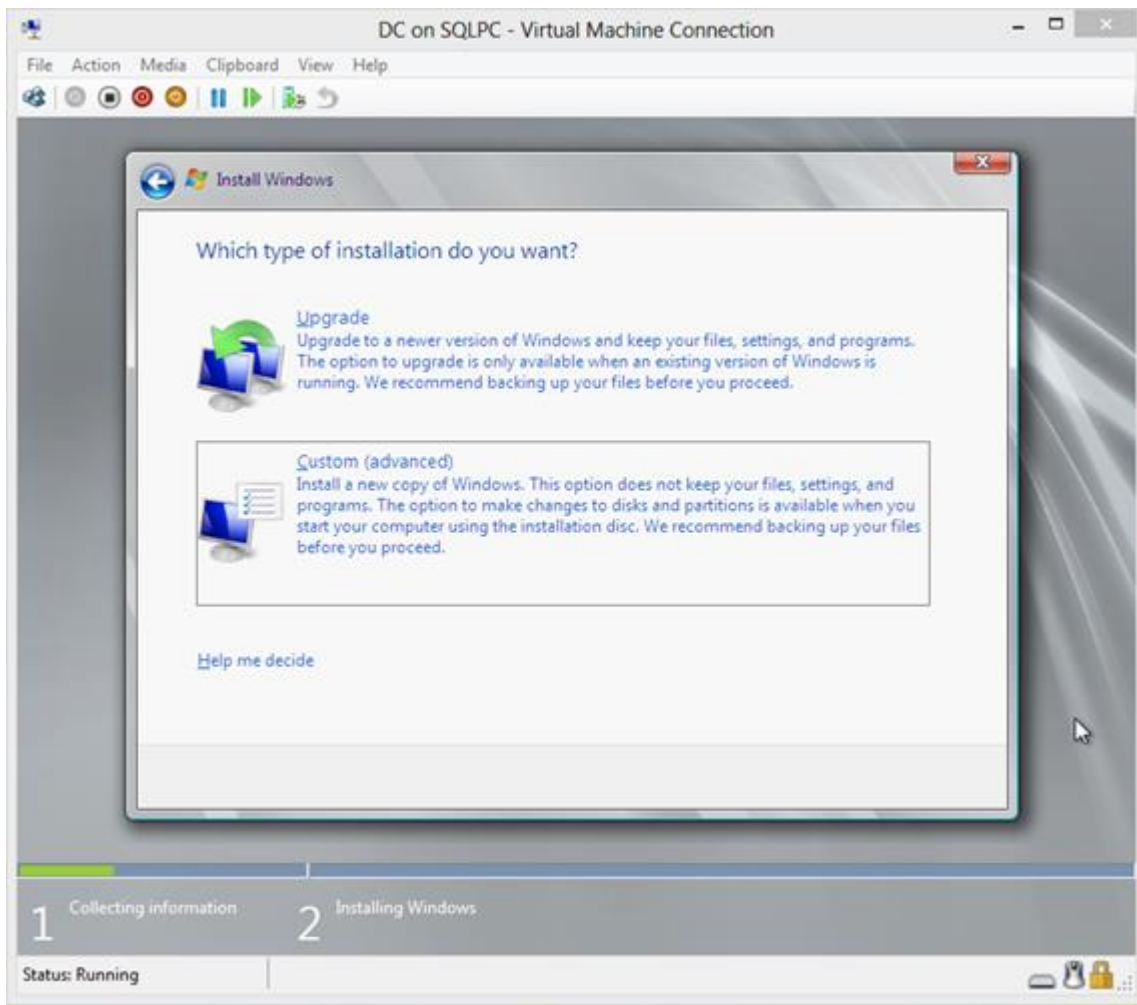
Select the operating system you want to install. Select windows server 2008 R2 Enterprise(Full installation) for DC(domain controller). Click Next.



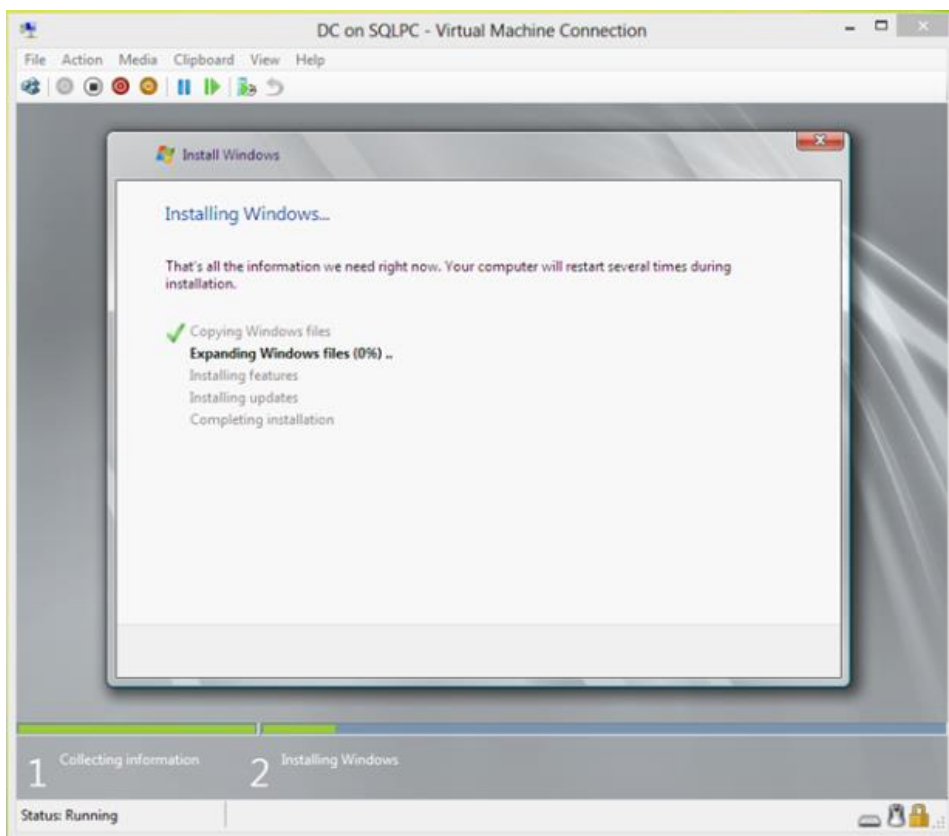
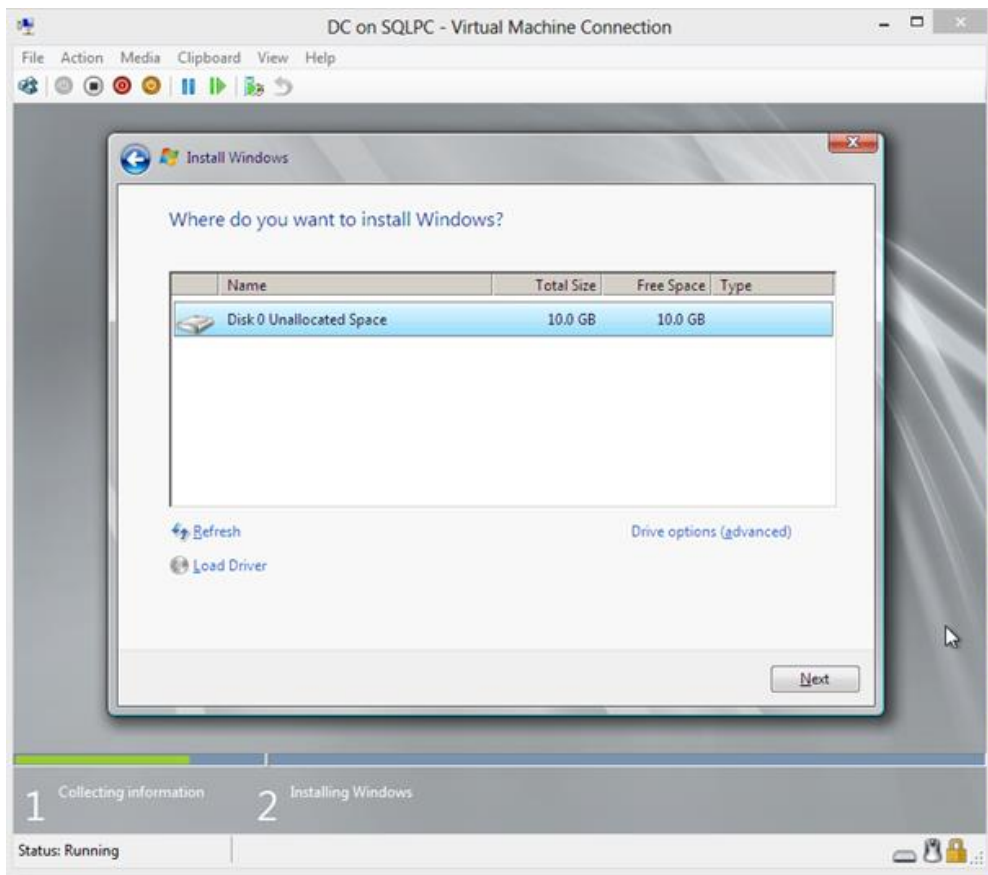
Accept the license term and click next



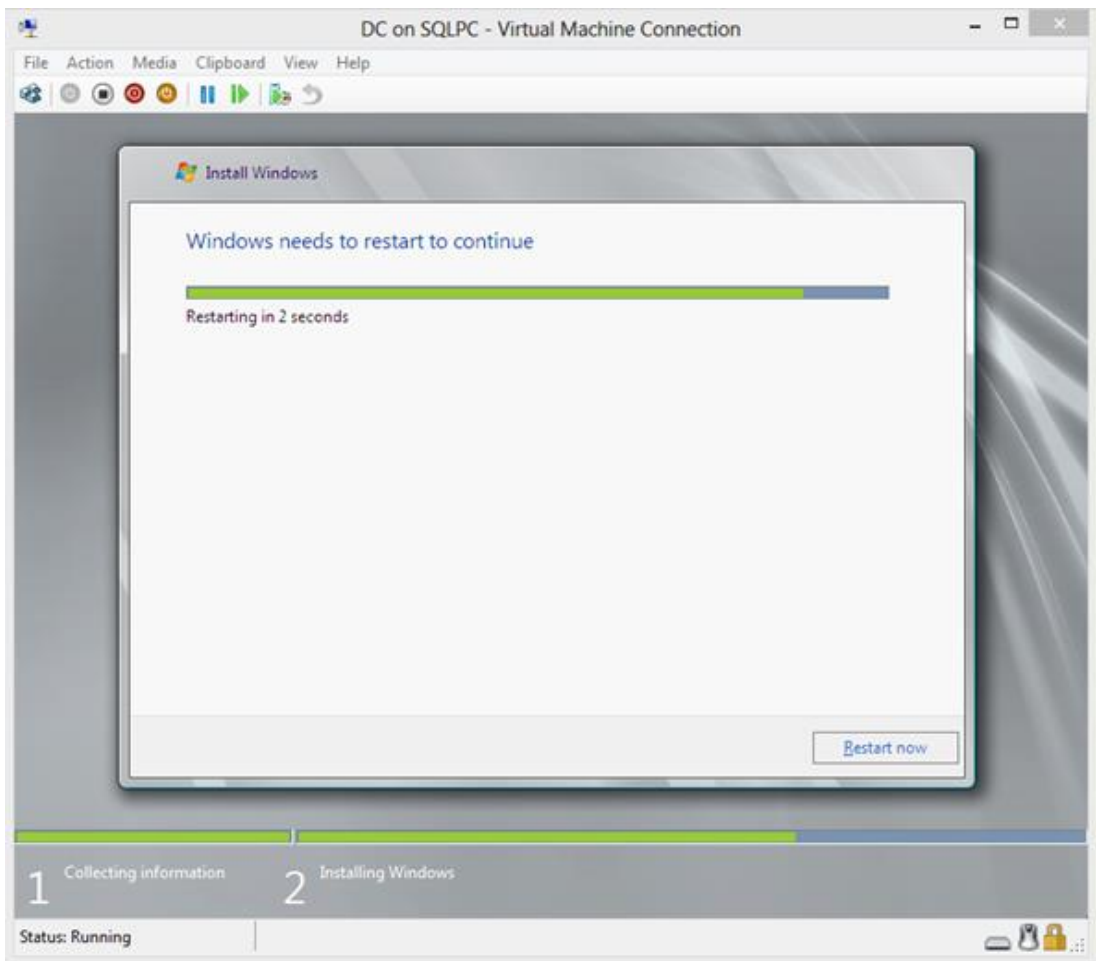
Select the type of installation as Custom(advanced)



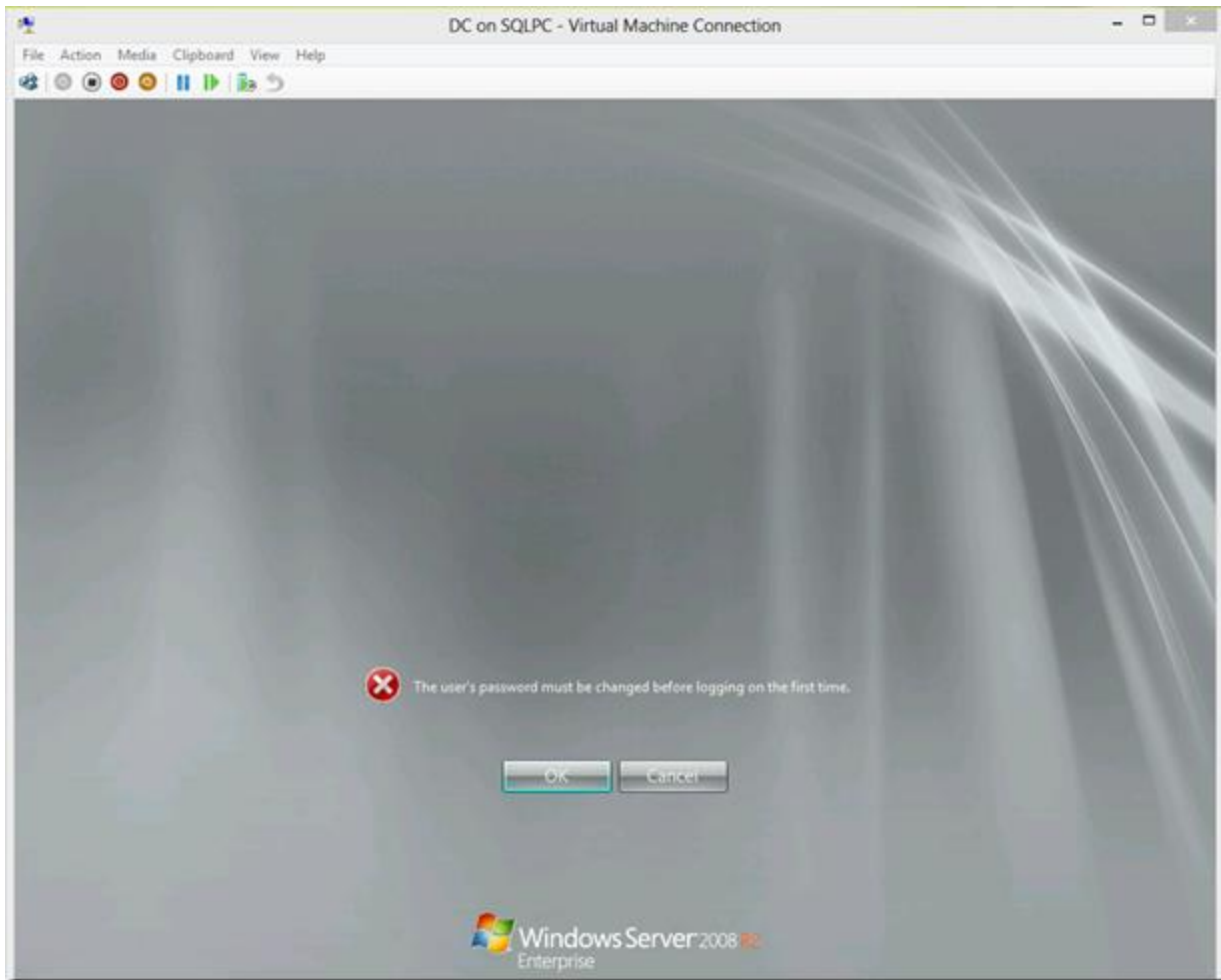
Click Next and continue



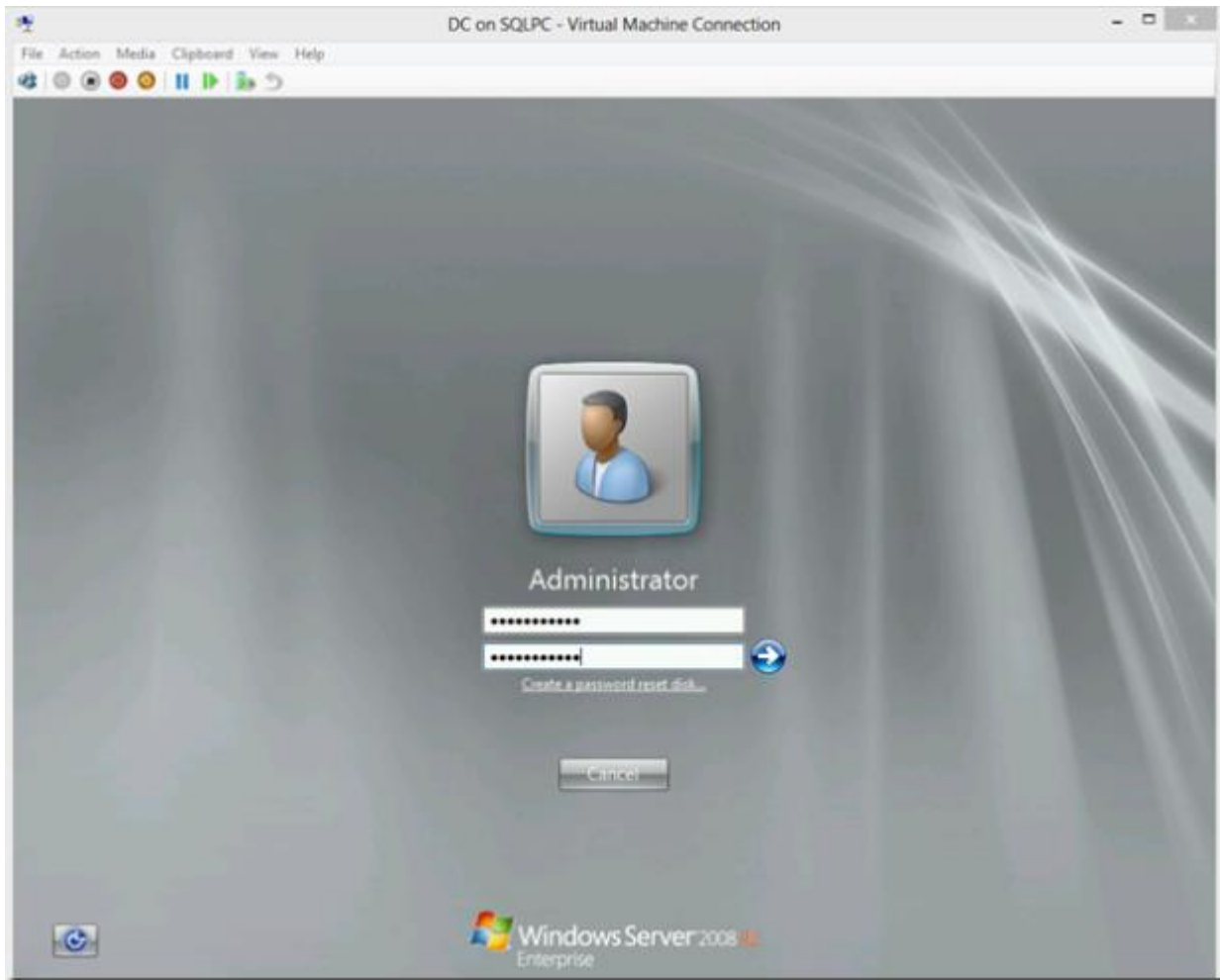
After installation restart your virtual machine to update with new setting



Give the password for login.

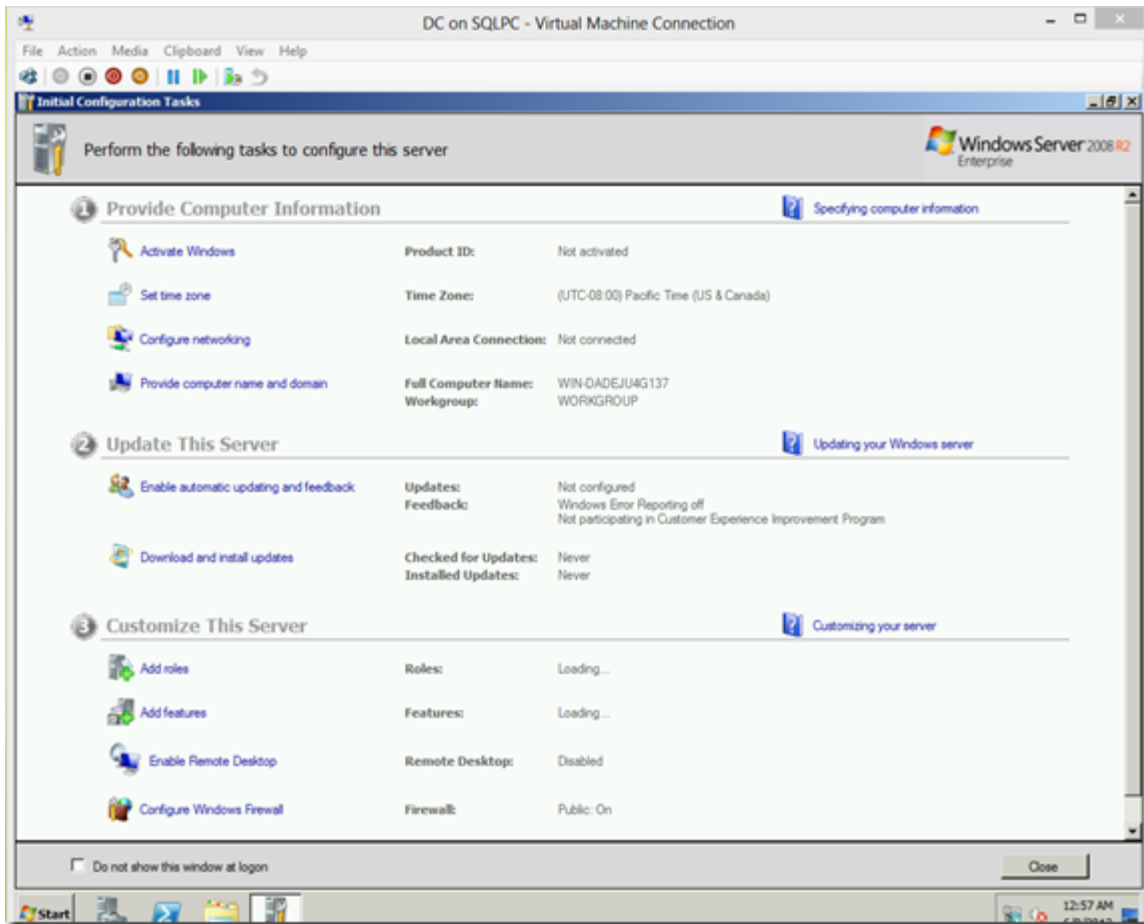


Domain controller is installed with OS.



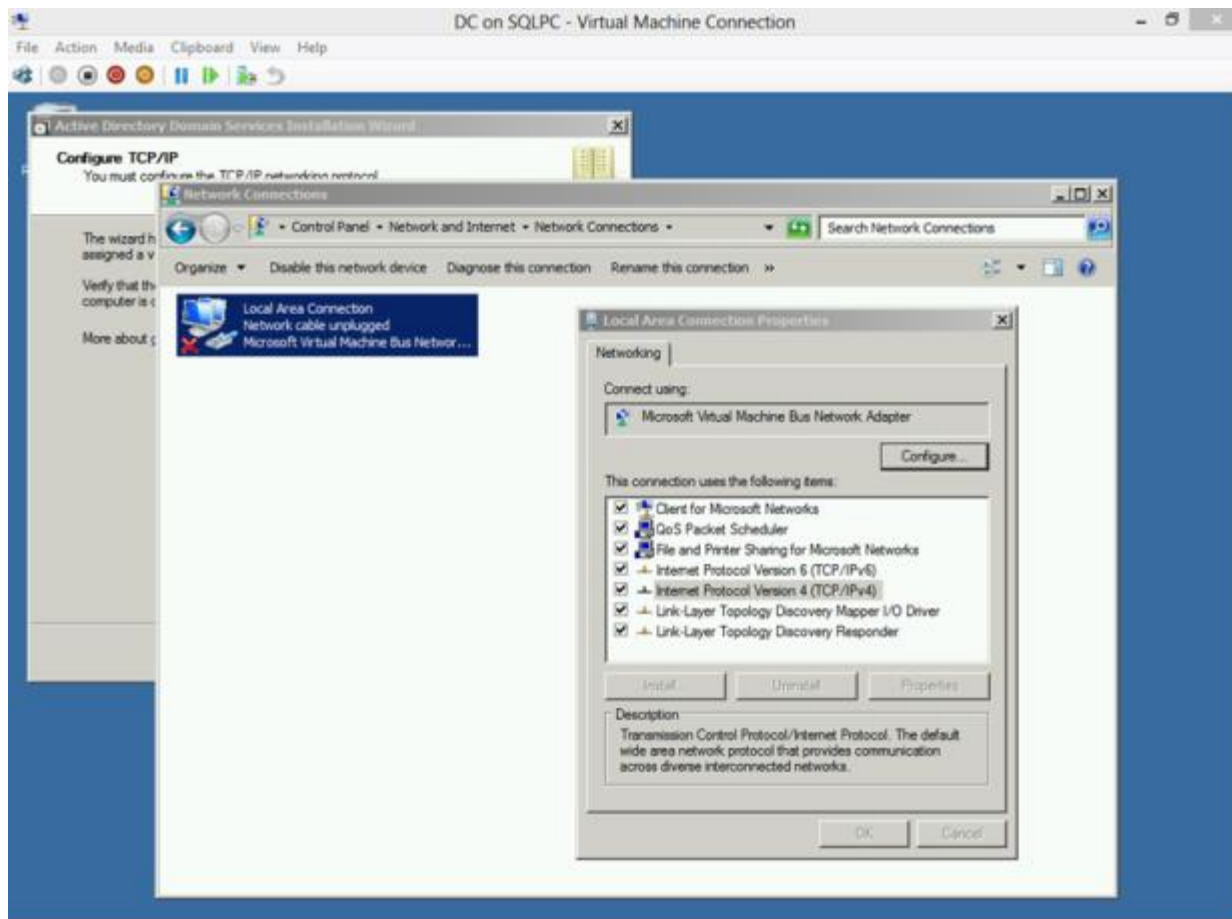
REPORT THIS AD

Perform the task which you want or else close the window.



Configure TCP/IP address and enable LAN connection.

To configure TCP/IP, open control panel->Network and internet->Network connections->click IPV4



REPORT THIS AD

Give IP address ,subnet mask and default gateway ,click ok.

Internet Protocol Version 4 (TCP/IPv4) Properties [?] [X]

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 0 . 1

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 192 . 168 . 0 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

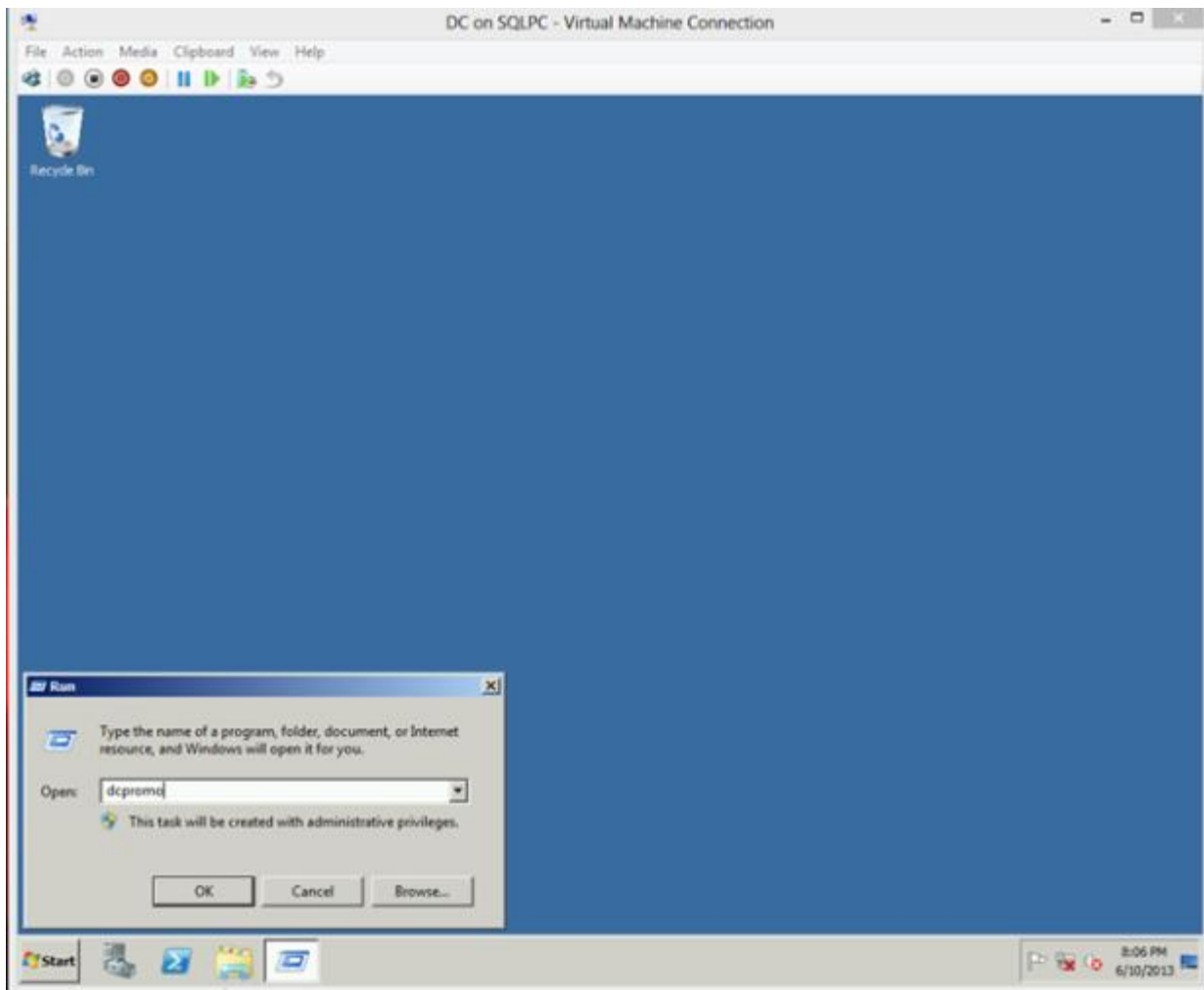
Alternate DNS server: . . .

☐ Validate settings upon exit

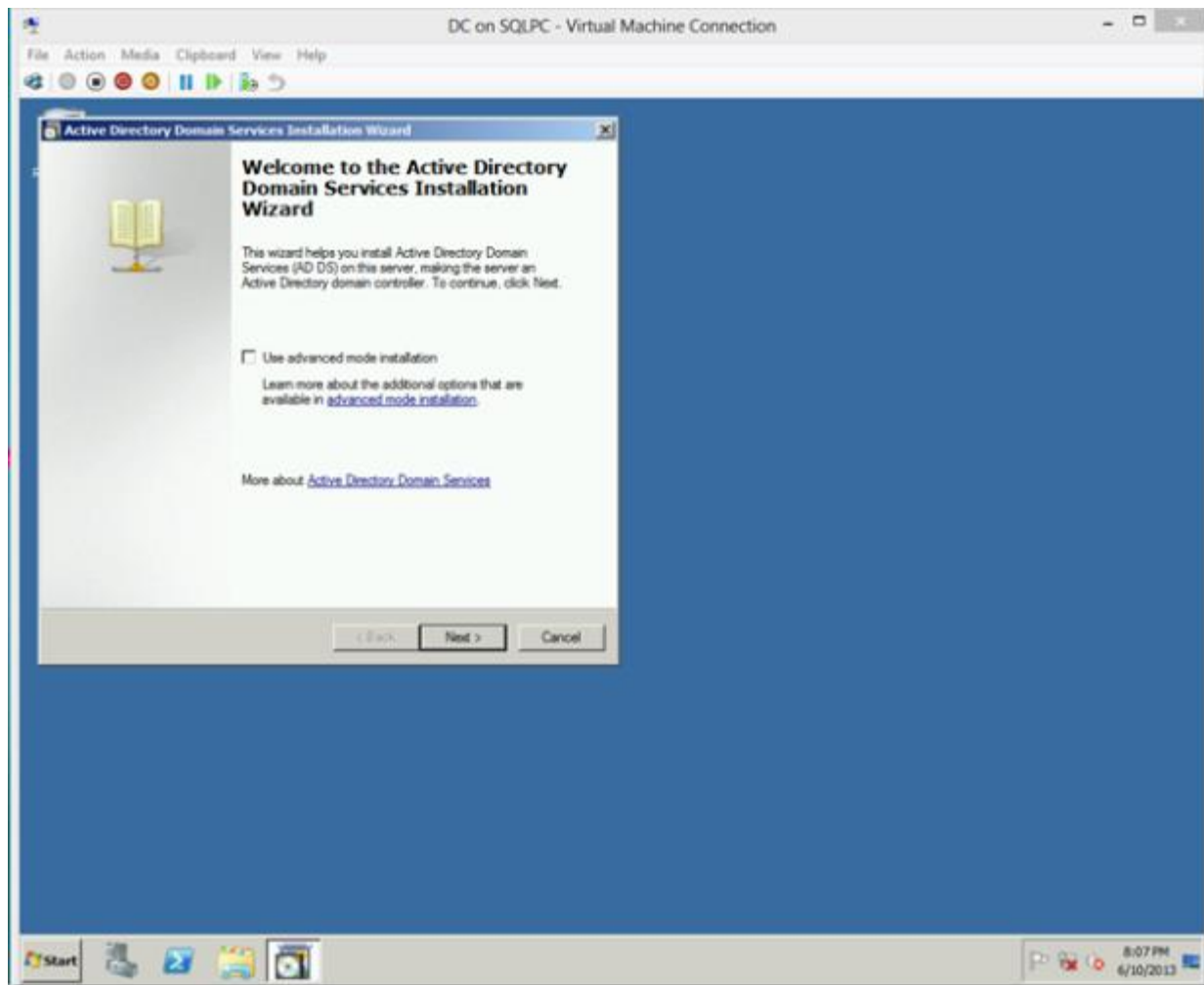
Advanced...

OK Cancel

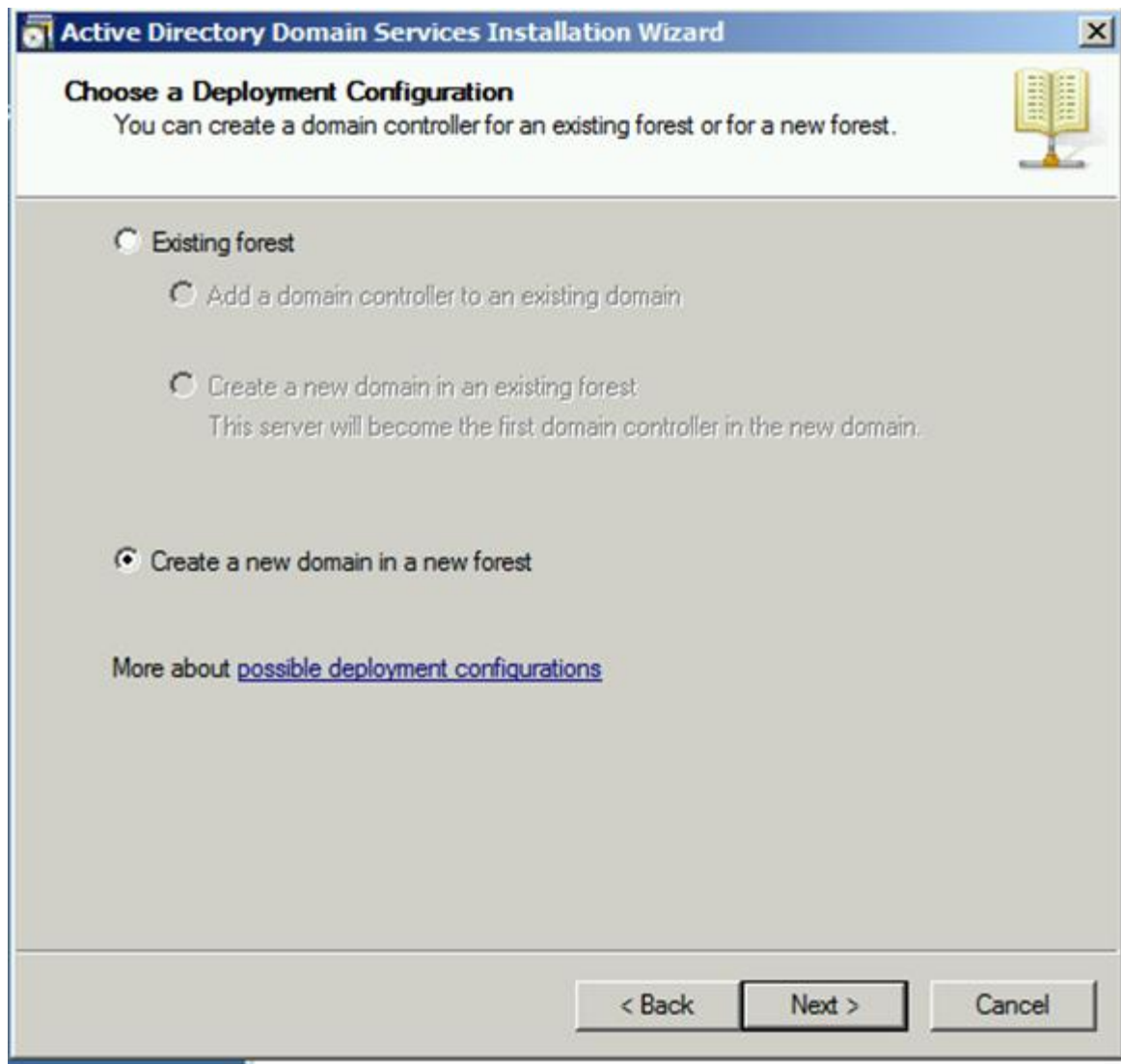
Open dcpromo .



Active directory domain service wizard will appear. Click Next and continue



Select create new domain in a new forest



Give forest root domain which acts as domain.

Active Directory Domain Services Installation Wizard

Name the Forest Root Domain

The first domain in the forest is the forest root domain. Its name is also the name of the forest.

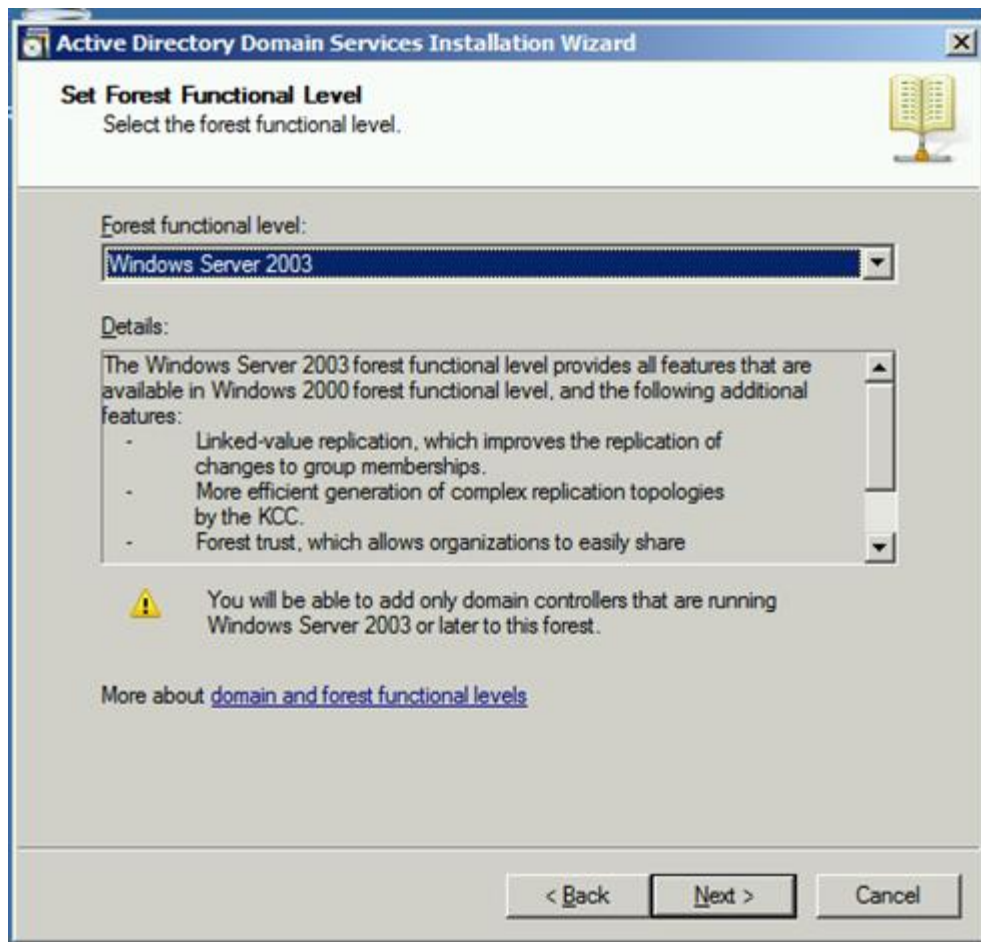
Type the fully qualified domain name (FQDN) of the new forest root domain.

FQDN of the forest root domain:

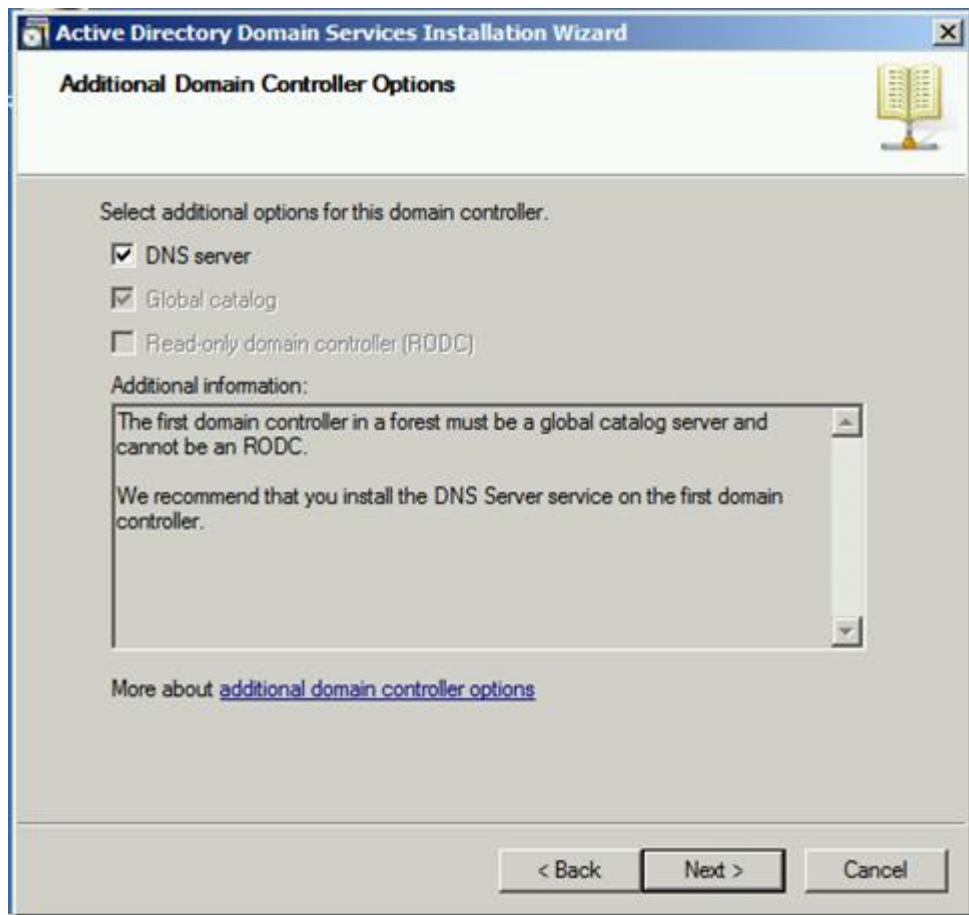
Example: corp.contoso.com

< Back Next > Cancel

Set forest functional level .Default Functional level is windows sever 2003.



Check DNS server and click Next.



Click Yes and continue.



Select the folder to store databases, log files and click Next.

Active Directory Domain Services Installation Wizard

Location for Database, Log Files, and SYSVOL

Specify the folders that will contain the Active Directory domain controller database, log files, and SYSVOL.

For better performance and recoverability, store the database and log files on separate volumes.

Database folder:

Log files folder:

SYSVOL folder:

More about [placing Active Directory Domain Services files](#)

Give the administrator password and click Next.

Active Directory Domain Services Installation Wizard

Directory Services Restore Mode Administrator Password

The Directory Services Restore Mode Administrator account is different from the domain Administrator account.

Assign a password for the Administrator account that will be used when this domain controller is started in Directory Services Restore Mode. We recommend that you choose a strong password.

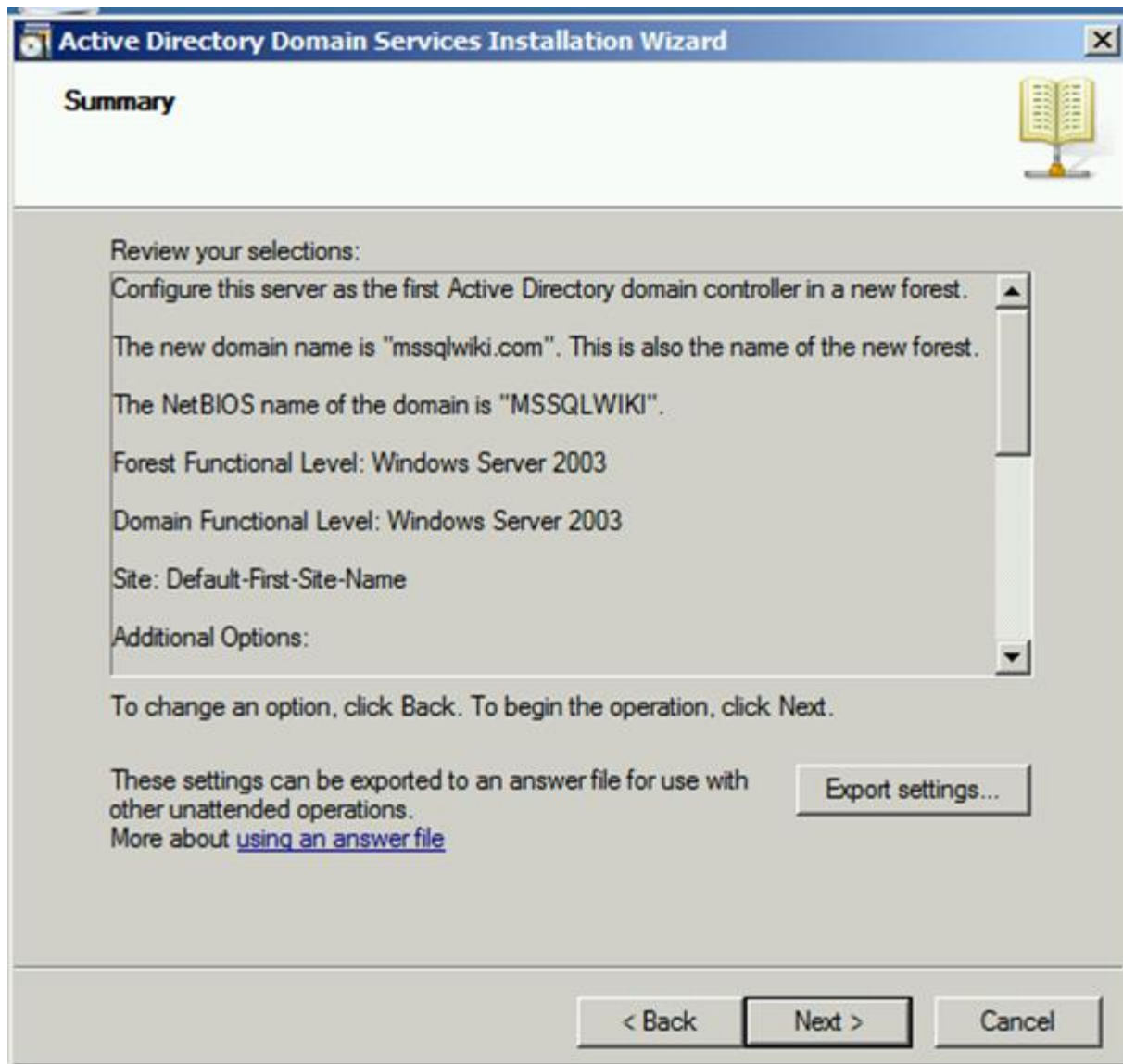
Password:

Confirm password:

More about [Directory Services Restore Mode password](#)

< Back Next > Cancel

Summary of your selections will appear, review your selection and click next.



DNS installation will be completed.

Active Directory Domain Services Installation Wizard

The wizard is configuring Active Directory Domain Services. This process can take from a few minutes to several hours, depending on your environment and the options that you selected.



Waiting for DNS installation to finish

Cancel

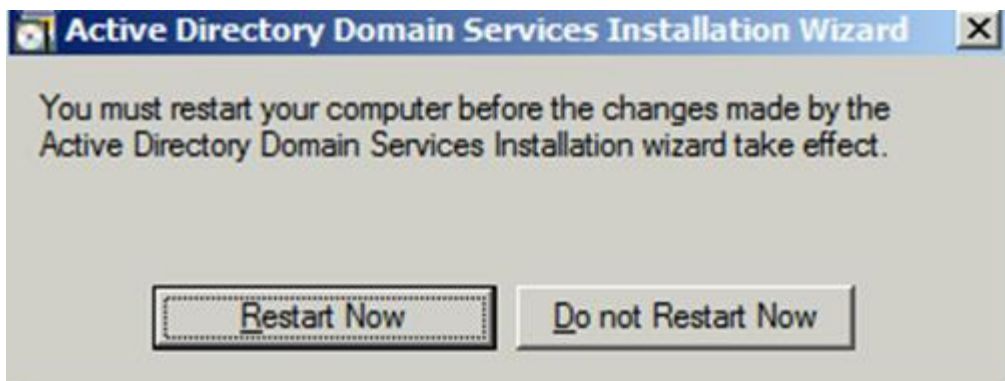


Reboot on completion

Click Finish and complete domain services installation wizard.

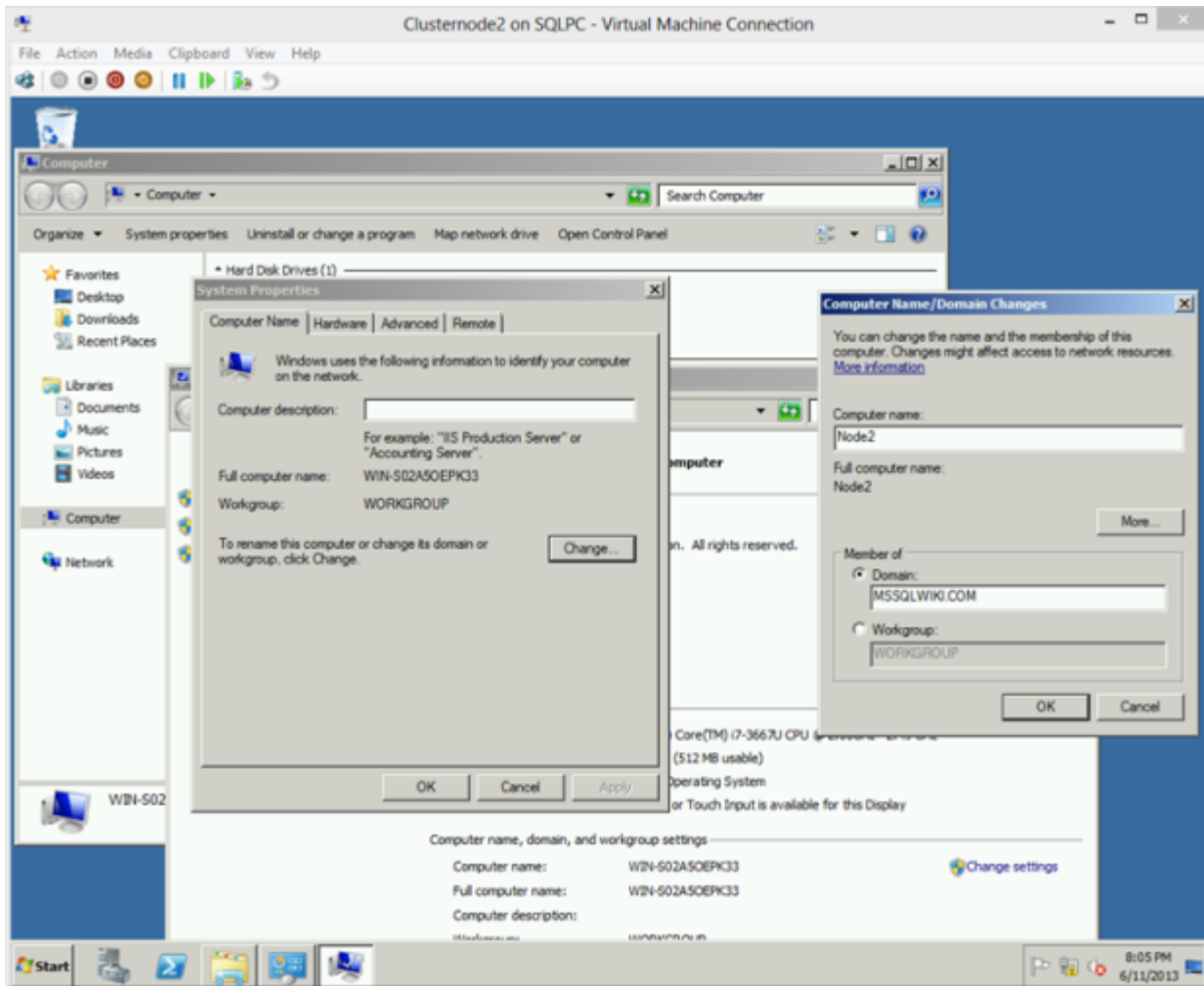


Restart your computer after installation.

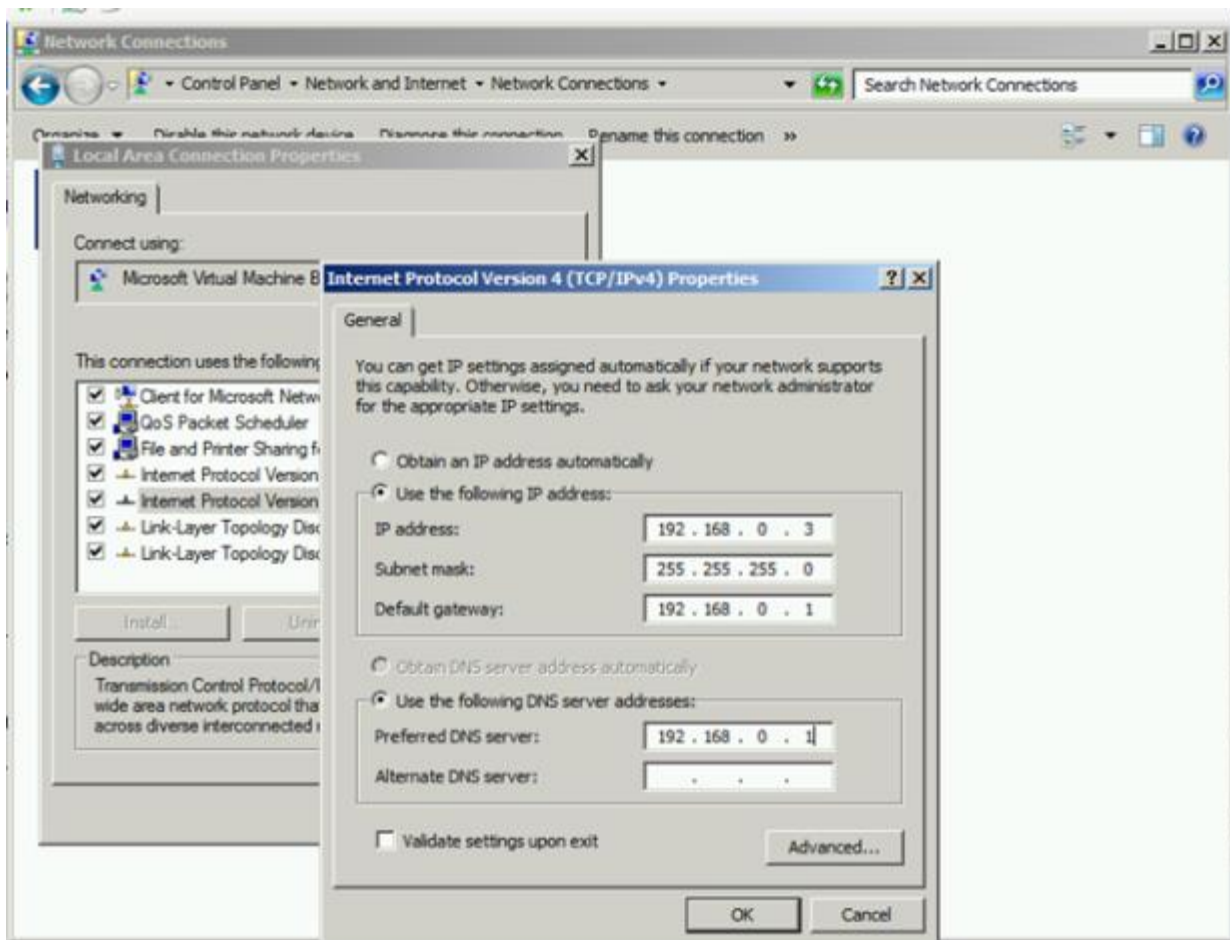


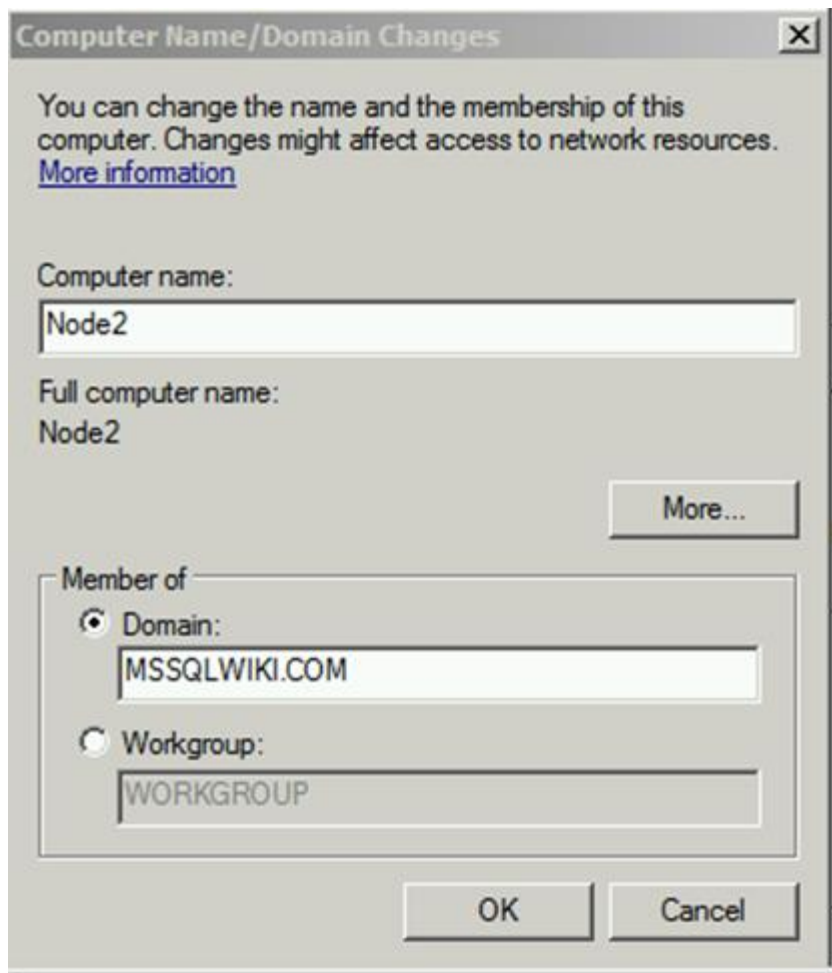
Now do the similar things for cluster nodes 1&2.

Select computer->properties->computer name. Change the computer name and give domain name which you gave in DC.

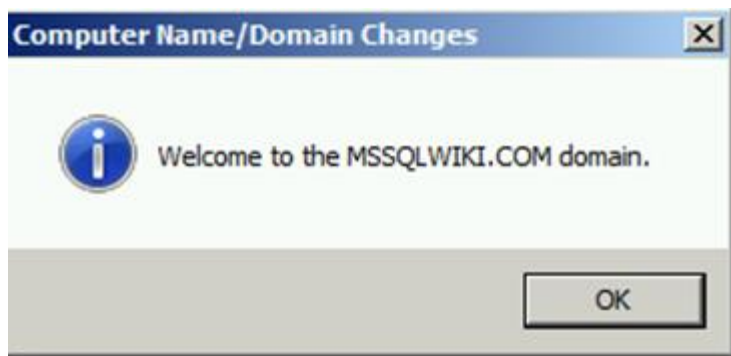


Configure IPV4 properties for cluster nodes.

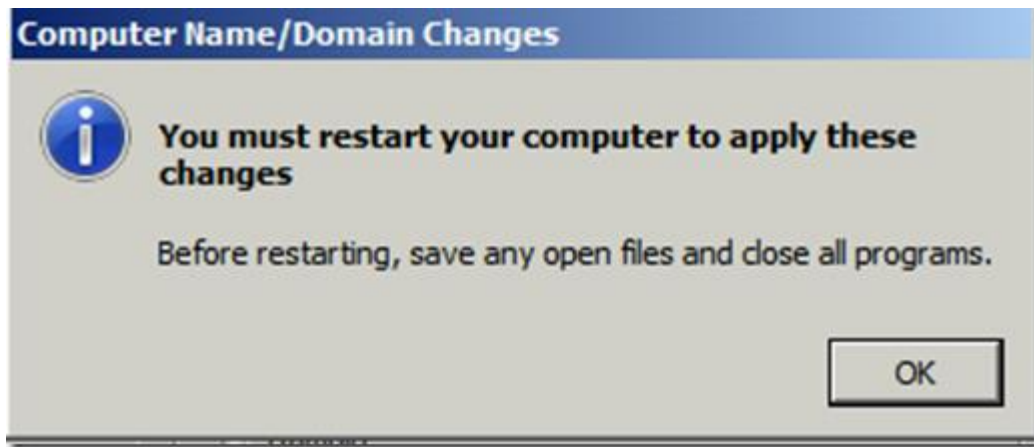




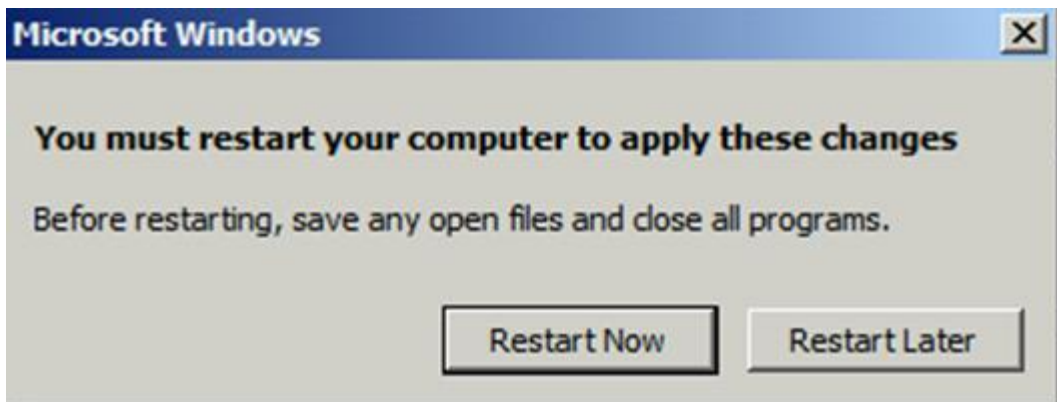
Click OK.



Restart your computer to apply the changes.



Click Ok,to restart.



Now we have domain controller and member servers. We have to install failover cluster feature and make this members servers as node of cluster which we are going to create.

To install the failover cluster feature and create a cluster follow the steps

in <https://sqlserverscribbles.com/2013/06/17/how-to-install-cluster-in-windows-2008-and-windows-2012/>

Once the cluster is created we have to create shared Disk for installing failover aware clustered application like SQL Server. Follow the steps in <https://sqlserverscribbles.com/2013/06/16/how-to-create-shared-disk-using-iscsi-software-target/> to create a shared disk using iSCSI target software.

How to install cluster in windows 2008 and windows 2012

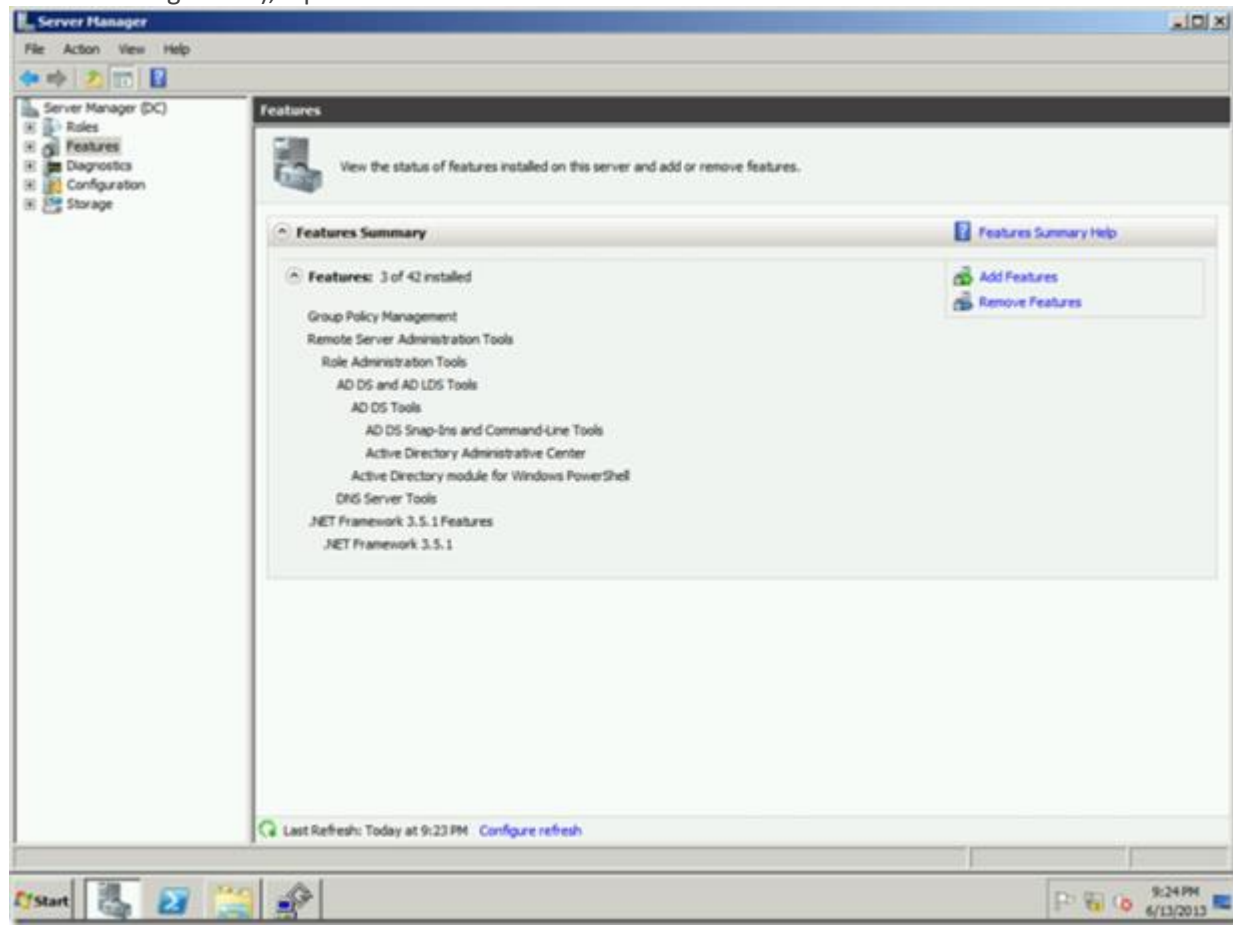
<https://sqlserverscribbles.com/2013/06/17/how-to-install-cluster-in-windows-2008-and-windows-2012/>

Failover cluster feature can be installed enterprise and datacenter edition of windows2008,windows2008R2 and windows2012.

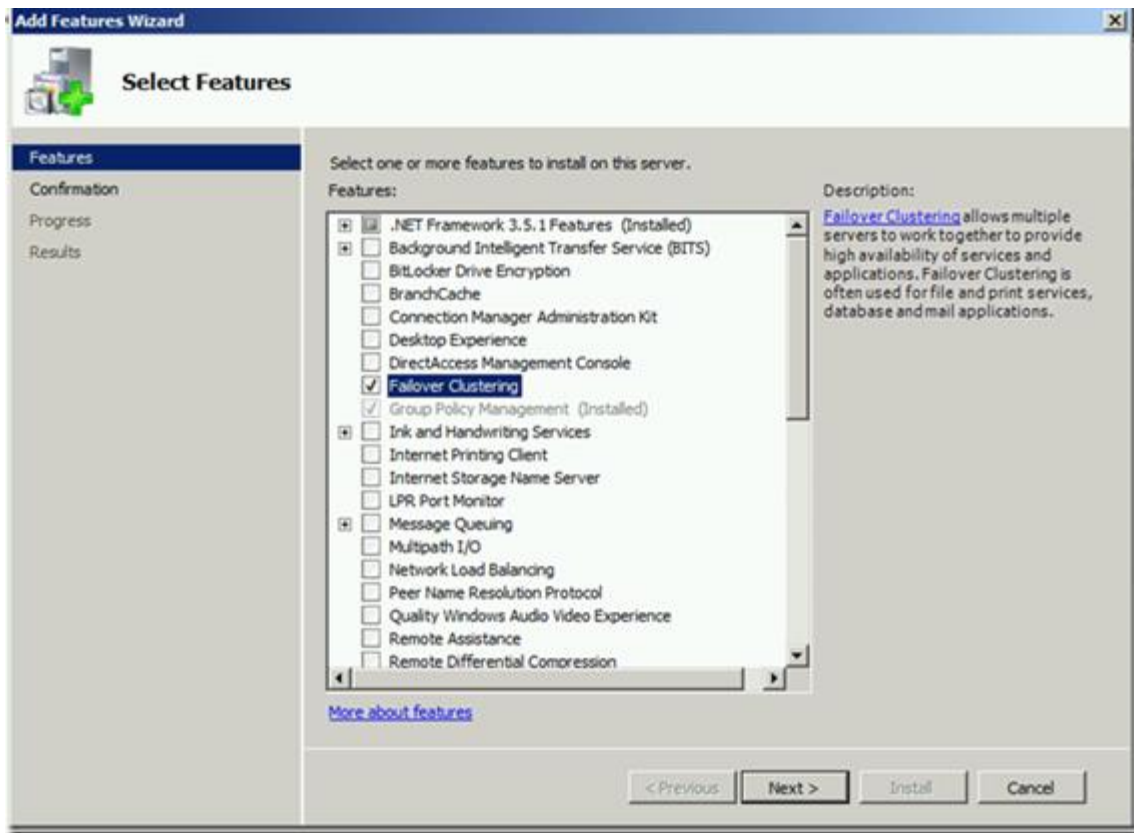
There are changes to the quorum configuration in windows2008,windows2008R2 and windows2012 cluster compared with earlier versions in which it is mandatory to have a shared disk to install the cluster (Exceptions Geo clusters). In this blog I will explain how to install failover cluster feature and create cluster. Shared disk is not mandatory for installing windows cluster but it is required for server applications like SQL Server when you are installing them on cluster.

If you would like to learn how to create shared disk for using in cluster follow the steps in [How to create shared disk using iSCSI Software Target](#)

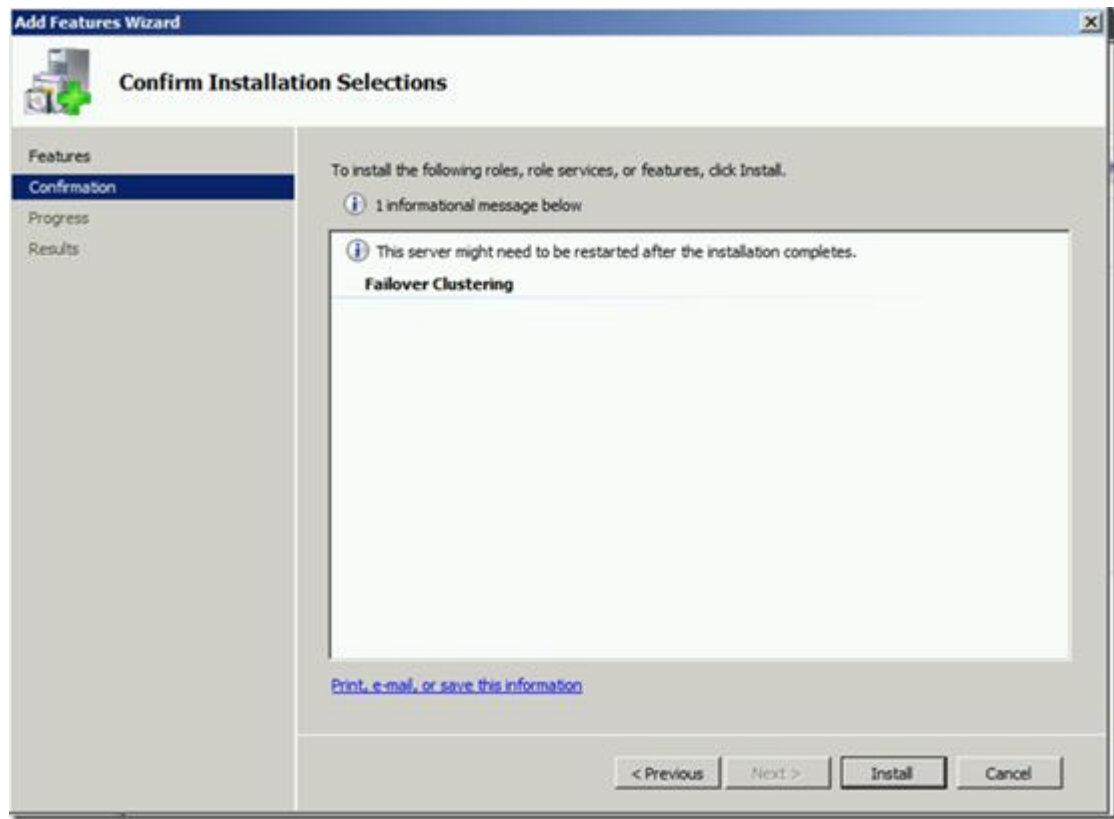
To install failover cluster feature Open Server Manager (Start—>Run—>Servermanager.msc),Expand Features.Select Add Features.



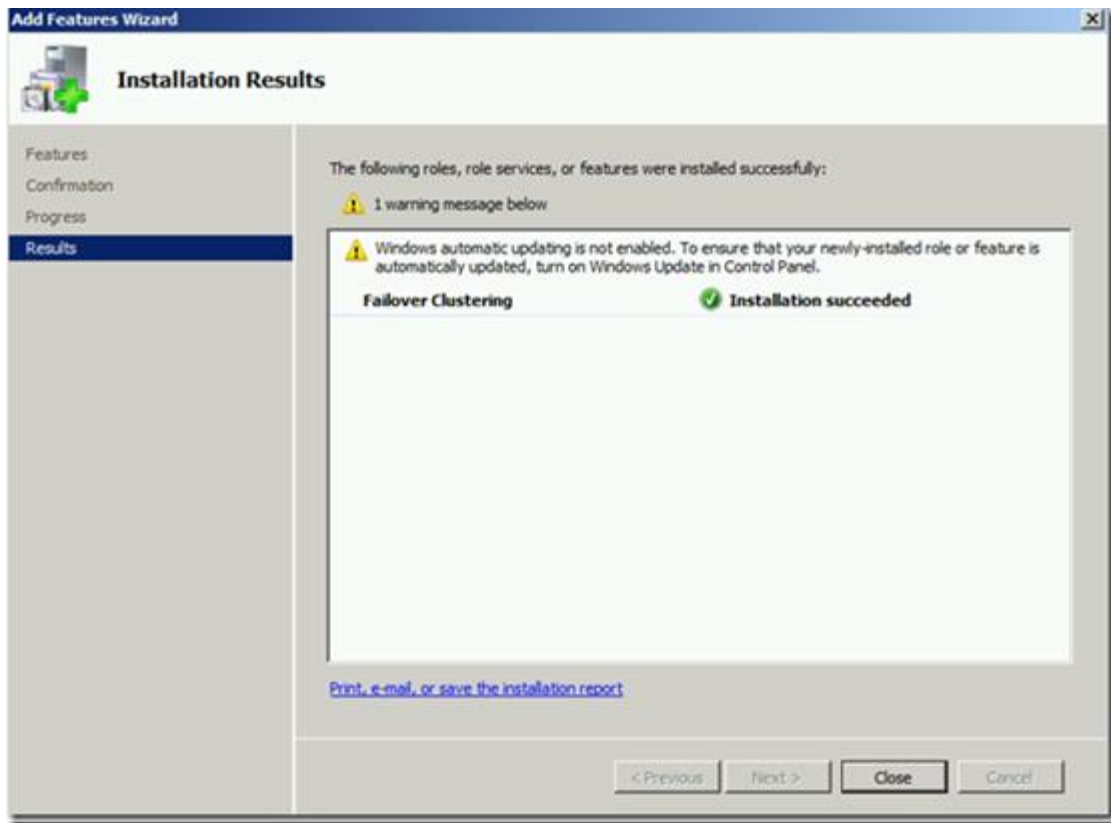
Then install Failover Clustering.



Confirm the installation and Click install.

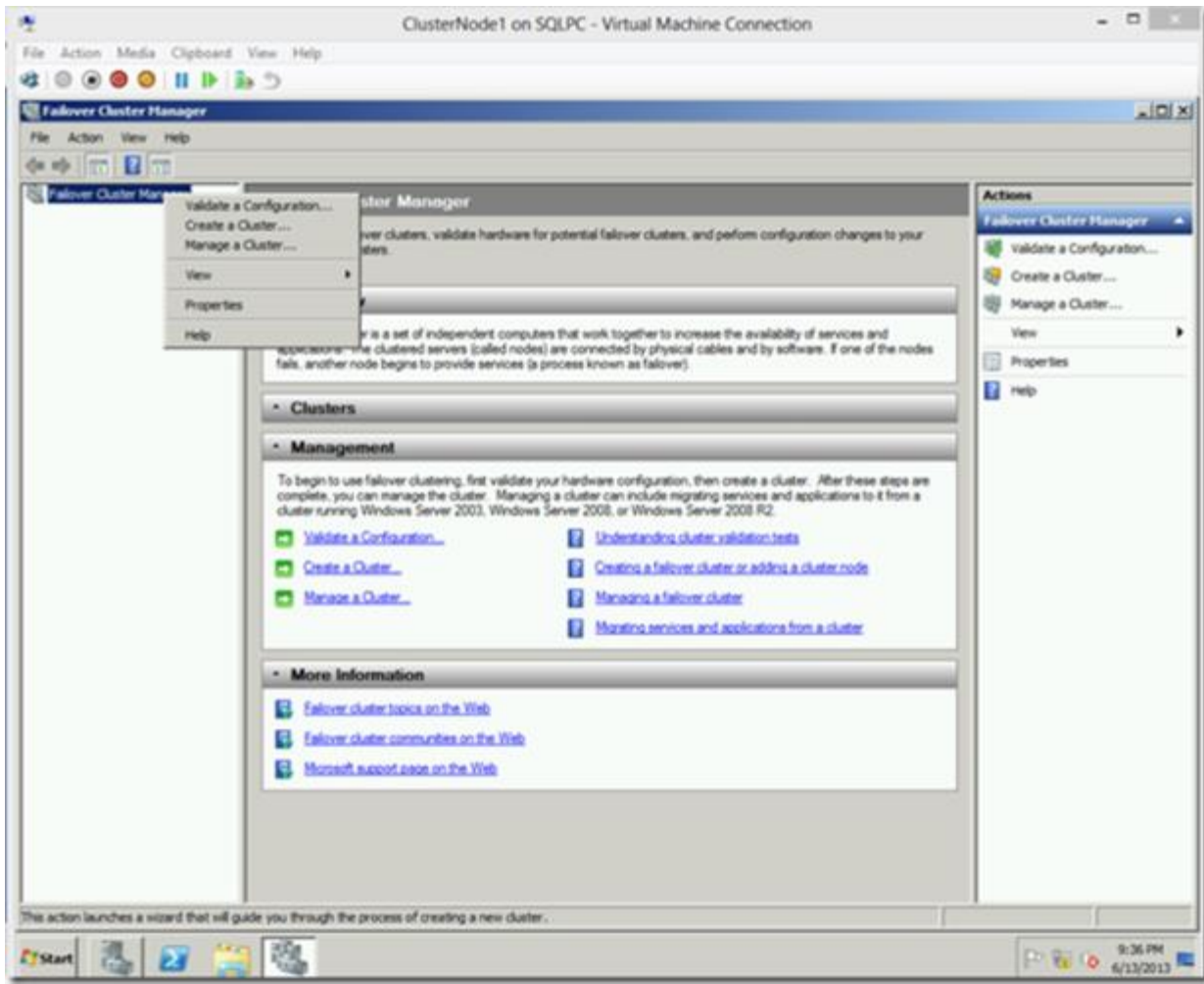


Click close and complete the installation.

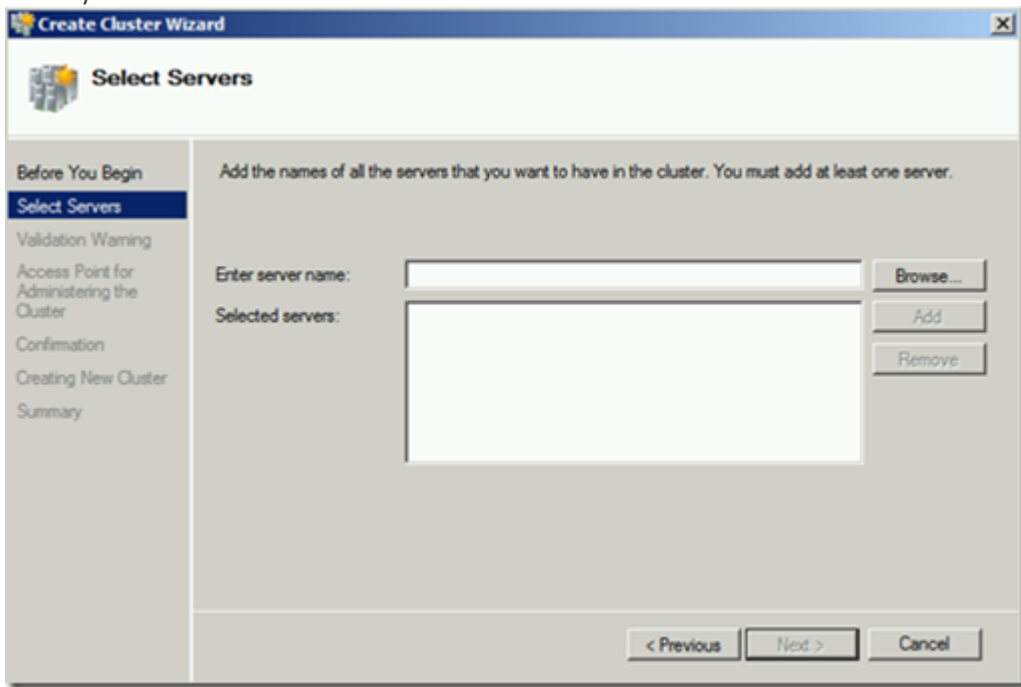


After installation ,open failover cluster manger.

Right click failover cluster manager->Create a cluster wizard.

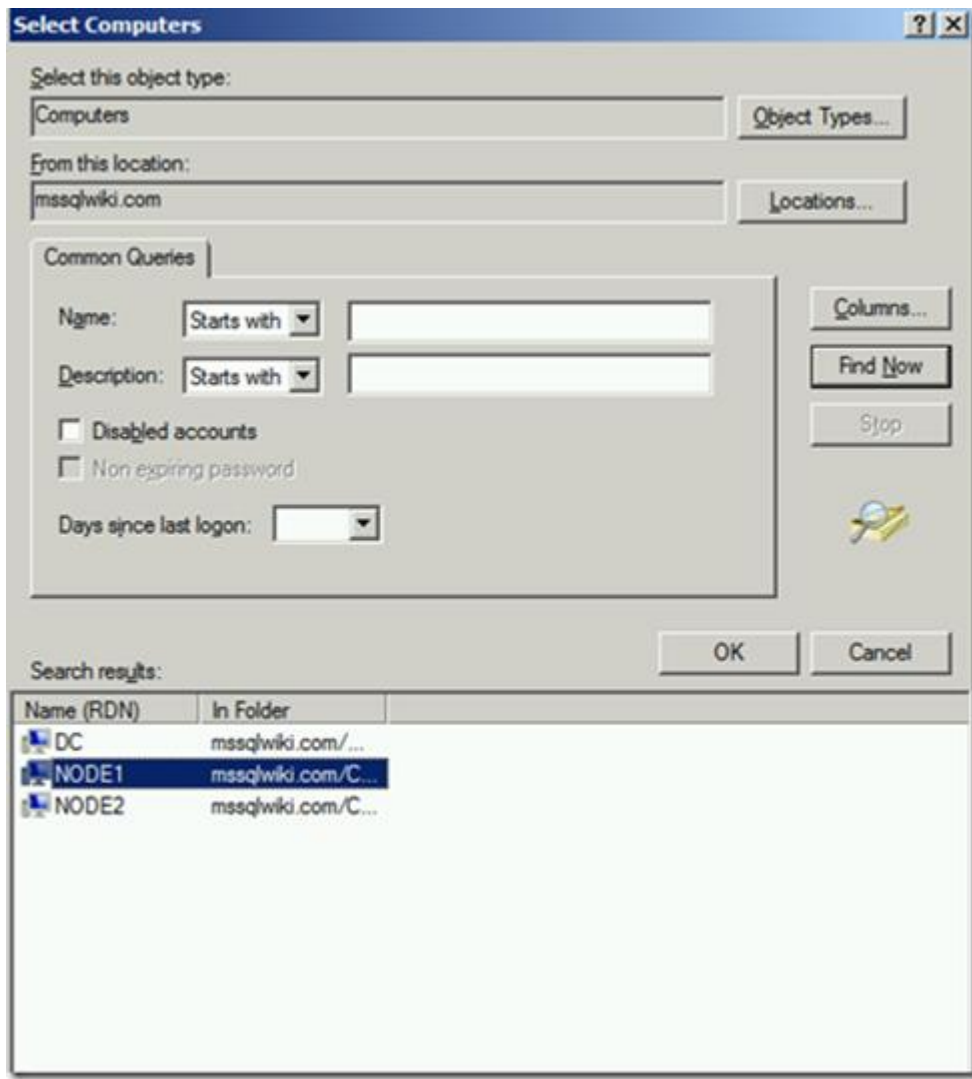


Enter Server name of all the nodes which will be part of cluster you are creating.If you do not know the server name you can click on browse and select the servers.

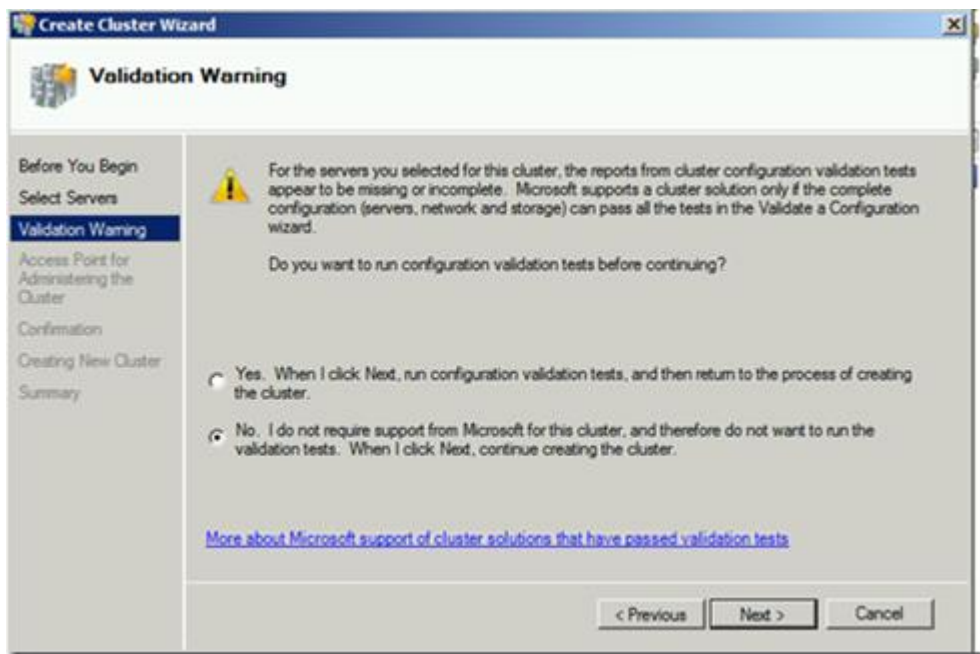


Below window appears when you click browse→Advanced→Find now. Select the systems which you want to add in the cluster.

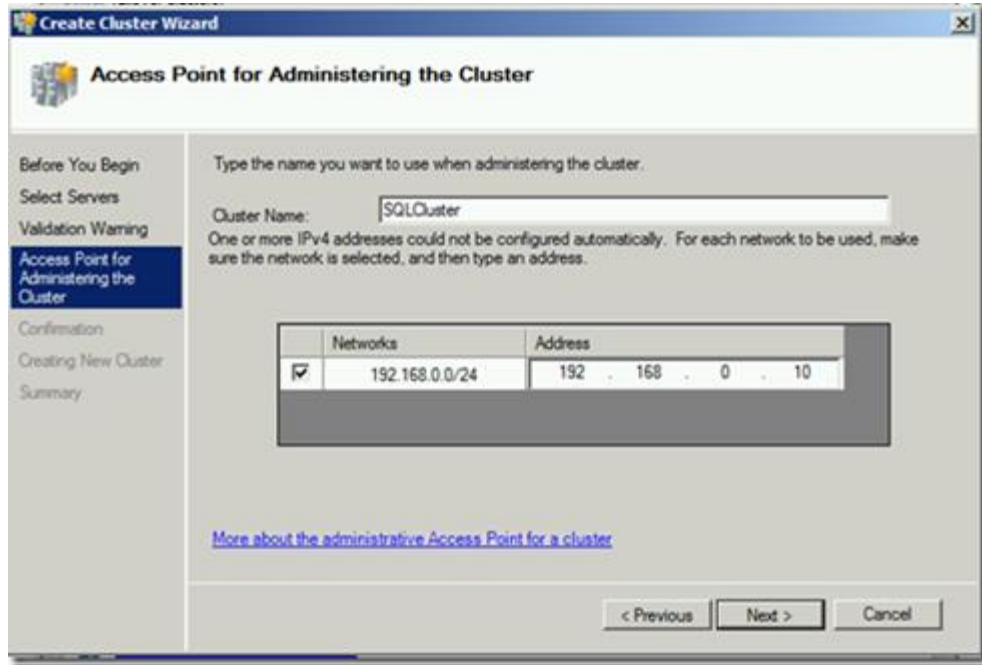
In the below example I have select Node1



Configure the validation warning and click Next.



Enter the IP address for cluster and Click Next.



Click **Finish** and complete the cluster creation.



Once you have created the cluster follow the steps in <http://blogs.msdn.com/b/clustering/archive/2010/05/14/10012930.aspx> to configure the Quorum setting for your cluster

How to create shared disk using iSCSI Software Target

<https://sqlserverscribbles.com/2013/06/16/how-to-create-shared-disk-using-iscsi-software-target/>

One of the mandatory requirement to install SQL Server and similar application in cluster is to have shared disk. I will explain how to create shared disk using iSCSI software target in this blog. You can download iSCSI Software Target from [This](#) link.

<https://www.microsoft.com/en-in/download/details.aspx?id=19867&ranMID=24542&ranEAID=TnL5HPStwNw&ranSiteID=TnL5HPStwNw->

[aZFud5PdSm40qn35_Q.D4Q&epi=TnL5HPStwNw-](#)

[aZFud5PdSm40qn35_Q.D4Q&irgwc=1&OCID=AID2000142_aff_7593_1243925&tduid=%28ir_mx9kkrf3rkkft1xpk0sohzzx32xihx9dkta2f9k00%29%287593%29%281243925%29%28TnL5HPStwNw-aZFud5PdSm40qn35_Q.D4Q%29%28%29&irclickid= mx9kkrf3rkkft1xpk0sohzzx32xihx9dkta2f9k00](#)

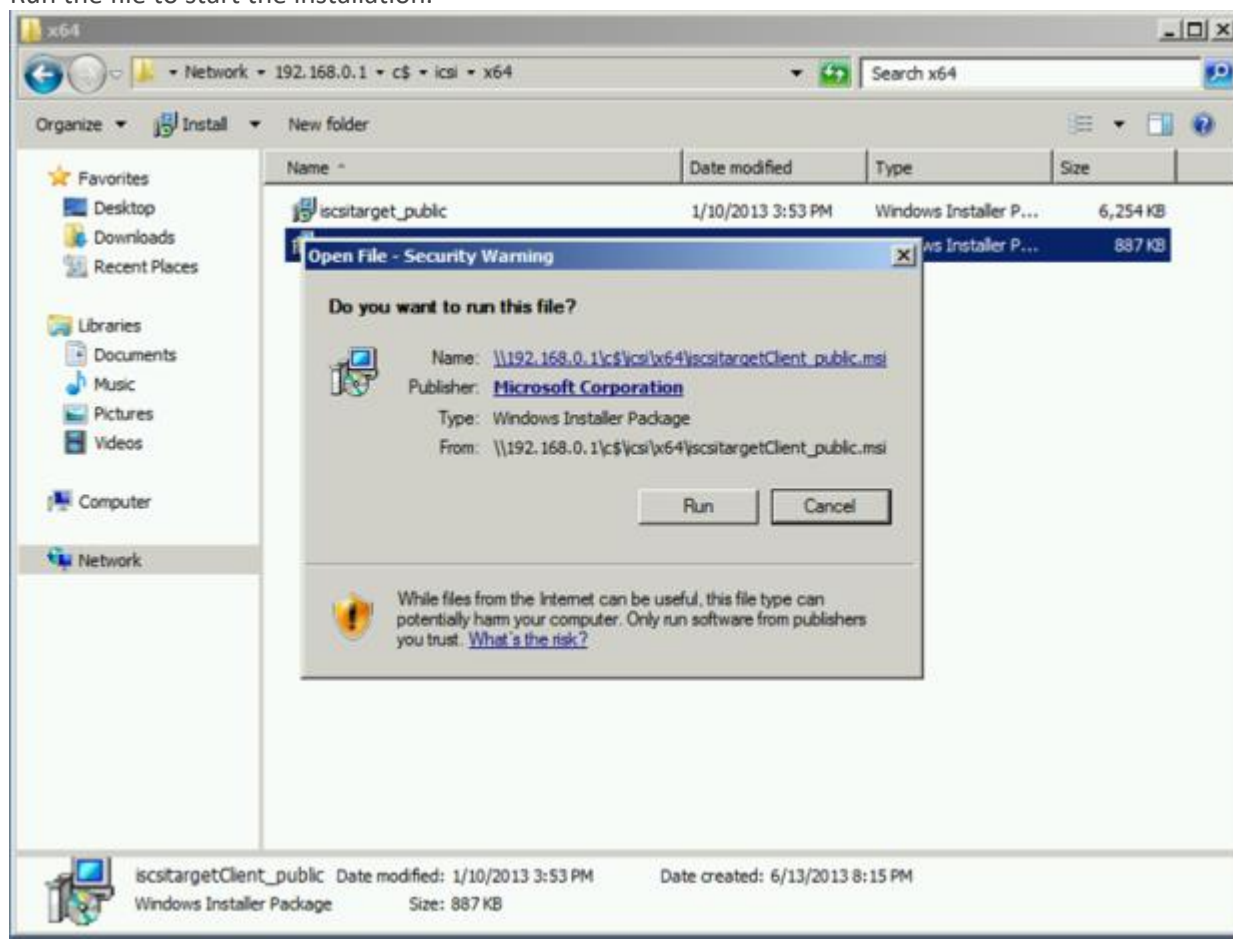
Installing Microsoft iSCSI software target.

When you invoke the self extracting package from downloaded above link it extracts X64 and x86 version of installer.

- Select iSCSI_target for installation on the system which will act as iSCSI target. We will create iSCSI target disks in this machine and these disks will be shared by nodes using iSCSI initiator.
- Select iSCSITargetClient_public and install in nodes in which you would like to have shared disk probably all nodes which is going to be part of cluster you are installing.

INSTALLATION IN CLUSTER NODES:(Follow same steps in all the nodes which will be part of cluster which you are installing)

Install iSCSITargetClient_public in all nodes which are going to be part of cluster which you are installing .
Run the file to start the installation.



Click Next and continue

Select the type of installation as **Typical**.



Specify users account that is member of administrators group in system which is acting iSCSI target and iSCSI initiator (current system).

Microsoft iSCSI Software Target Client Setup Wizard

User Name and Password

Type the user name and password for the VSS/VDS hardware providers.

Specify a domain user account that is a member of the Administrators group on the Microsoft iSCSI Software Target and the iSCSI initiator computer. The computers must belong to a domain and the user account must be in the format Domain\UserName.

User name:

Password:

< Back Next > Cancel

Configure other settings.

Microsoft iSCSI Software Target Client Setup

Configure Microsoft Update

Use Microsoft Update to help keep your computer secure and up-to-date.

Microsoft Update offers security and important updates for Windows and other Microsoft products, including the Microsoft iSCSI Software Target Client. Updates are delivered using your Automatic Updates setting, or you can visit the Microsoft Update Web site.

☒ Use Microsoft Update when I check for updates (recommended).

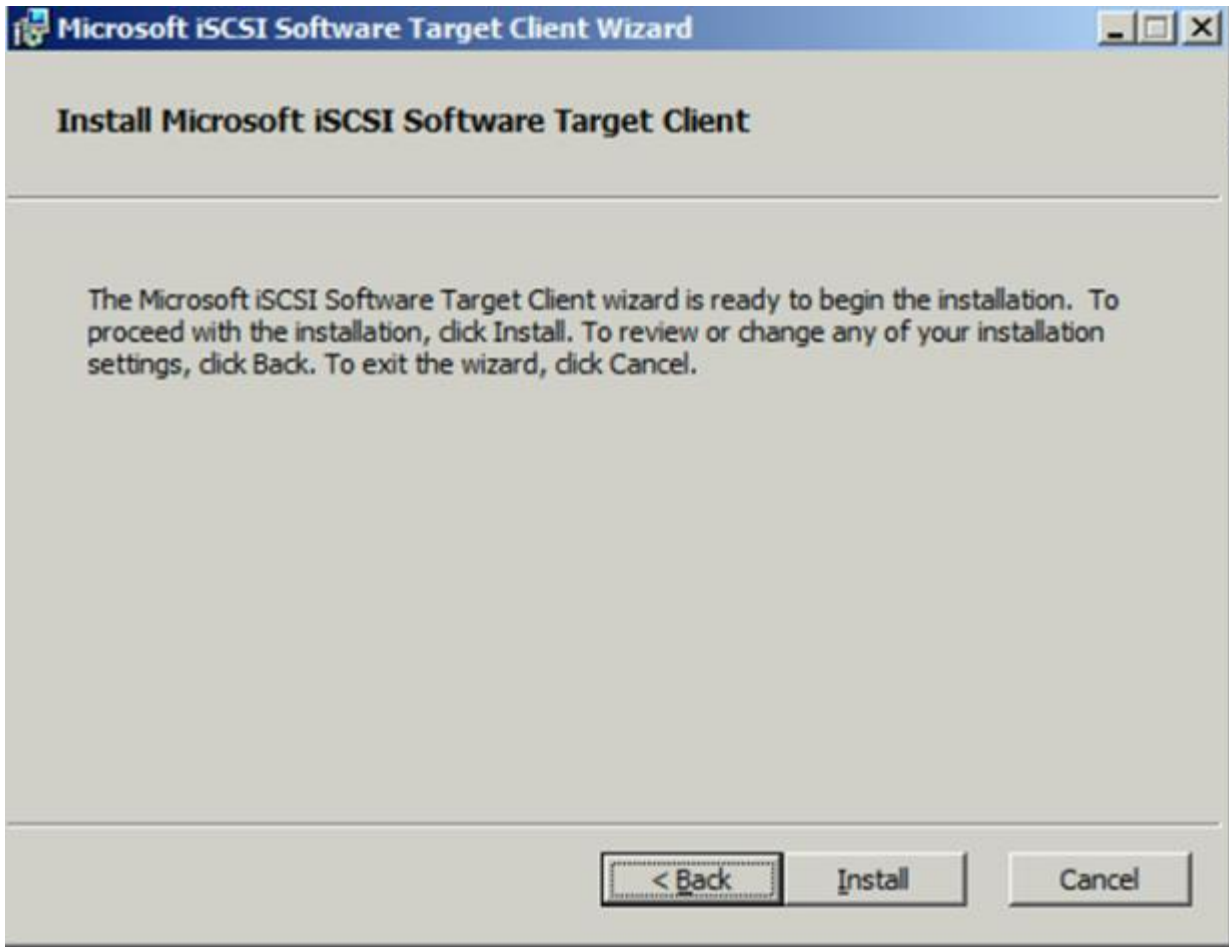
☐ I don't want to use Microsoft Update.

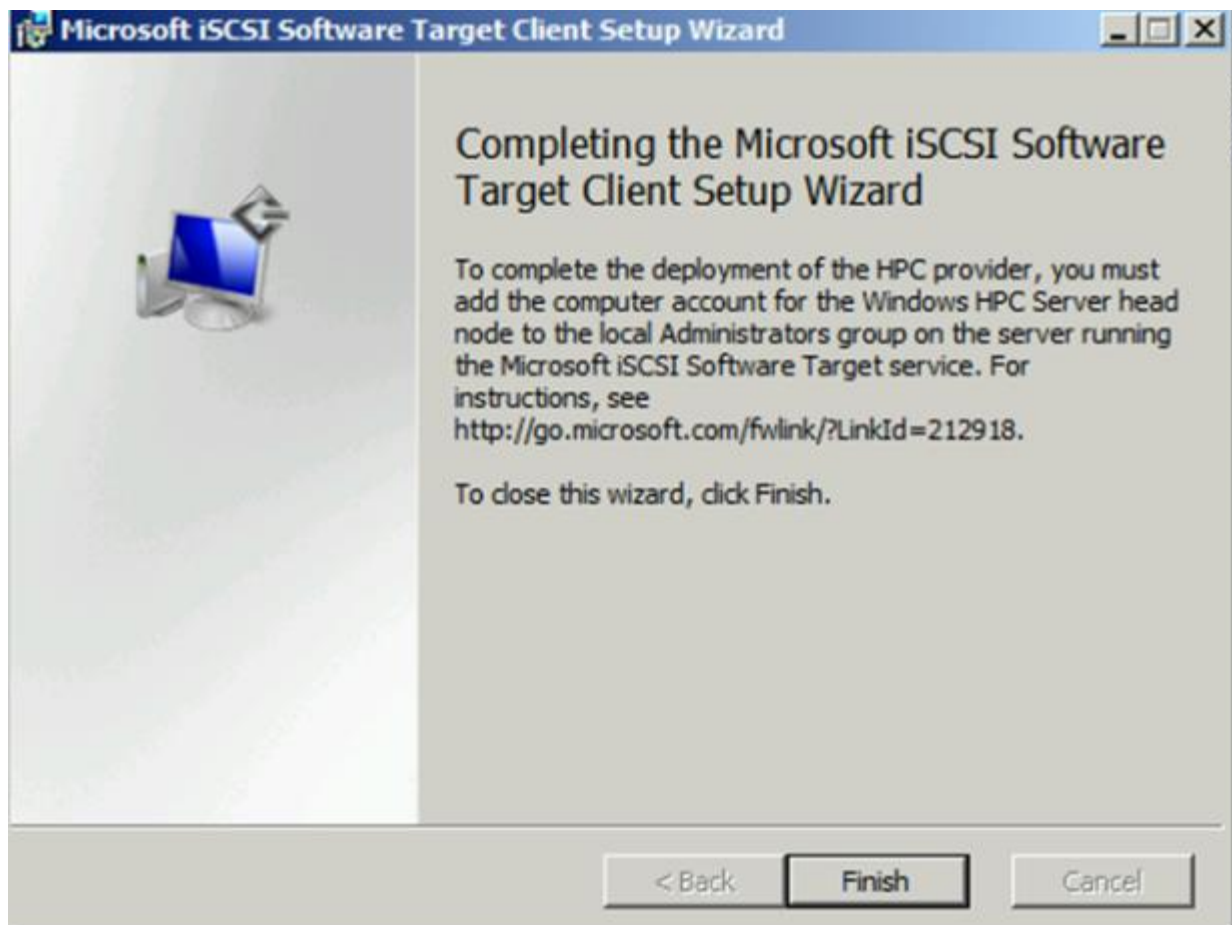
See the Microsoft Update FAQ at <http://go.microsoft.com/fwlink/?LinkID=57191>.

Read the Microsoft Update Privacy Statement at <http://go.microsoft.com/fwlink/?LinkID=57190>.

< Back Next > Cancel

Start the installation by clicking Install button.



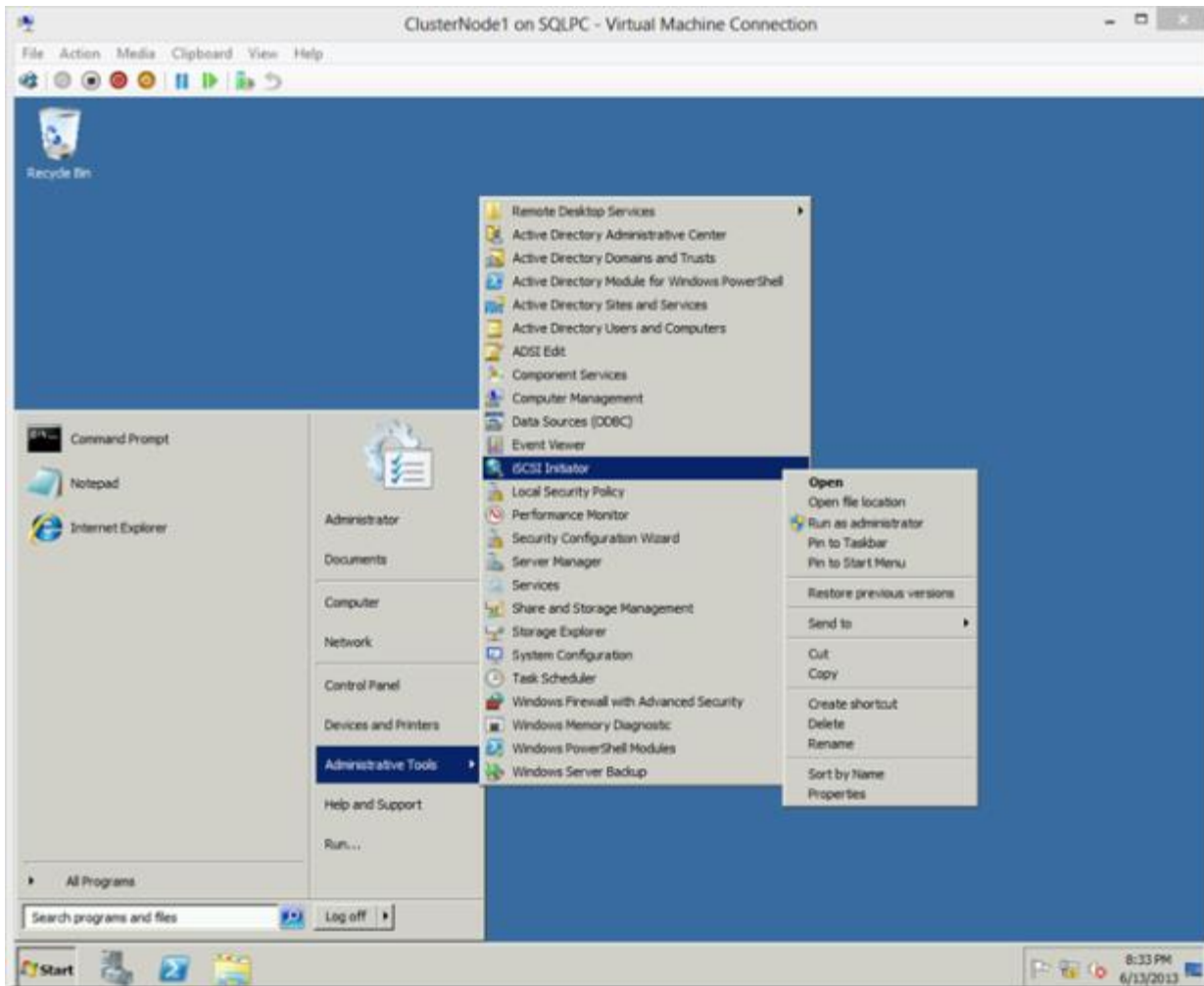


Click finish to complete the installation

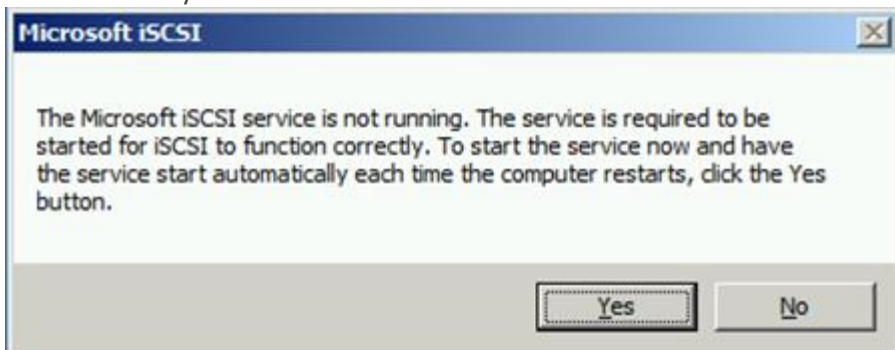
Install the iSCSI_target on the system which will act as iSCSI target probably file server (or) Domain controller if you are installing failover clustering for testing.

CONFIGURE iSCSI INITIATOR IN CLUSTER NODE'S:

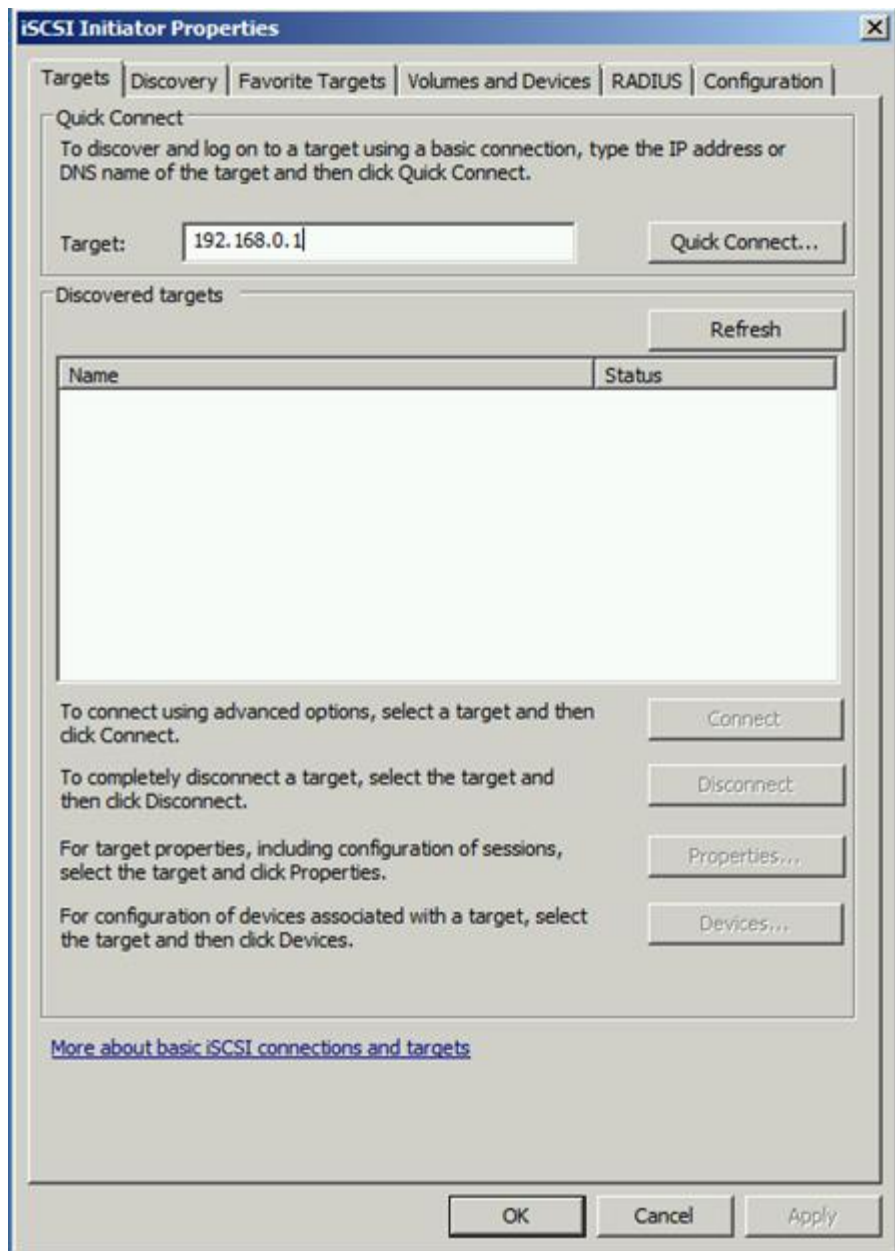
After installation, Open iSCSI initiator.



- Select yes and continue.



- Type the target server name and click on **Quick Connect**



(Or)

- Click **Discover panel** and select the target sever.

iSCSI Initiator Properties

Targets | **Discovery** | Favorite Targets | Volumes and Devices | RADIUS | Configuration

Target portals

The system will look for Targets on following portals:

Address	Port	Adapter	IP address
---------	------	---------	------------

Refresh

To add a target portal, click Discover Portal.

Discover Portal...

To remove a target portal, select the address above and then click Remove.

Remove

iSNS servers

The system is registered on the following iSNS servers:

Name

Refresh

To add an iSNS server, click Add Server.

Add Server...

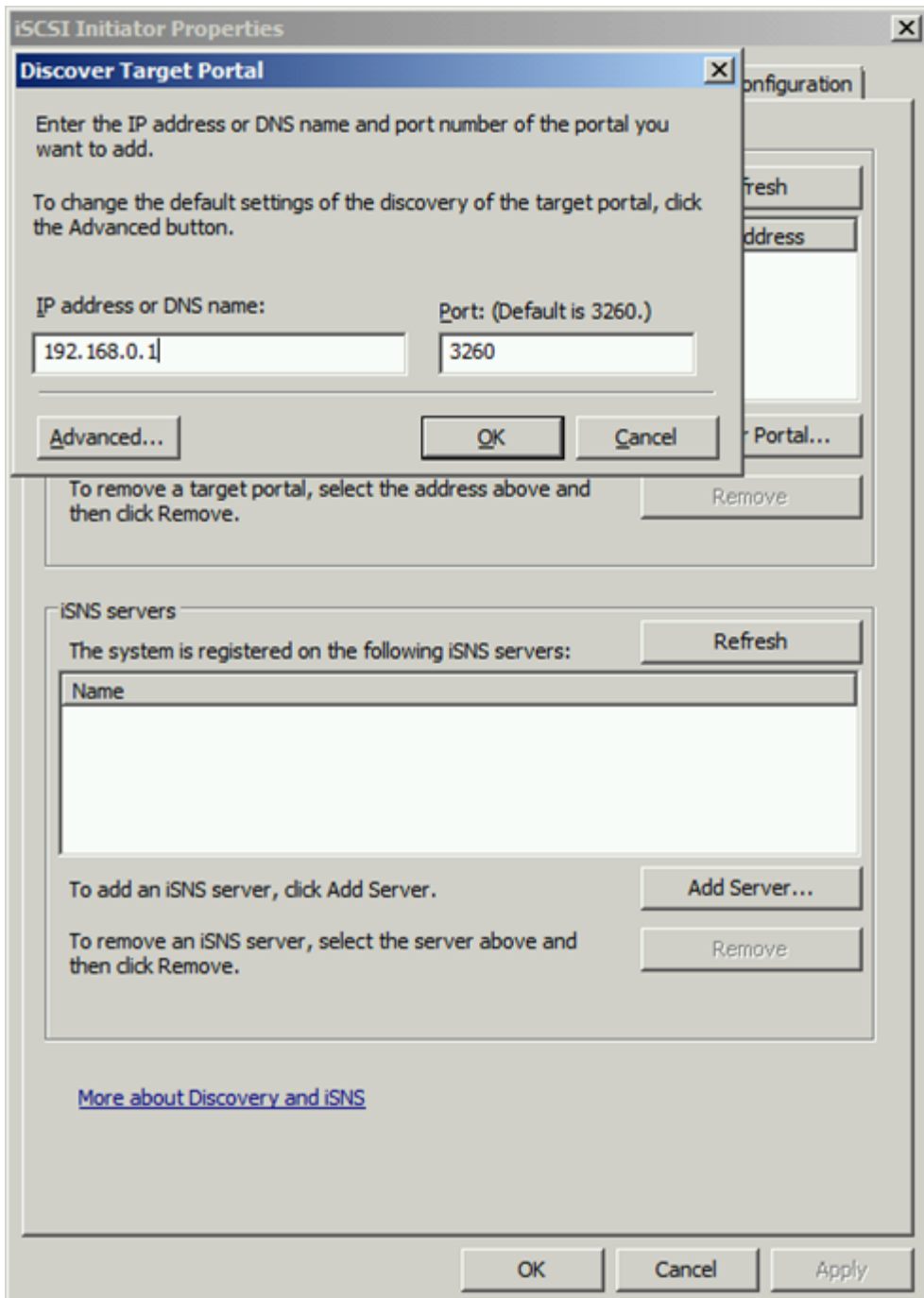
To remove an iSNS server, select the server above and then click Remove.

Remove

[More about Discovery and iSNS](#)

OK Cancel Apply

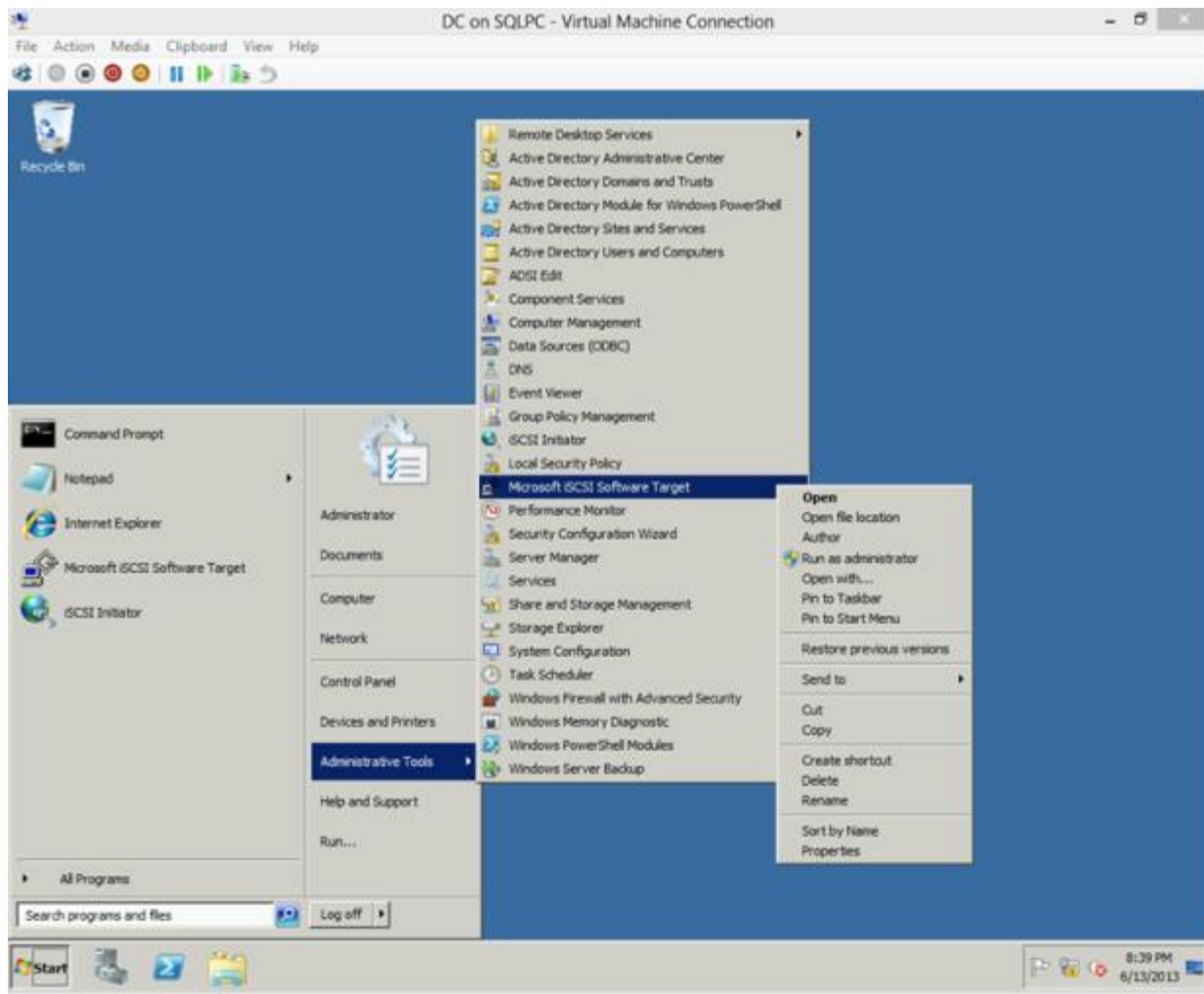
- Give the server IP address. Click ok.



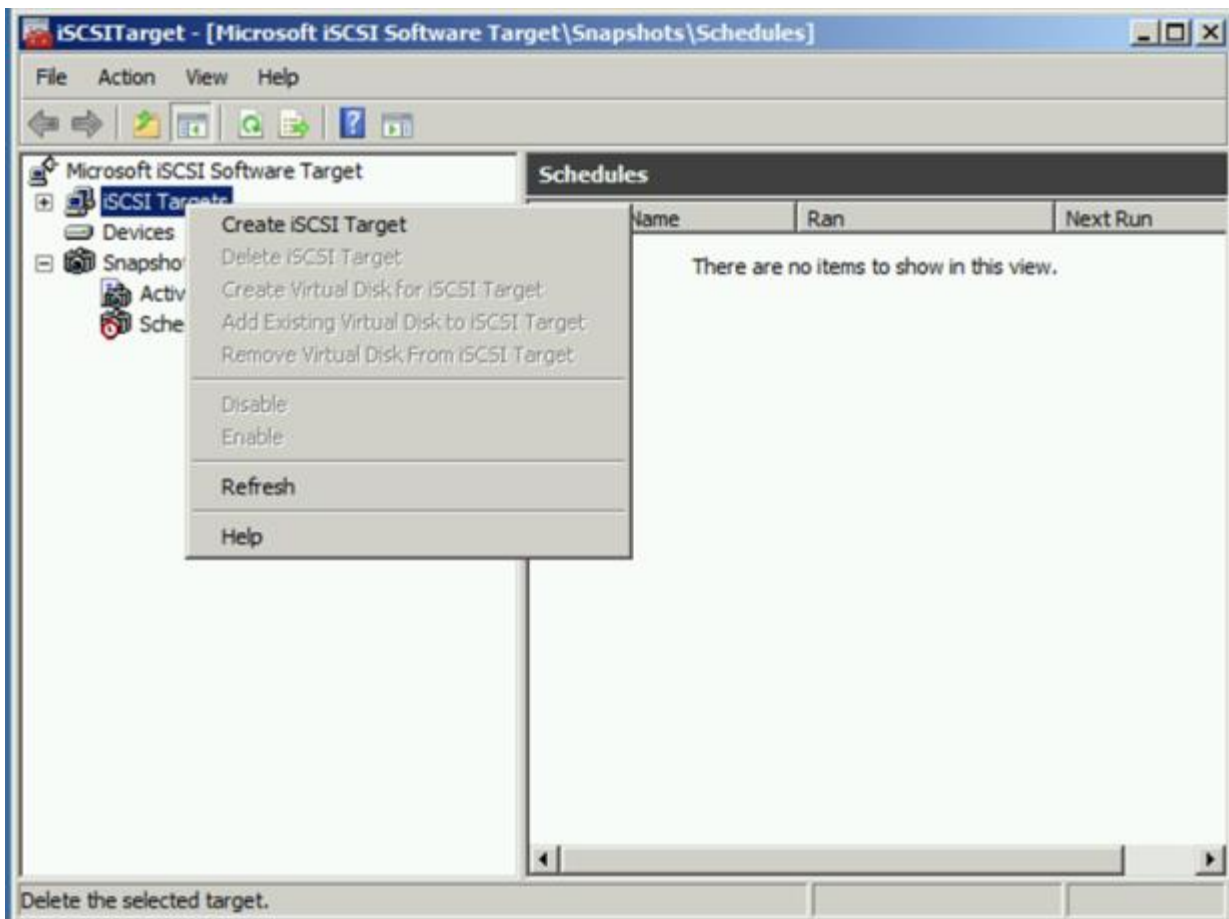
Similarly Follow the above steps and select the target in all nodes of cluster.

Configuring iSCSI target

Now after selecting the target, we have to add all the iSCSI clients in iSCSI target. Open MS iSCSI Target in system which is acting as iSCSI target.



Right click iSCSI target->create iSCSI target.



Create target wizard will open.



Specify the target node name,click Next.

Create iSCSI Target Wizard

iSCSI Target Identification
Specify a name and description to identify the iSCSI target you want to create.

iSCSI target name:

Description:

<Previous Next > Cancel

Browse and select the iSCSI initiator,click ok.

Add iSCSI Initiator

You can type or select an iSCSI initiator from the list, which includes all initiators connected to the Microsoft iSCSI Software Target in the past seven days and initiators that registered with the iSNS server.

iSCSI Initiator Name:

iSCSI Initiators:

IQN
iqn.1991-05.com.microsoft.node1.mssqlwiki.com
iqn.1991-05.com.microsoft.node2.mssqlwiki.com

Refresh OK Cancel

iSCSI Initiator Selection

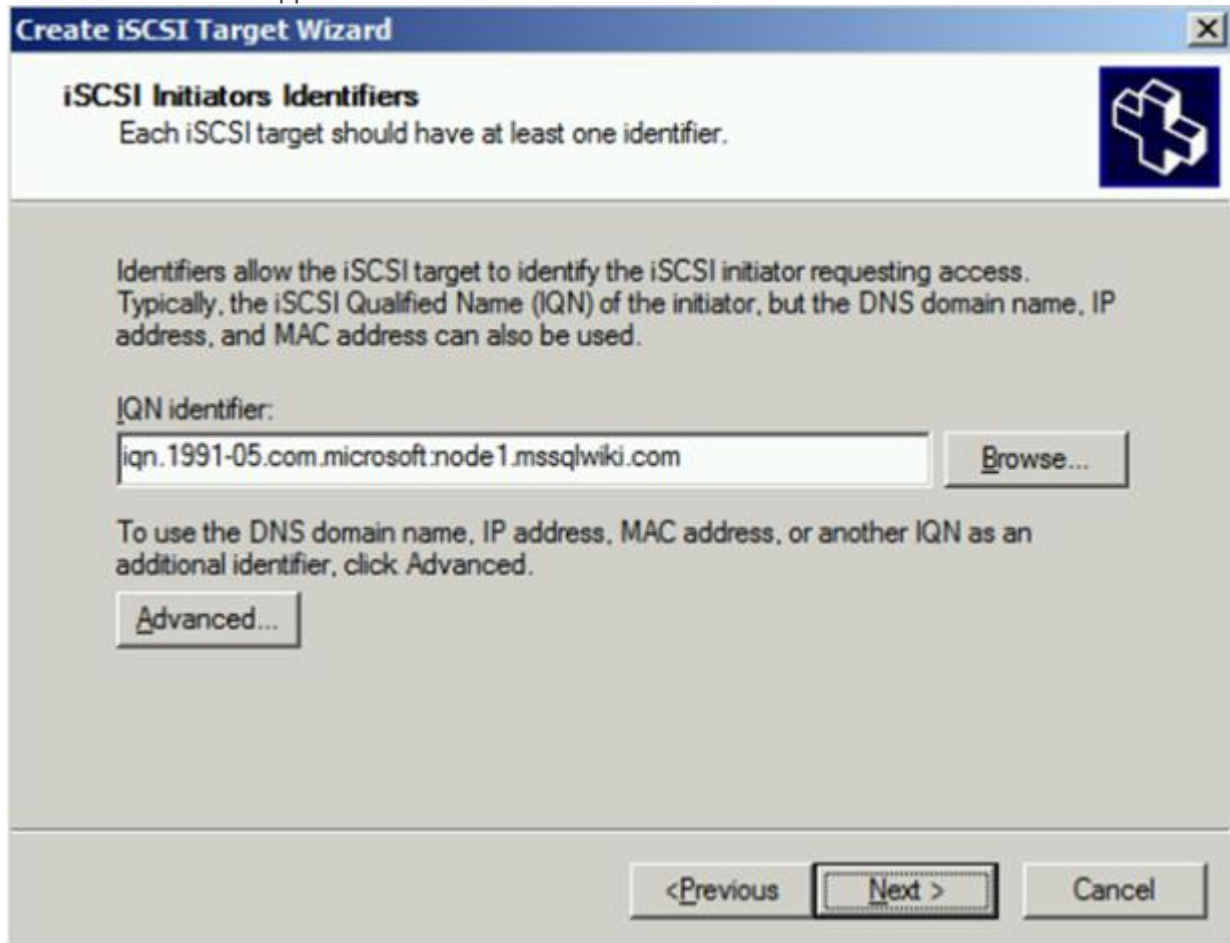
requesting access.
the DNS domain name, IP

Browse...

another IQN as an

Next > Cancel

Selected initiator will appear. Click Next.



Create iSCSI Target Wizard

iSCSI Initiators Identifiers

Each iSCSI target should have at least one identifier.

Identifiers allow the iSCSI target to identify the iSCSI initiator requesting access. Typically, the iSCSI Qualified Name (IQN) of the initiator, but the DNS domain name, IP address, and MAC address can also be used.

IQN identifier:

To use the DNS domain name, IP address, MAC address, or another IQN as an additional identifier, click **Advanced...**

Click **Finish** and complete the target creation.



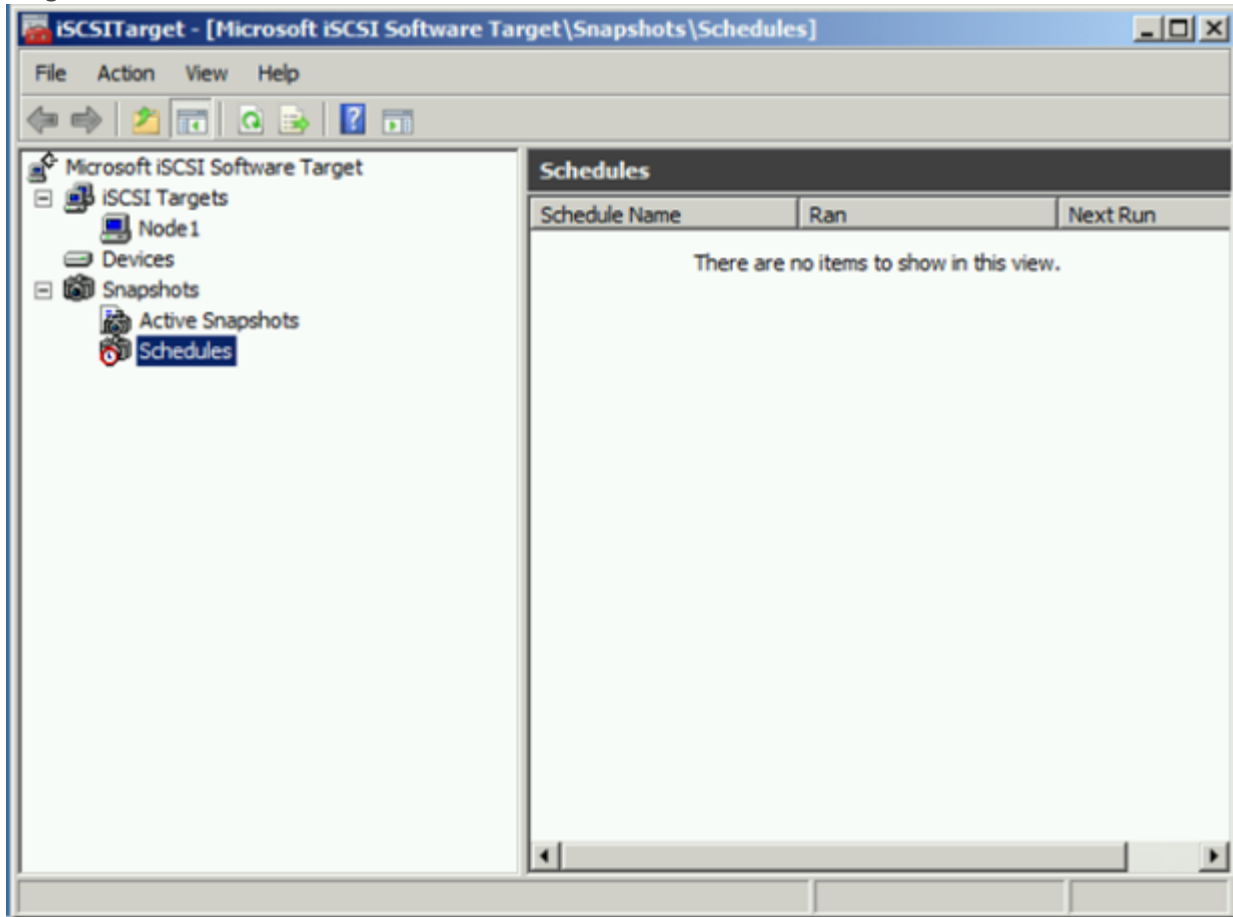
Create iSCSI Target Wizard

Completing the Create iSCSI Target Wizard

You have successfully completed the Create iSCSI Target Wizard.

To close this wizard and create the target, click **Finish**.

Target **Node1** is added.

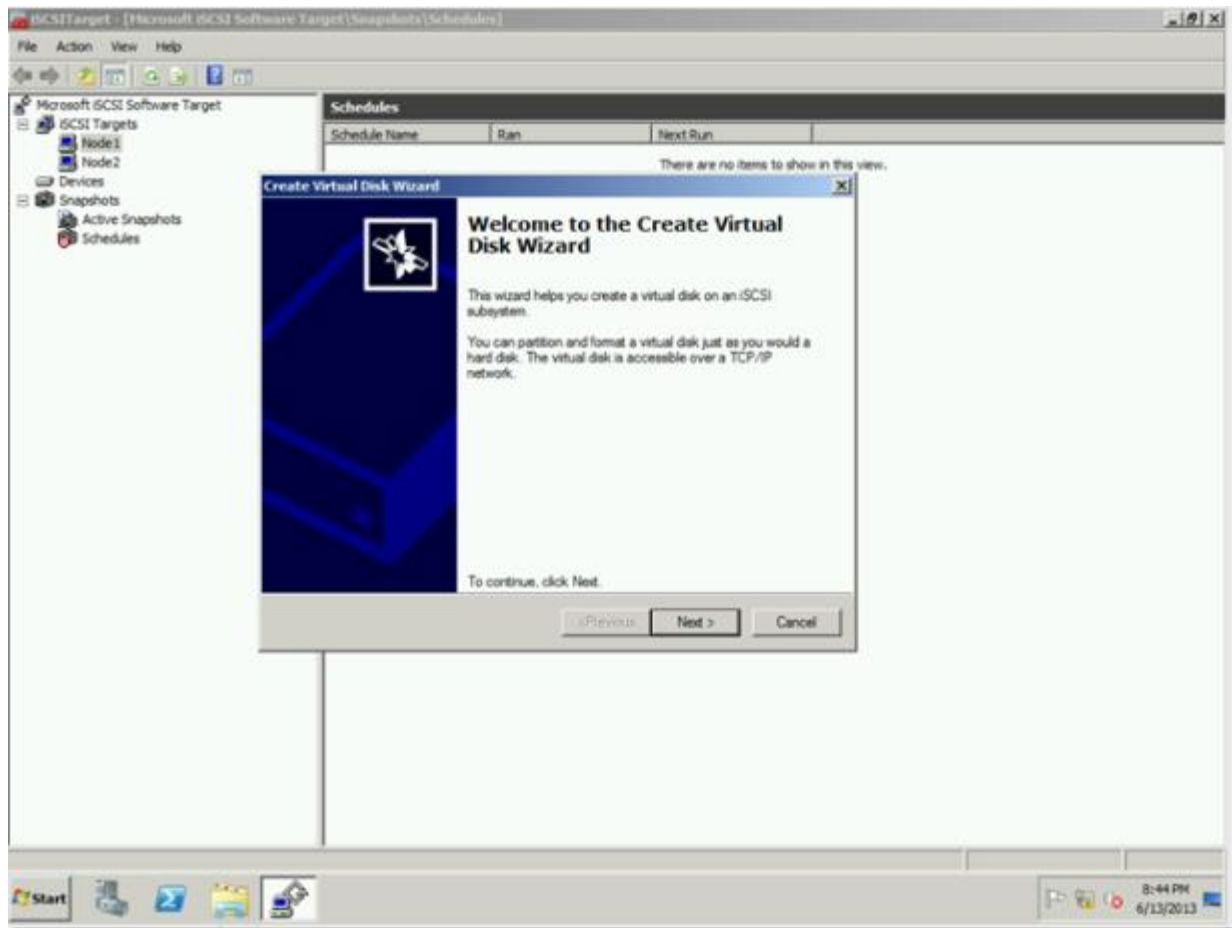


Follow the above steps in all nodes.

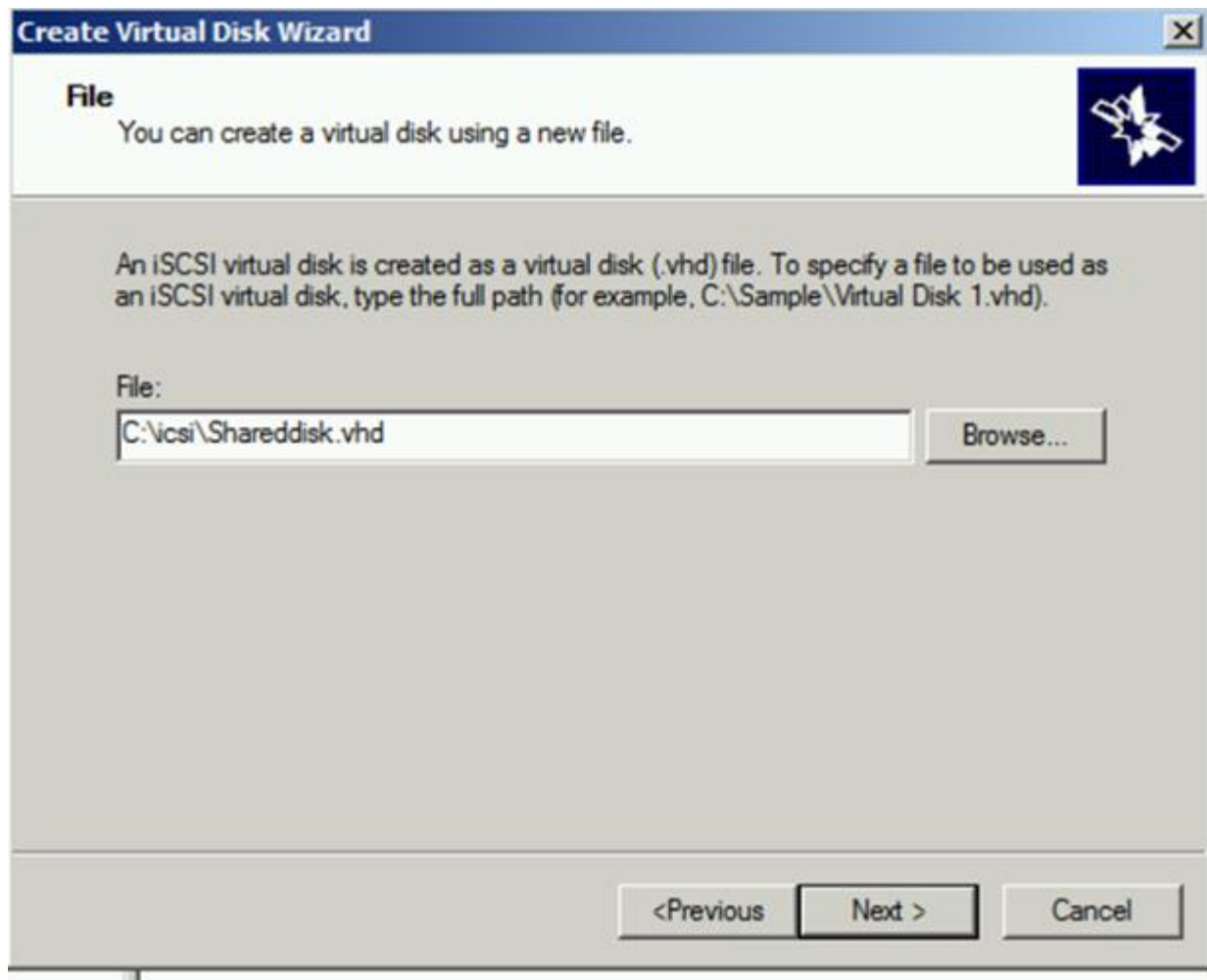
CREATING VIRTUAL DISK FOR ALL NODES:

To create virtual disk in iSCSI target right click **Node1**->**Create virtual disk**.

Click **Next**.



Specify the file name which will be used as virtual disk.Specify it with **.vhd** extension.Click Next.



Specify the **size** for your virtual disk. Click **Next**.

Create Virtual Disk Wizard [X]

Size
Specify how much space on the volume to use for the virtual disk.

Current drive capacity: 9.90 GB

Currently available free space: 1.57 GB

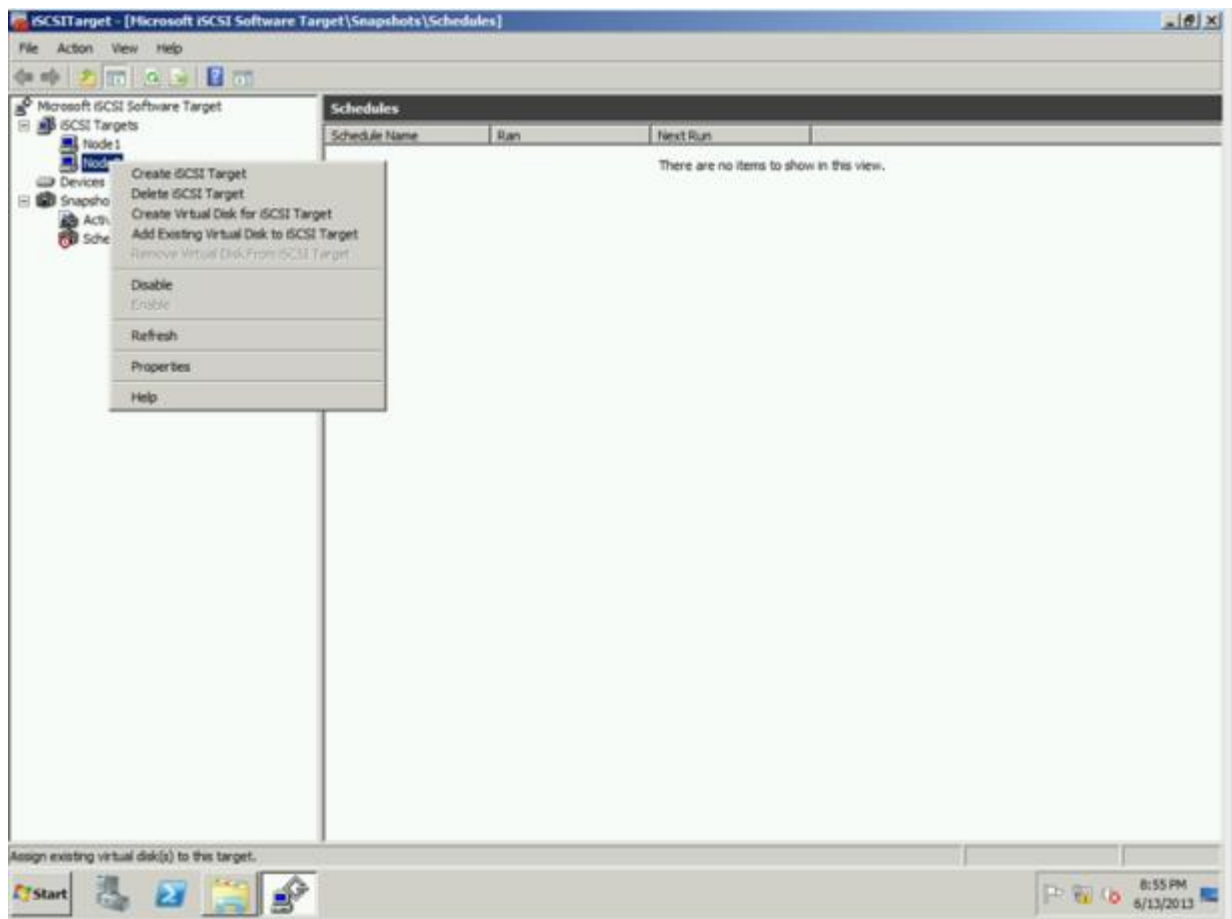
Size of virtual disk (MB):

<Previous Next > Cancel

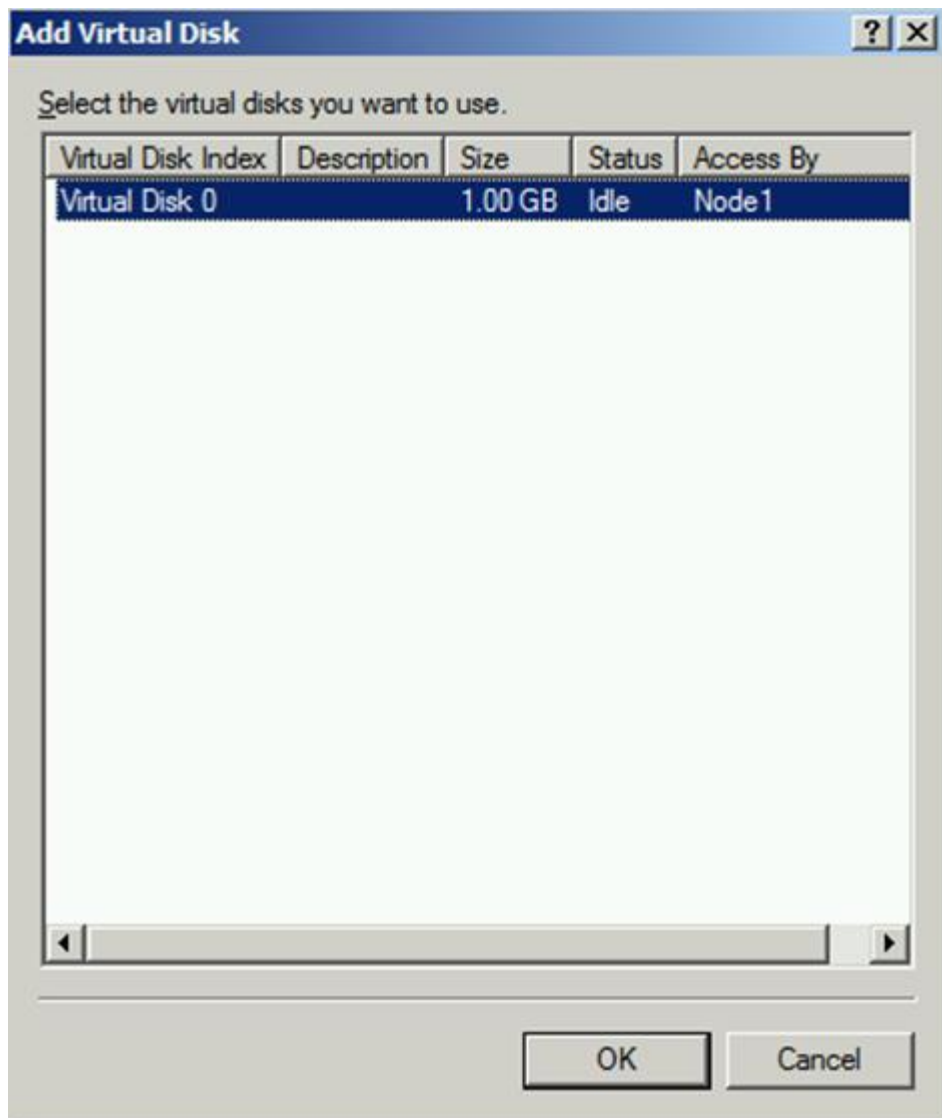
Click **Finish**.



For node2, Right click **Node2**->**Add Existing virtual disk** to iSCSI target.

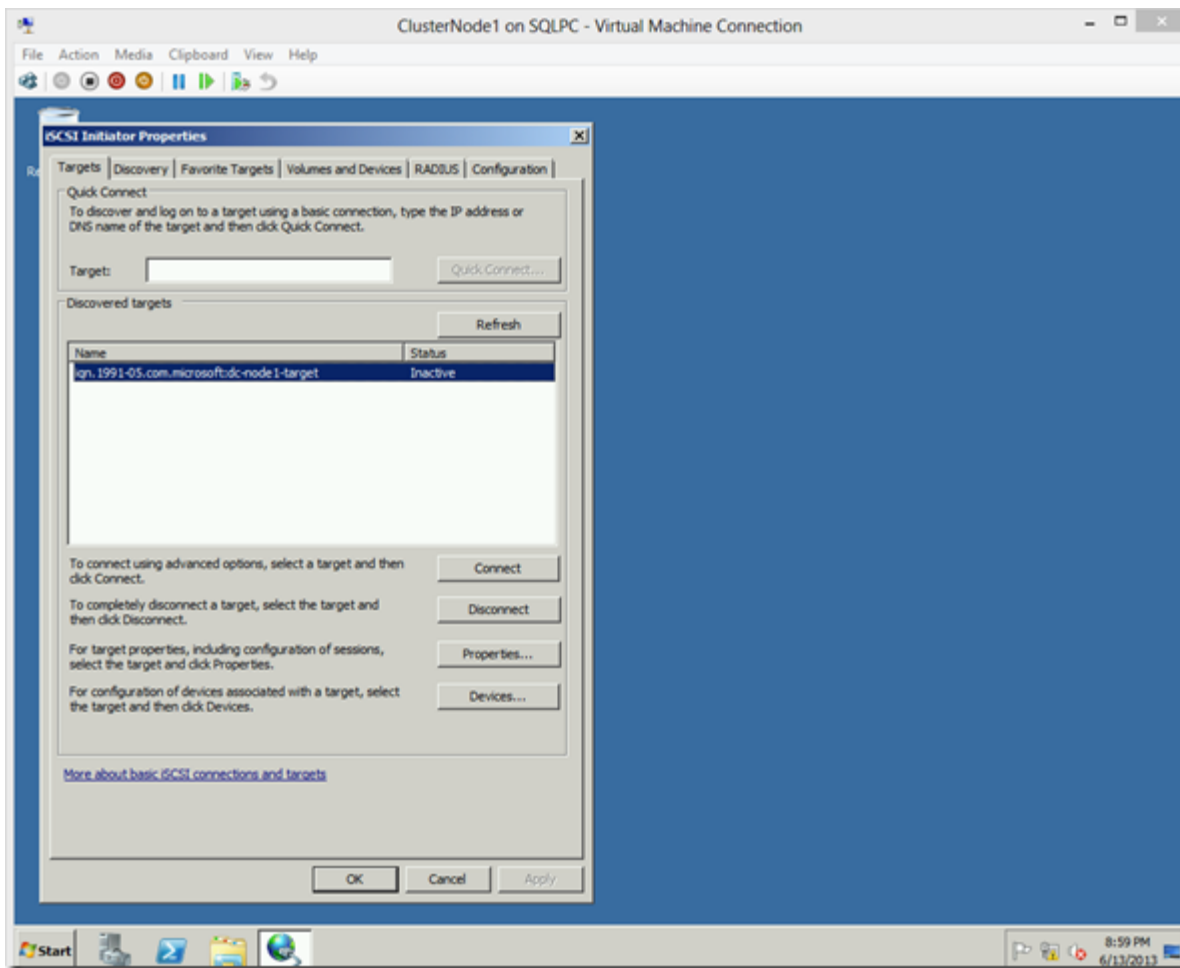


Click OK.

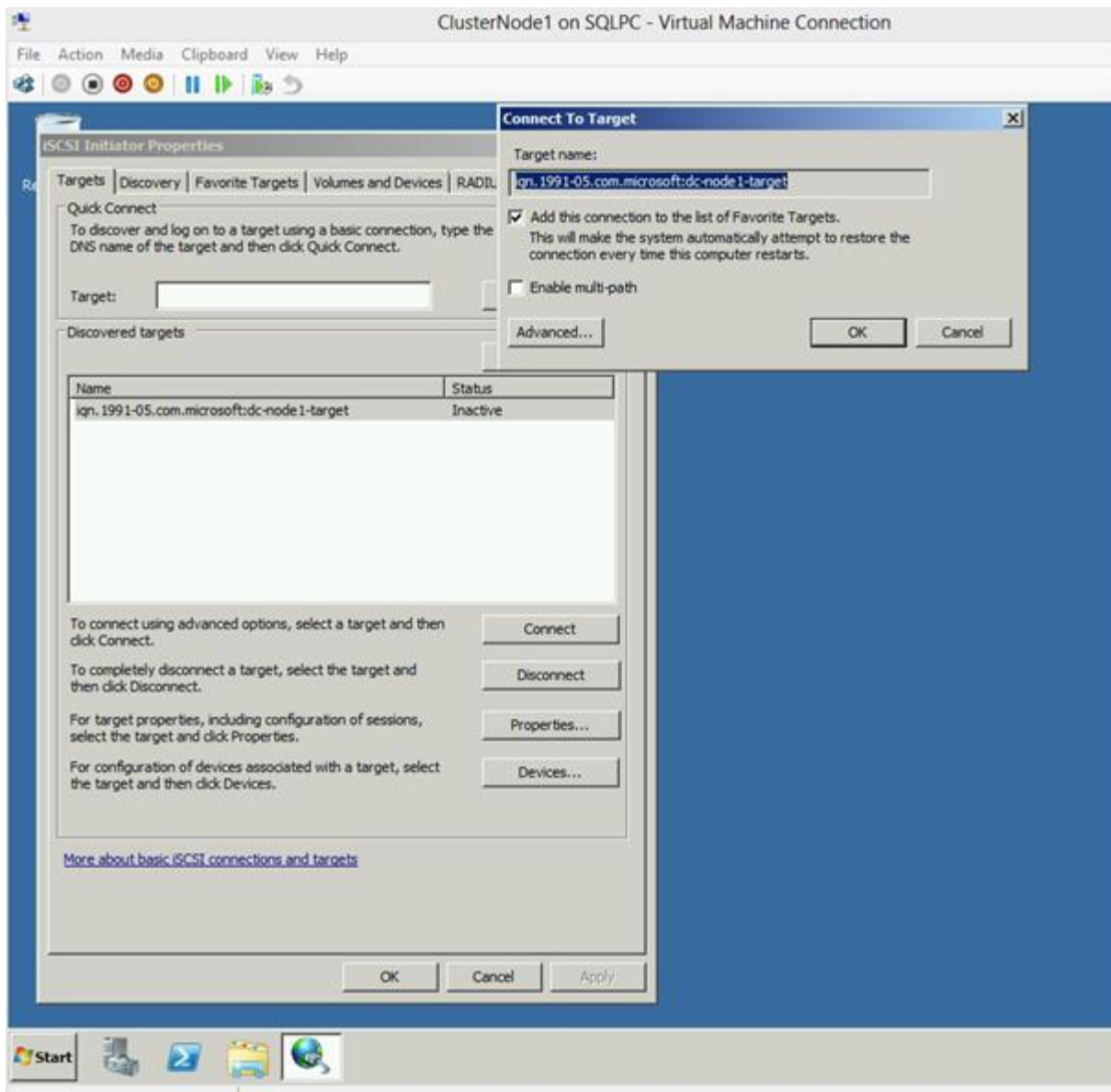


CLUSTER NODES:

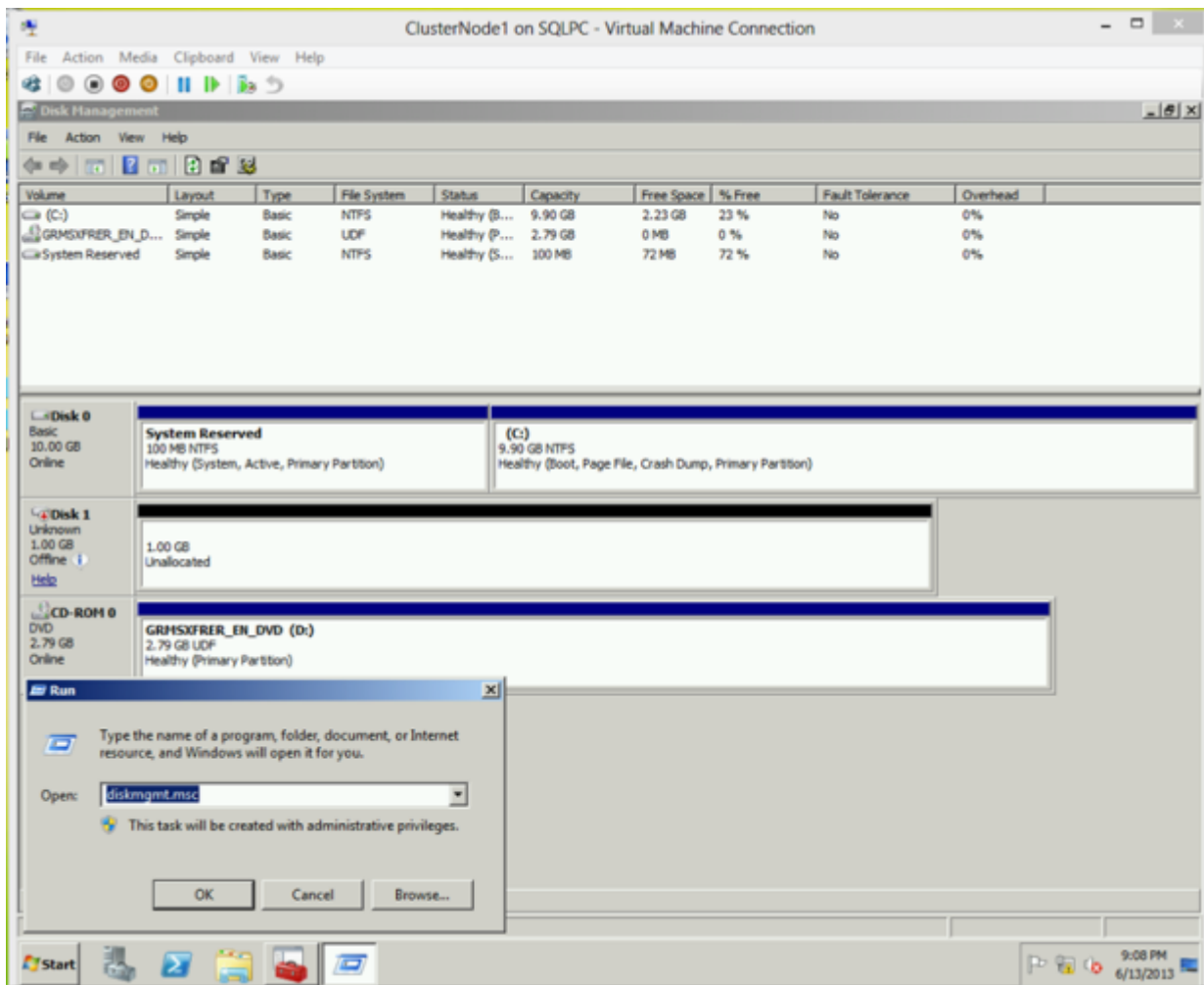
Now after adding the nodes as target in DC, we have to initiate the server in cluster nodes. Select **iSCSI initiator**. Click **Refresh**, if you didn't get any discovered target.



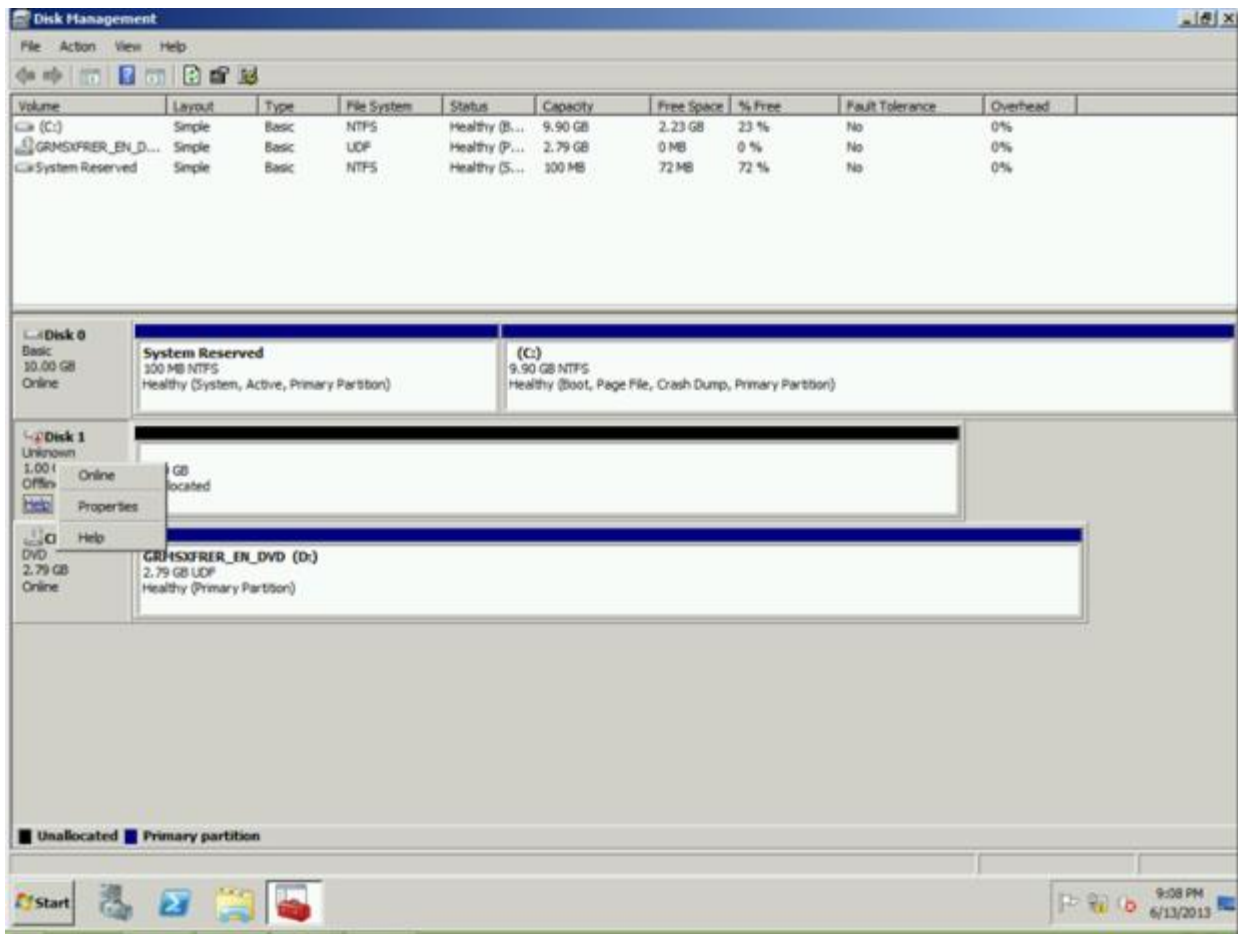
Select the target and **connect** it. Click OK.



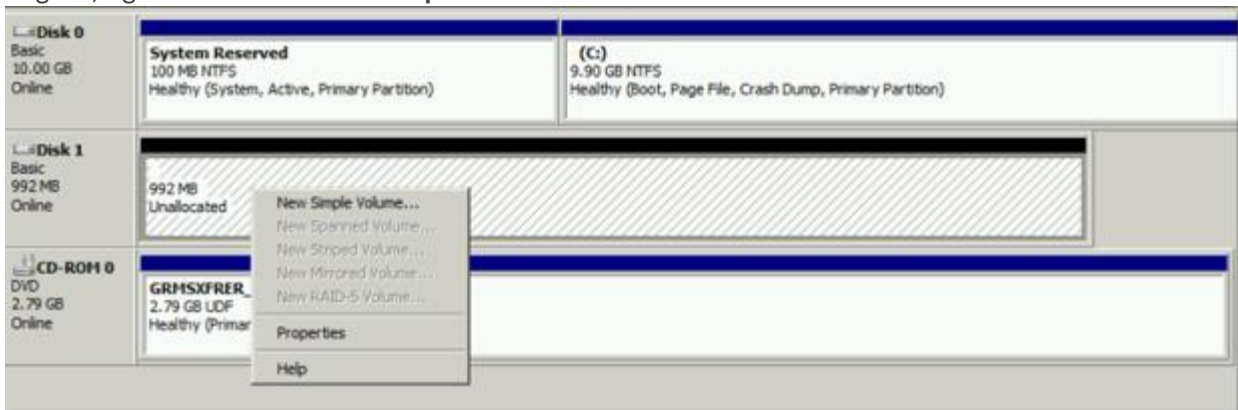
Open Disk management in cluster nodes:



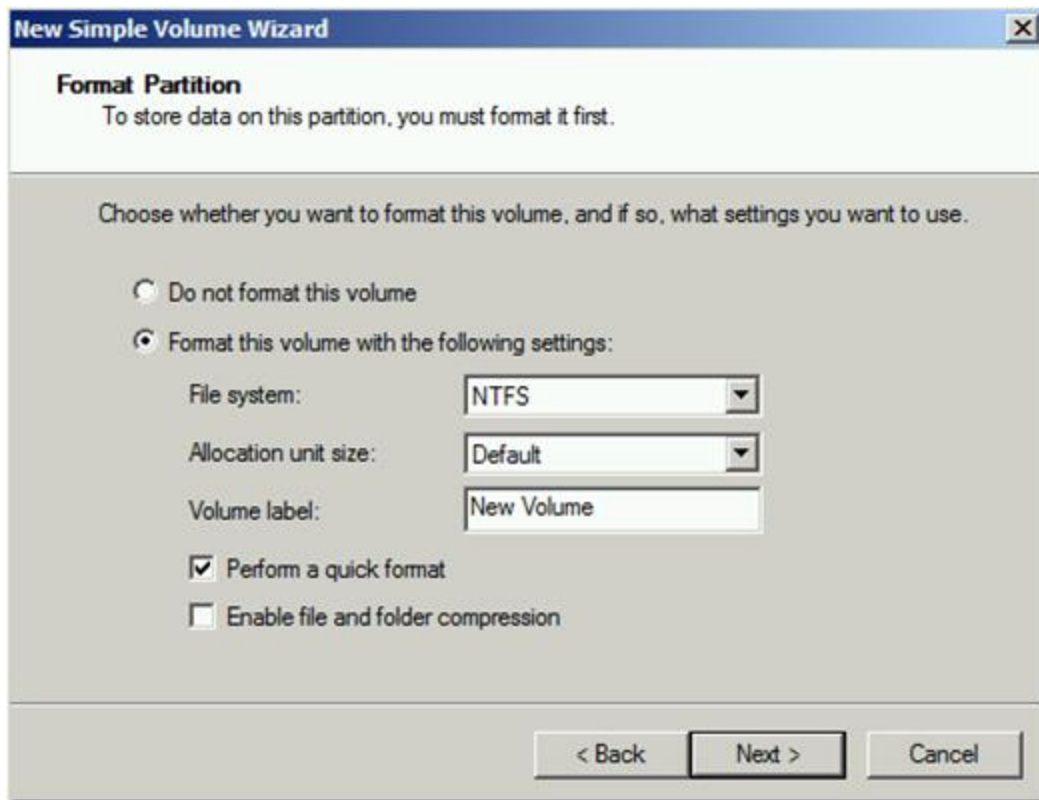
Right click **Disk1**->**Online**



Again ,Right click **Disk1**->**New Simple Volume**.

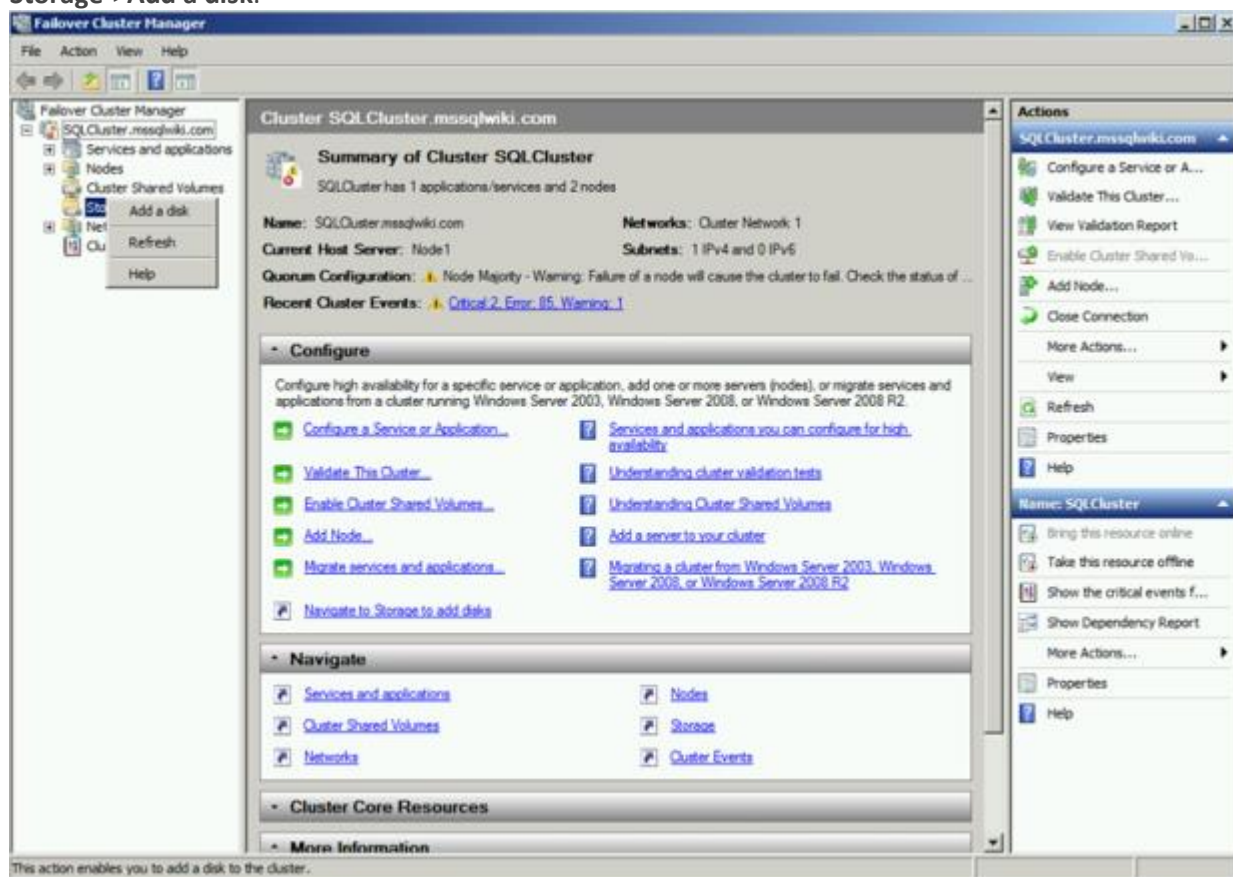


New Simple Volume Wizard->Format this volume with the following setting.Click Next.

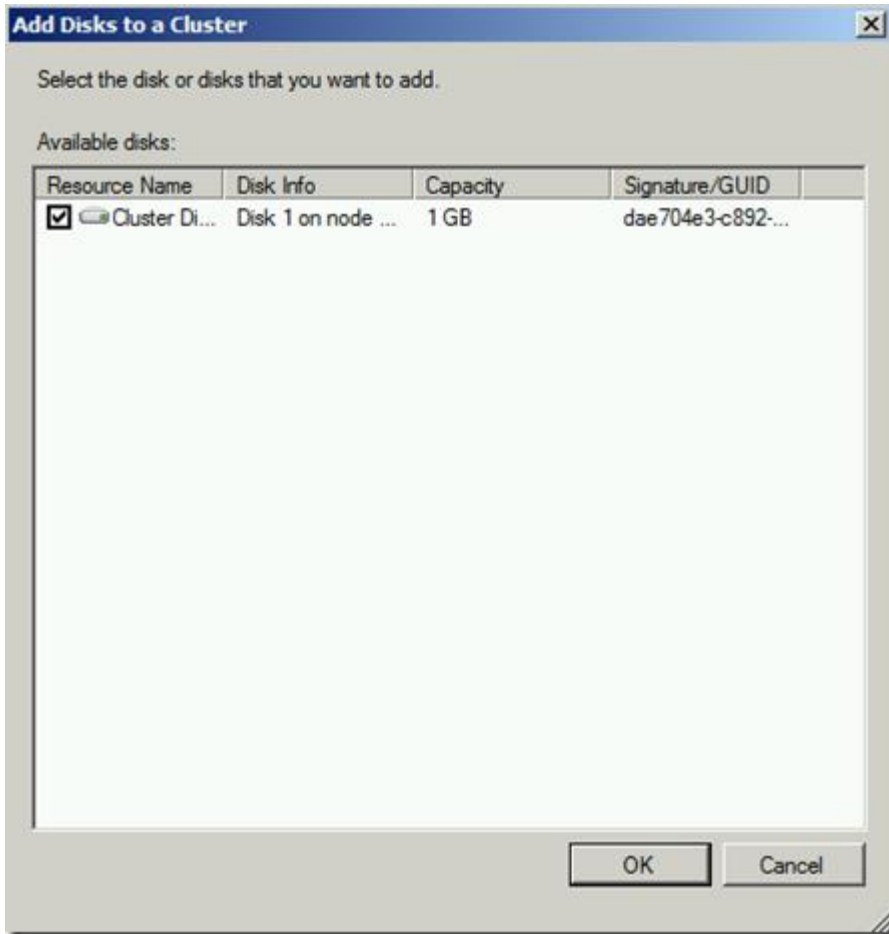


Then **Finish** the setting.

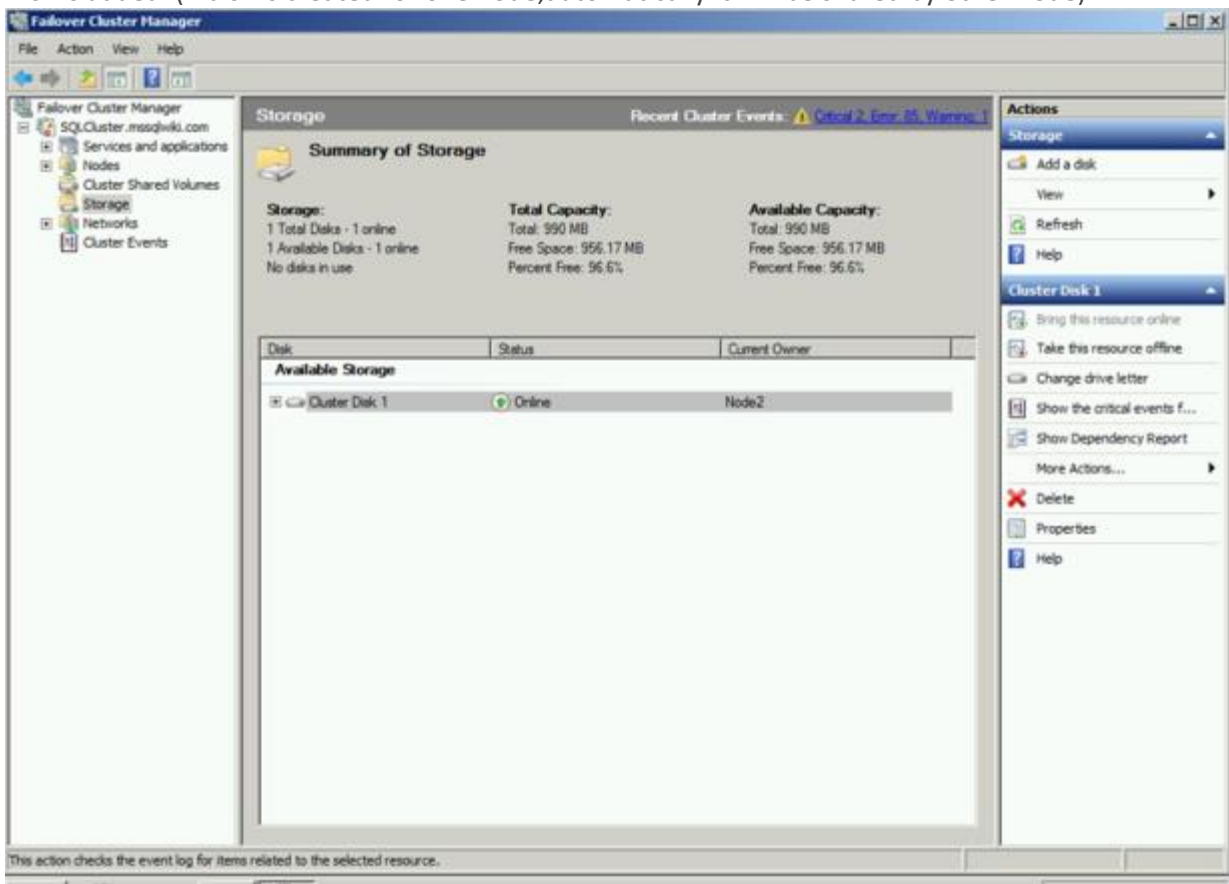
Open **Failover cluster manager**. Expand **SQL Cluster.mssql.wiki.com**.
Storage->Add a disk.



Click **OK** and add disk to a cluster.



Disk is added. (If disk is created for one node, automatically it will be shared by other node)



Now the shared disk is ready for use and can be used for installation of SQL Server cluster (or) any other cluster aware application which needs shared disk.

HOW TO INSTALL SQL Server CLUSTER IN HYPER-V

<https://sqlserverscribbles.com/2013/06/28/how-to-install-sql-server-cluster-in-hyper-v/>

To install SQL Server failover cluster we need to have windows cluster configured with a shared disk.

Step 1

If you do not have a windows cluster configured you can create using Hyper-v

Follow the steps in **"HOW TO CREATE CLUSTER USING HYPER-V"** to create Domain controller and two member servers which will act as nodes of the cluster.

Once the domain controller and two member servers are created proceed to step 2

Step 2

Once the domain controller and nodes are created install failover cluster feature and create cluster

To install the failover cluster feature and create a cluster follow the steps in **"How to install cluster in windows 2008 and windows 2012"**

Step 3

Once the cluster is created we have to create shared Disk for installing failover aware clustered application like SQL Server.

Follow the steps in **"How to create shared disk using iSCSI Software Target"** to create a shared disk using iSCSI target software.

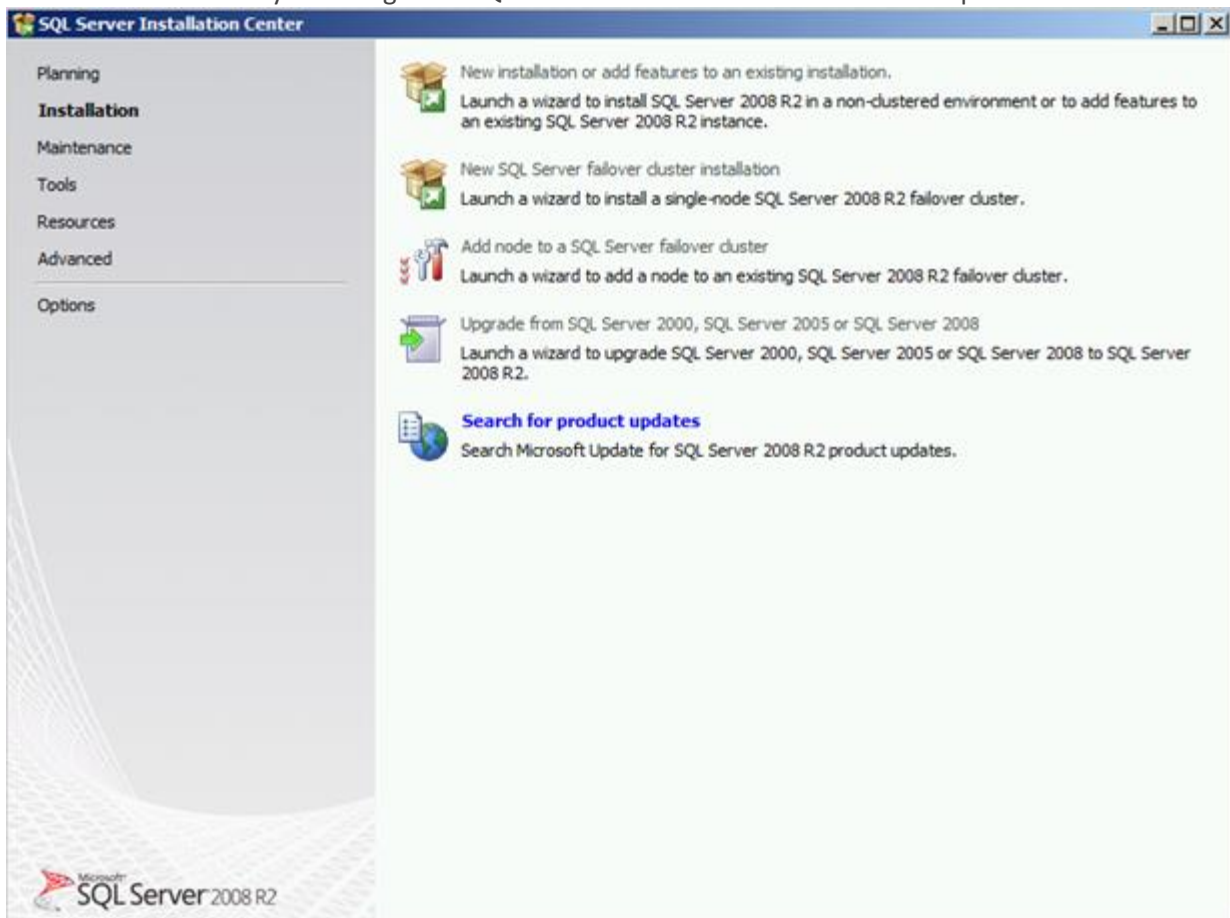
Once shared disk is configured invoke Setup from SQL server installation media .Make sure the node in which you perform the installation has shared disk online.

Name ^	Date modified	Type	Size
1033_ENU_LP	6/18/2013 9:45 PM	File folder	
ia64	6/18/2013 9:47 PM	File folder	
MasterDataServices	6/18/2013 9:47 PM	File folder	
redist	6/18/2013 9:47 PM	File folder	
resources	6/18/2013 9:47 PM	File folder	
StreamInsight	6/18/2013 9:47 PM	File folder	
x64	6/18/2013 9:48 PM	File folder	
x86	6/18/2013 9:49 PM	File folder	
autorun	3/31/2010 2:21 AM	Setup Information	1 KB
MediaInfo	4/5/2010 4:53 PM	XML Document	1 KB
Microsoft.VC80.CRT.manifest	3/31/2010 4:07 AM	MANIFEST File	1 KB
msvcr80.dll	3/31/2010 4:09 AM	Application extension	618 KB
setup	4/3/2010 11:56 AM	Application	130 KB
setup.exe	3/30/2010 11:52 PM	XML Configuration File	1 KB
sqmapi.dll	4/3/2010 11:47 AM	Application extension	137 KB

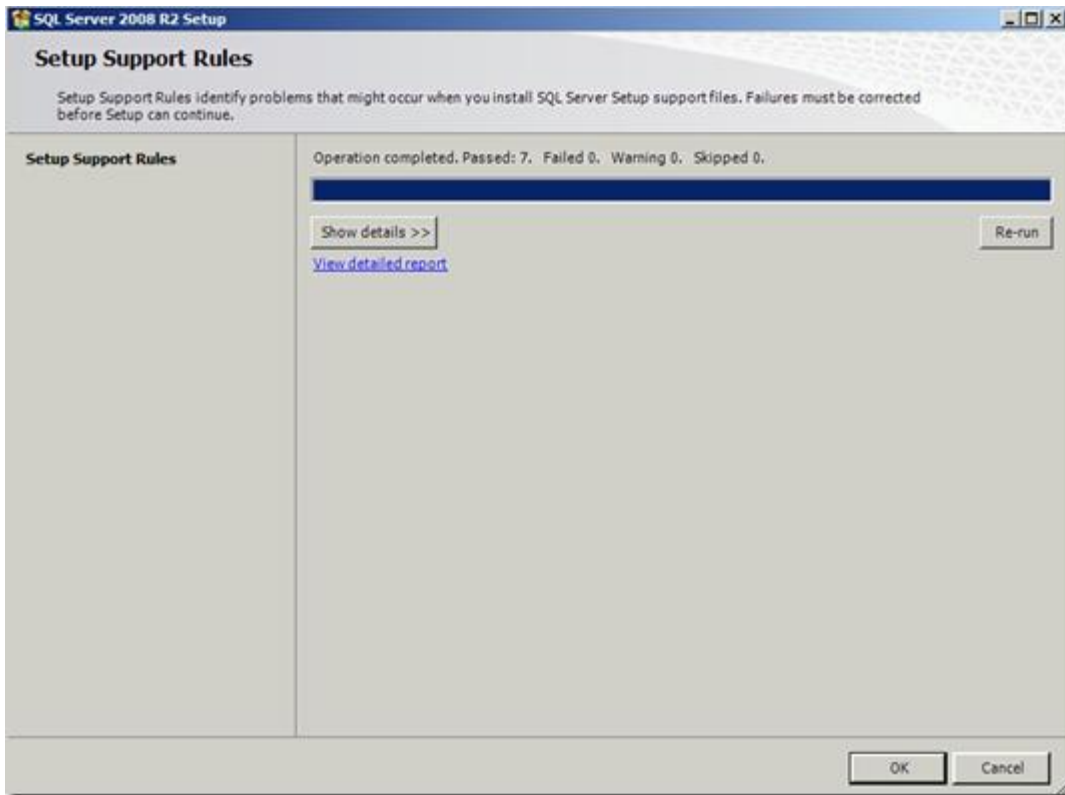
REPORT THIS AD

Run the setup.

Start the installation by selecting **New SQL Server Failover Cluster Installation** option.

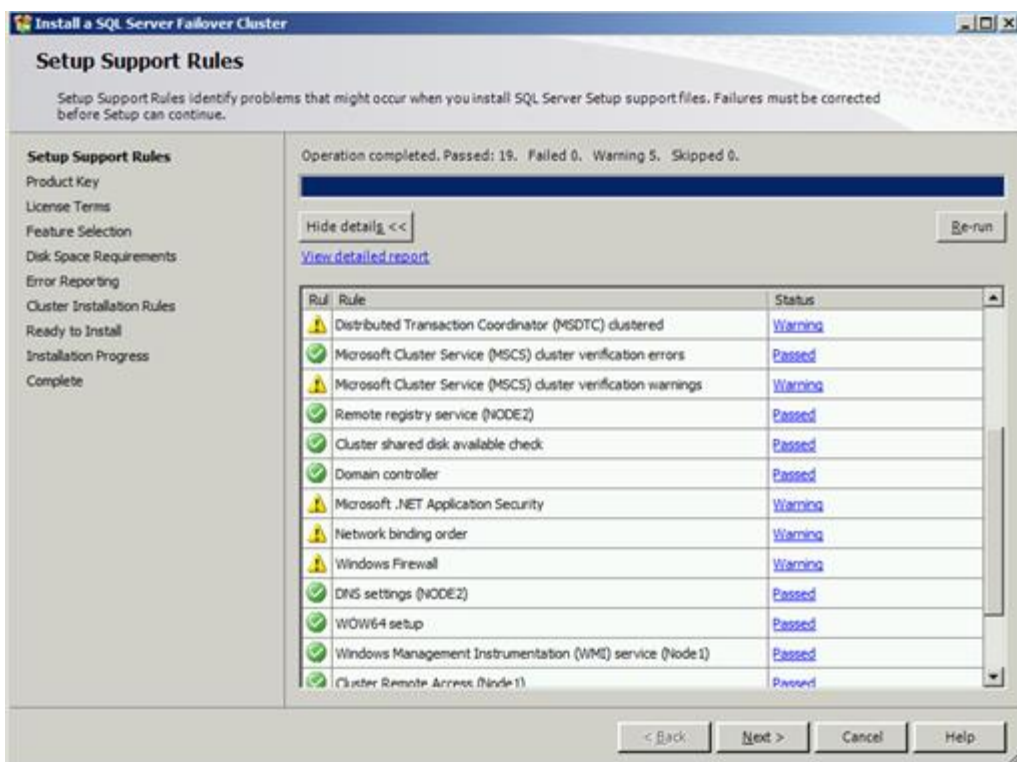


Setup support rules will analyze and identify the problems that might occur during setup installation.

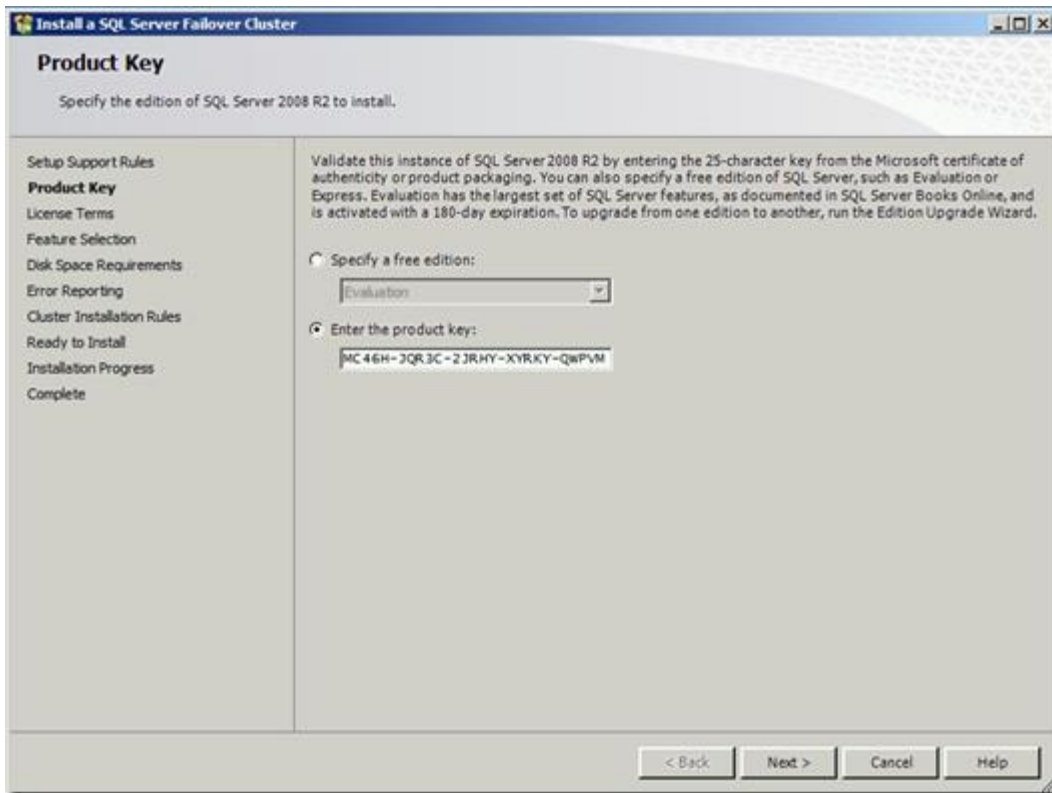


REPORT THIS AD

After setup validation ,you will get details about operations completed,Click Next and continue

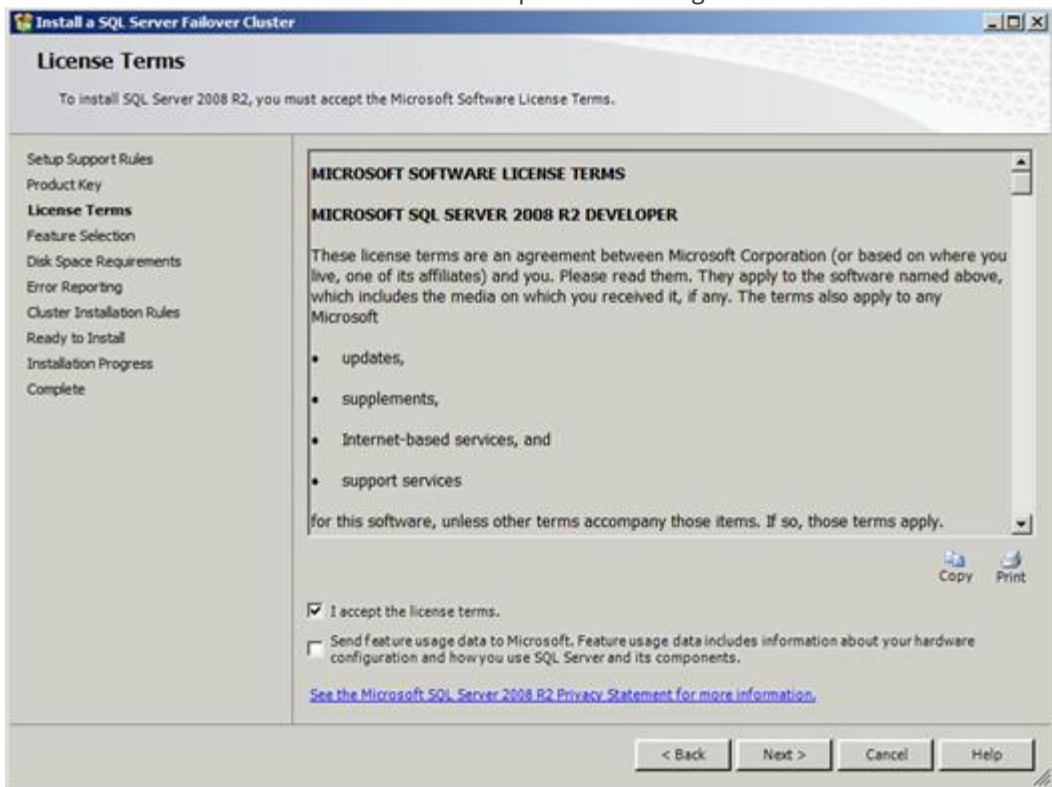


On the Product key page, specify the Product ID key of the product. Click Next.

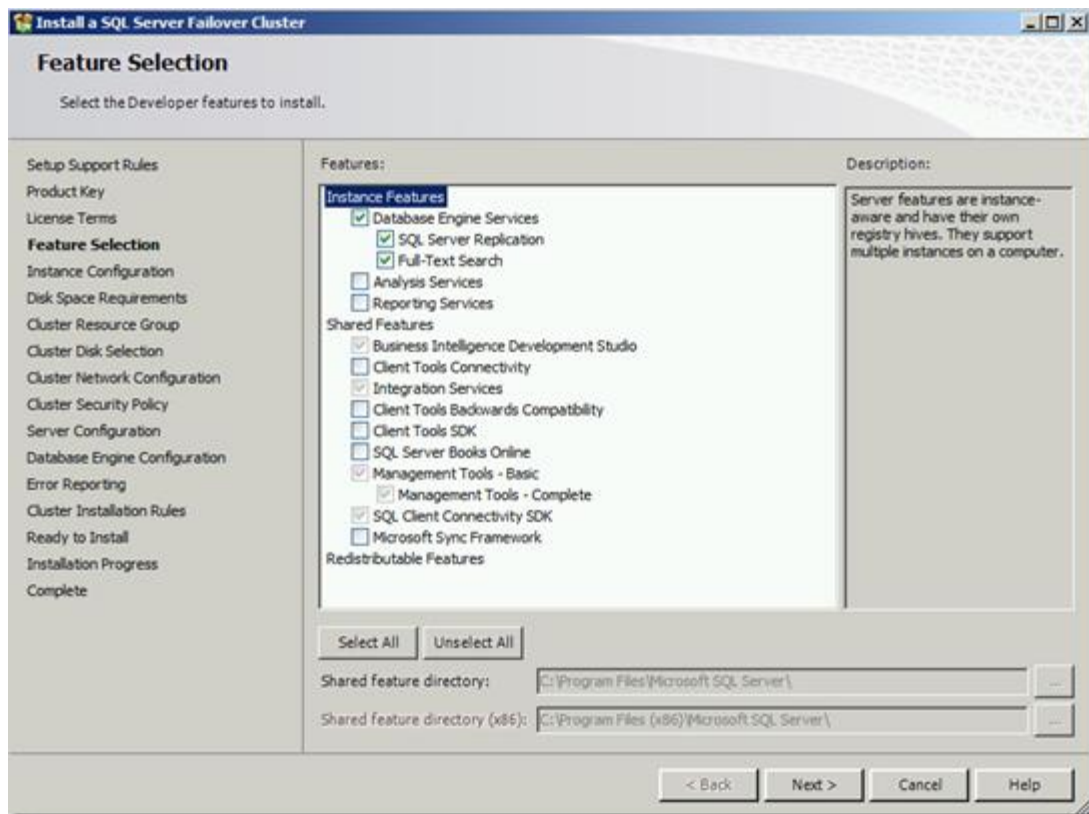


REPORT THIS AD

Then read and select the check box to accept terms and agreements. Click Next.

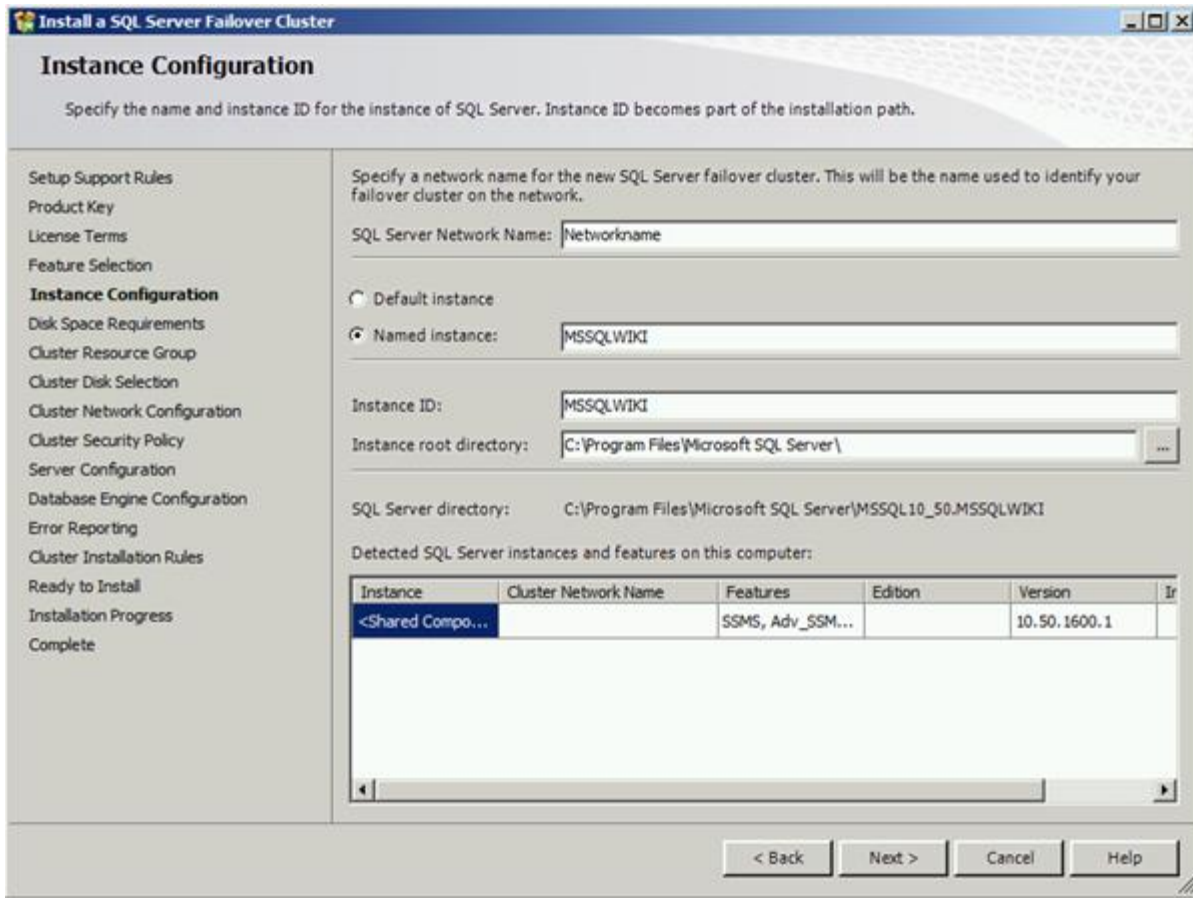


Select the **developer** features which you want and click Next.

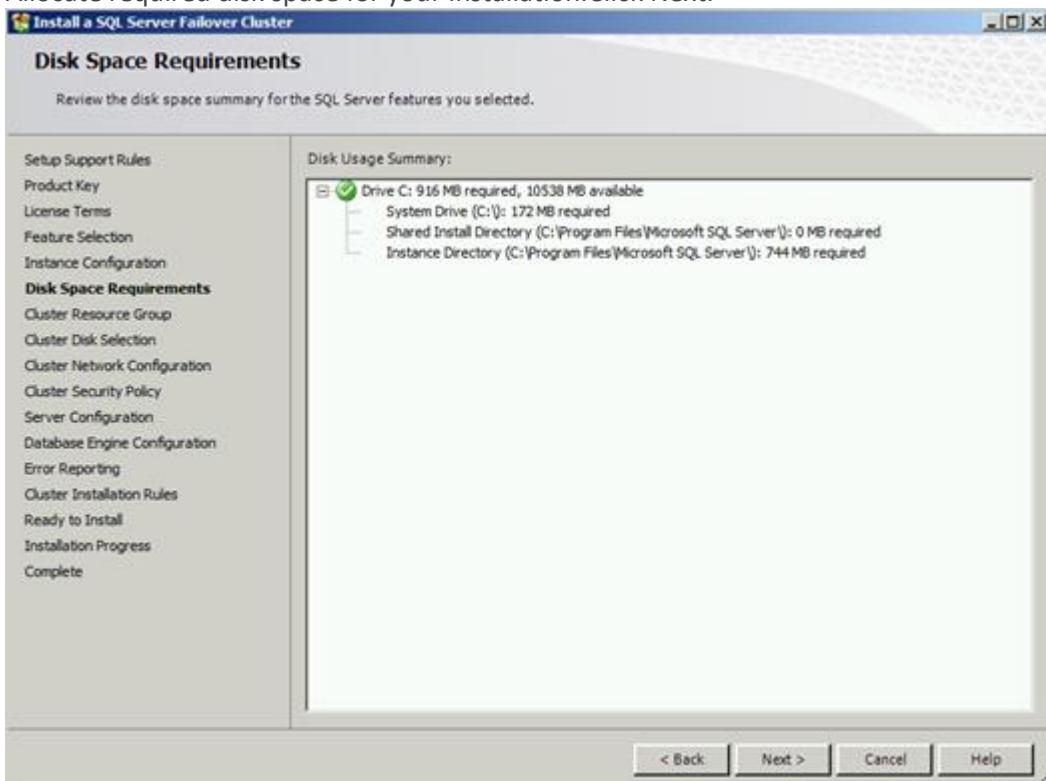


REPORT THIS AD

Perform instance configuration. Give SQL server network name. Either you could have default instance or named instance. But you could have only one default instance. Click Next.

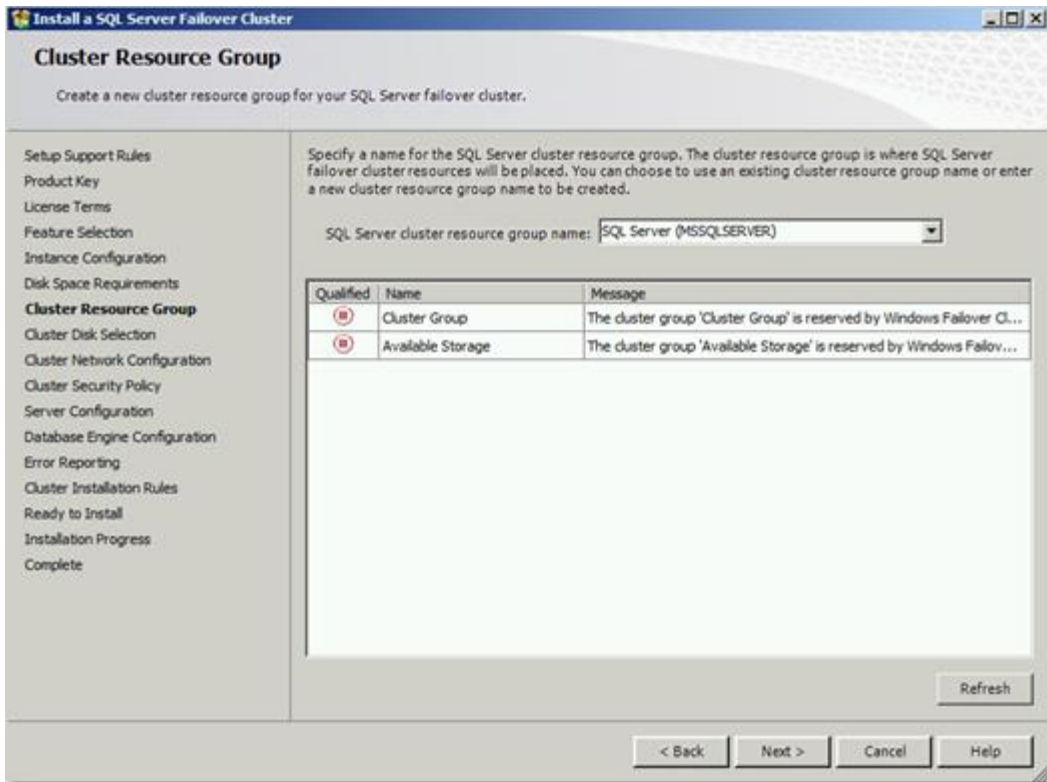


Allocate required disk space for your installation. Click Next.

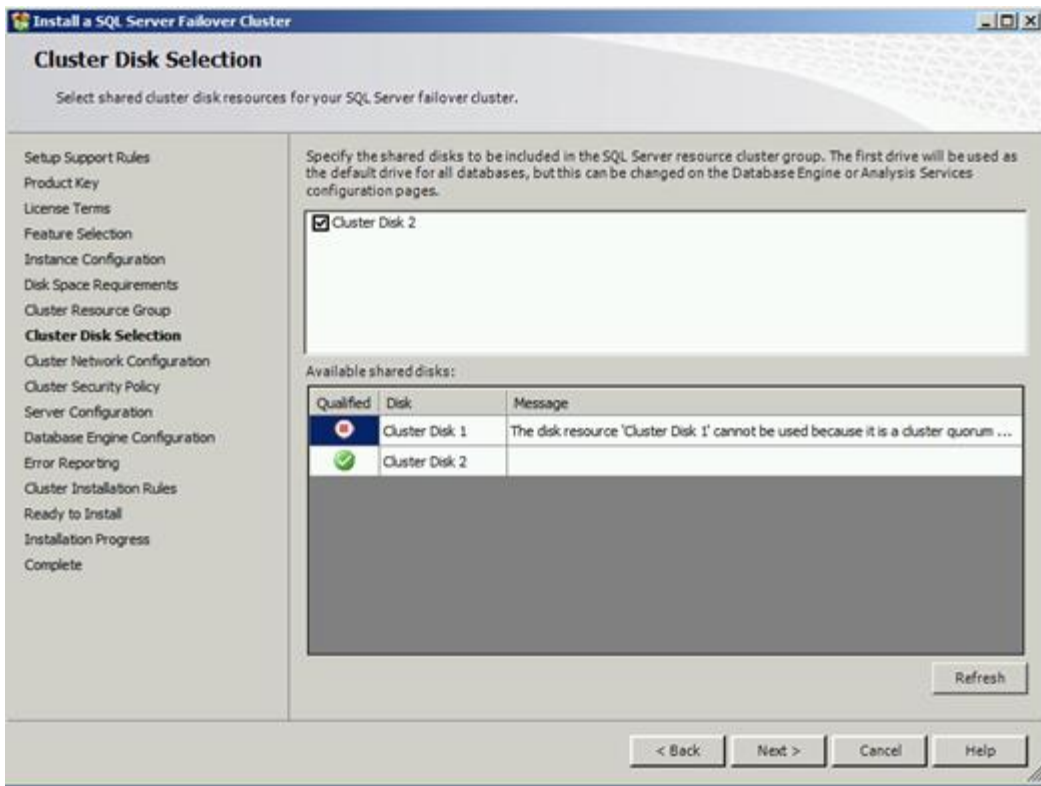


REPORT THIS AD

Select the SQL Server cluster resource group name. Click Next and continue.



Cluster disk selection will help us to view the disk we have selected for SQL Server failover cluster.



REPORT THIS AD

Configure the network resource for your SQL Server failover cluster. Select DHCP check box if you are going to use DHCP network. Otherwise give the IP address for your network (IP address which you provide not be in use currently).

Install a SQL Server Failover Cluster

Cluster Network Configuration

Select network resources for your SQL Server failover cluster.

Setup Support Rules
Product Key
License Terms
Feature Selection
Instance Configuration
Disk Space Requirements
Cluster Resource Group
Cluster Disk Selection
Cluster Network Configuration
Cluster Security Policy
Server Configuration
Database Engine Configuration
Error Reporting
Cluster Installation Rules
Ready to Install
Installation Progress
Complete

Specify the network settings for this failover cluster:

☒ IP Type	DHCP	Address	Subnet Mask	Network
☒ IPv4	☐	192.168.0.101	255.255.255.0	Cluster Network 1

Refresh

< Back Next > Cancel Help

Select the database engine domain group and SQL Server agent domain group. These groups have to be pre created in AD before the installation and startup account of SQL Server services will be added to these groups. You can also use service SIDs option.

Install a SQL Server Failover Cluster

Cluster Security Policy

Configure the security policy for your SQL Server failover cluster.

Setup Support Rules
Product Key
License Terms
Feature Selection
Instance Configuration
Disk Space Requirements
Cluster Resource Group
Cluster Disk Selection
Cluster Network Configuration
Cluster Security Policy
Server Configuration
Database Engine Configuration
Error Reporting
Cluster Installation Rules
Ready to Install
Installation Progress
Complete

Specify global or local security domain groups for the clustered services that will be installed as part of your SQL Server failover cluster. All resource permissions are controlled by domain-level groups that include SQL Server service accounts as members.

☐ Use service SIDs (recommended)
☒ Use domain group

Database Engine domain group: MSSQLWIKI\enginewiki

SQL Server Agent domain group: MSSQLWIKI\sqlwiki

< Back Next > Cancel Help

Specify the account name and password for SQL server agent and SQL server database engine. Click Next.

Install a SQL Server Failover Cluster

Server Configuration

Specify the service accounts and collation configuration.

Service Accounts | Collation

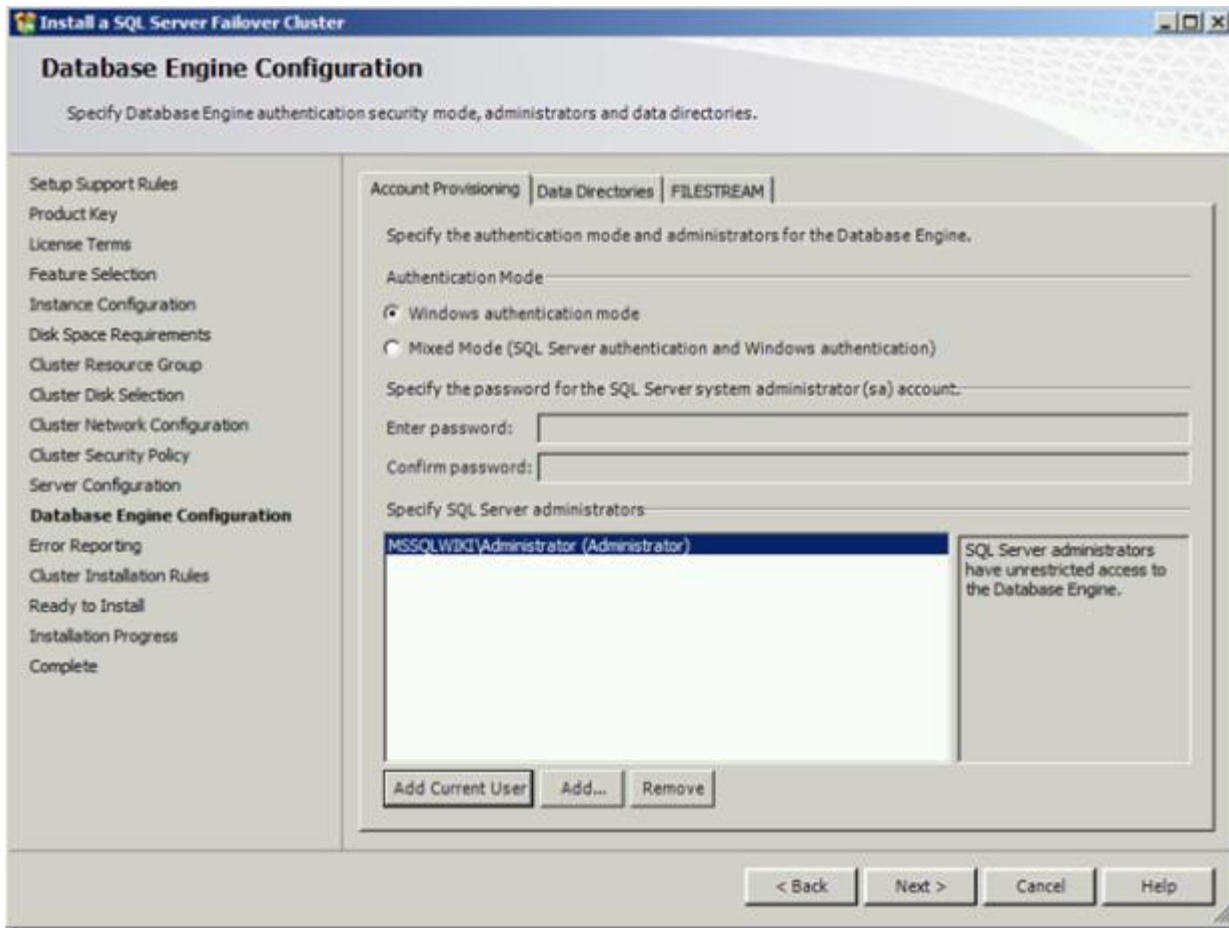
Microsoft recommends that you use a separate account for each SQL Server service.

Service	Account Name	Password	Startup Type
SQL Server Agent	mssqlwiki\administrator	*****	Manual
SQL Server Database Engine	mssqlwiki\administrator	*****	Manual
SQL Full-text Filter Daemon Launcher	NT AUTHORITY\LOCAL S...		Manual
SQL Server Browser	NT AUTHORITY\LOCAL S...		Automatic

☐ Use the same account for all SQL Server services

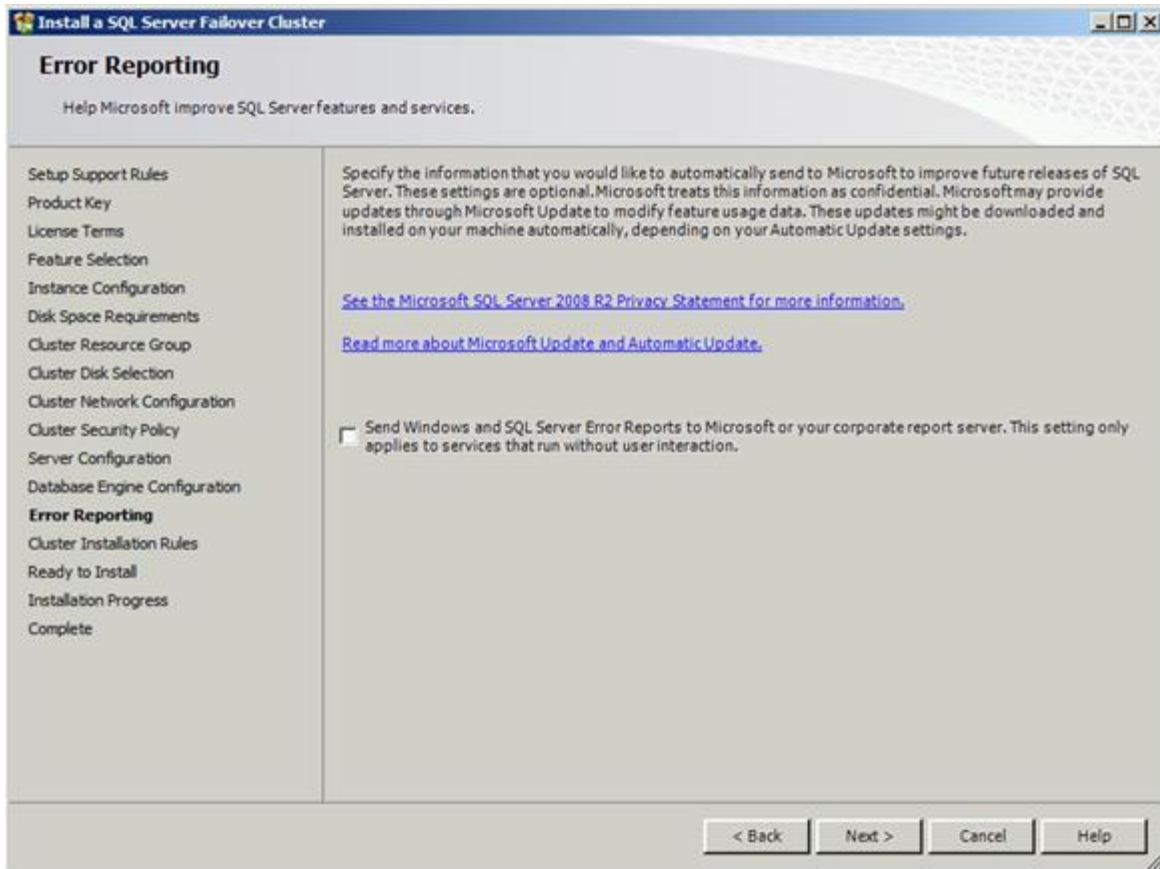
< Back Next > Cancel Help

Specify the authentication mode for your SQL Server failover cluster. Click Next and continue your installation. Verify the cluster installation rules.

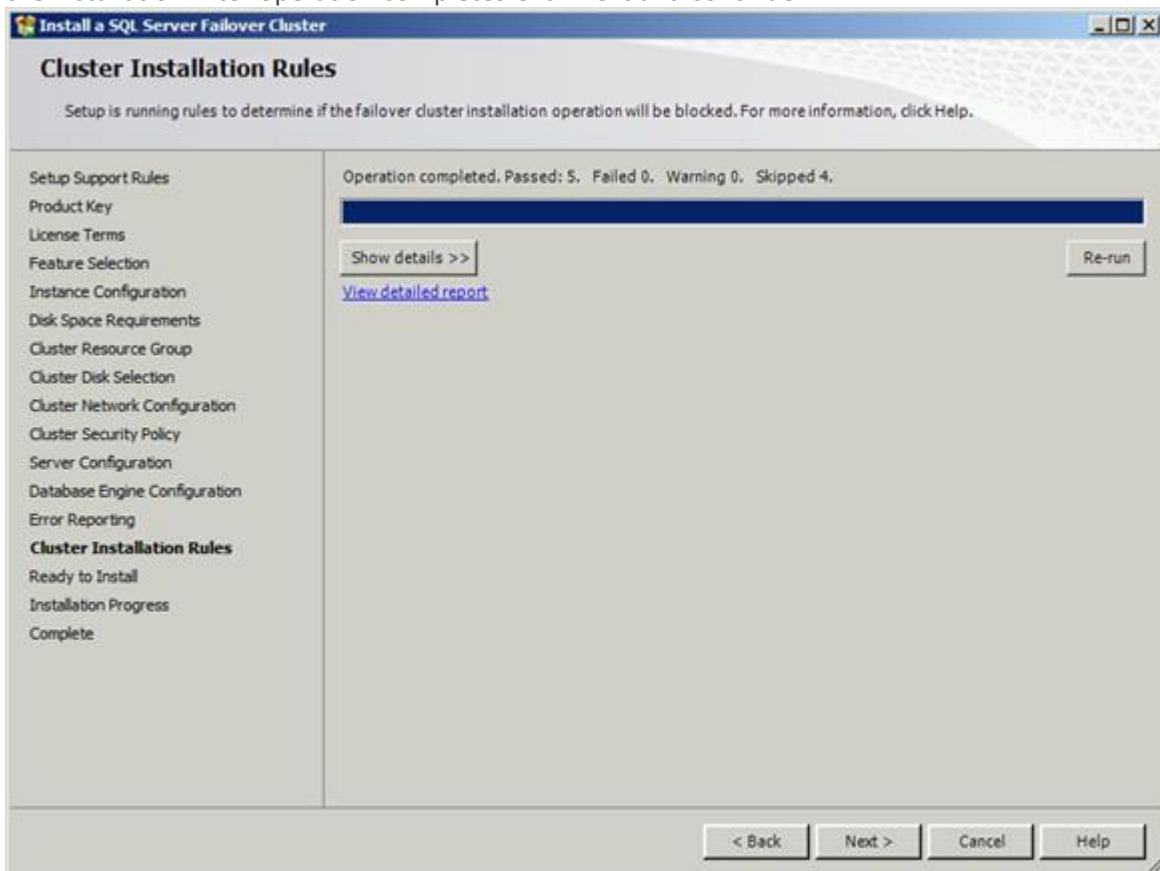


REPORT THIS AD

Enable error reporting service if you want or else,click Next and continue.

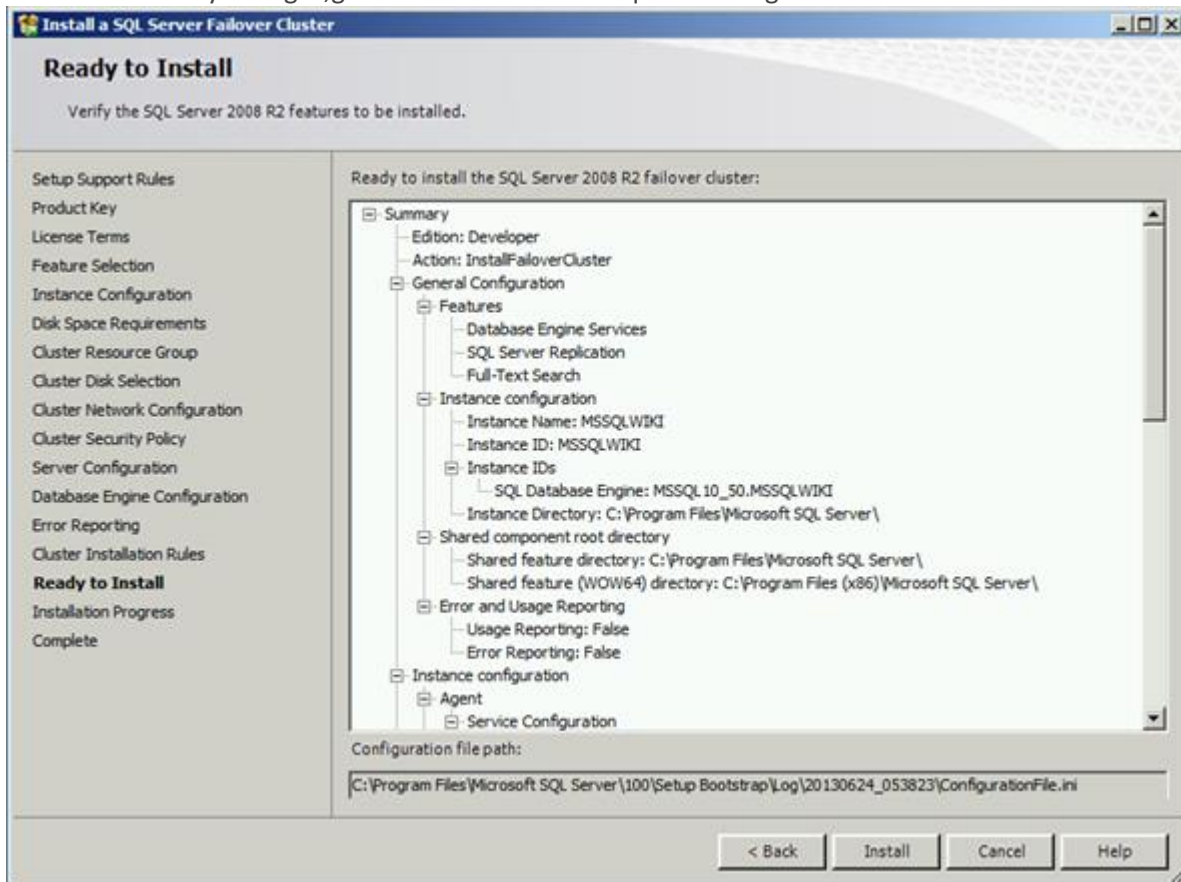


Cluster installation rules will run the setup to determine whether any operations will block the installation. After operation completes, click Next and continue.

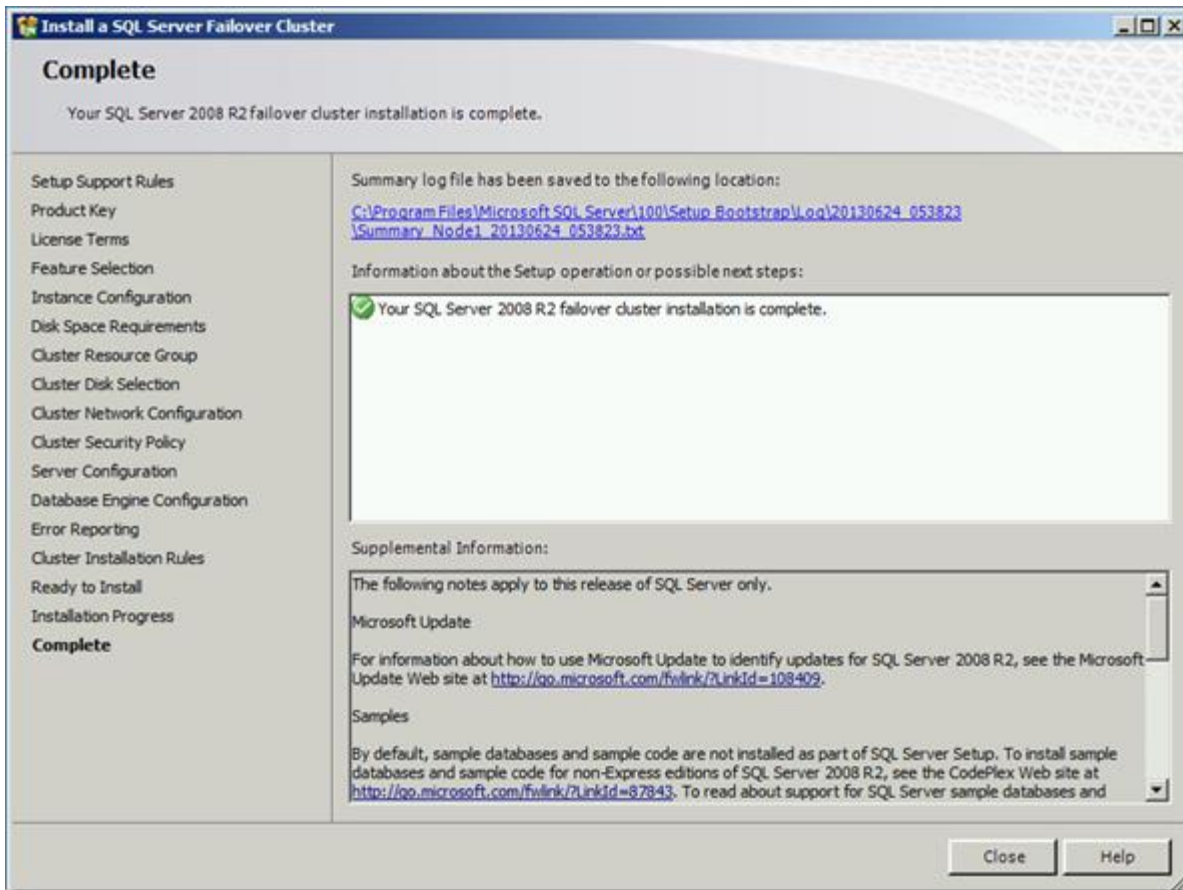


REPORT THIS AD

Ready to install page will display the summary of options which we have selected during the setup process. If you want to make any changes, go back and make the required changes or else Click Install and start the installation.

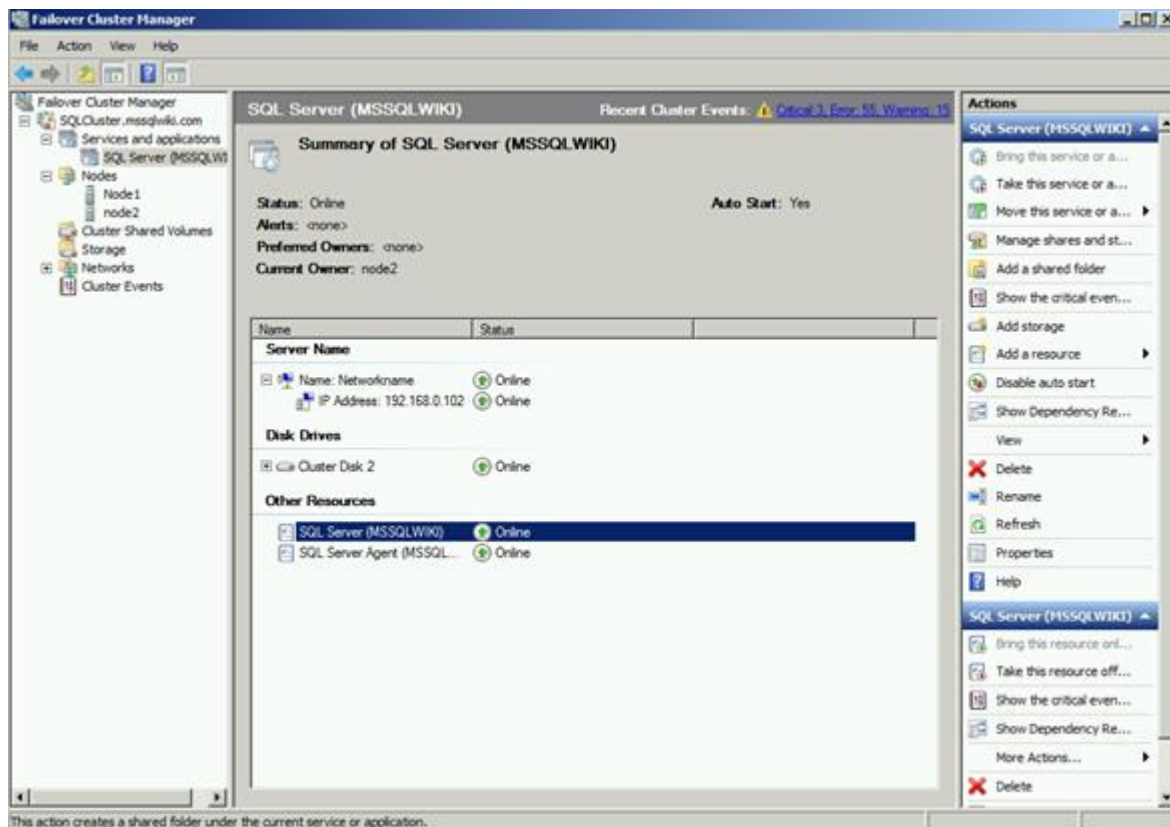


Click Close and Complete the installation process.



REPORT THIS AD

Successfully we have installed the SQL server failover cluster in first node.



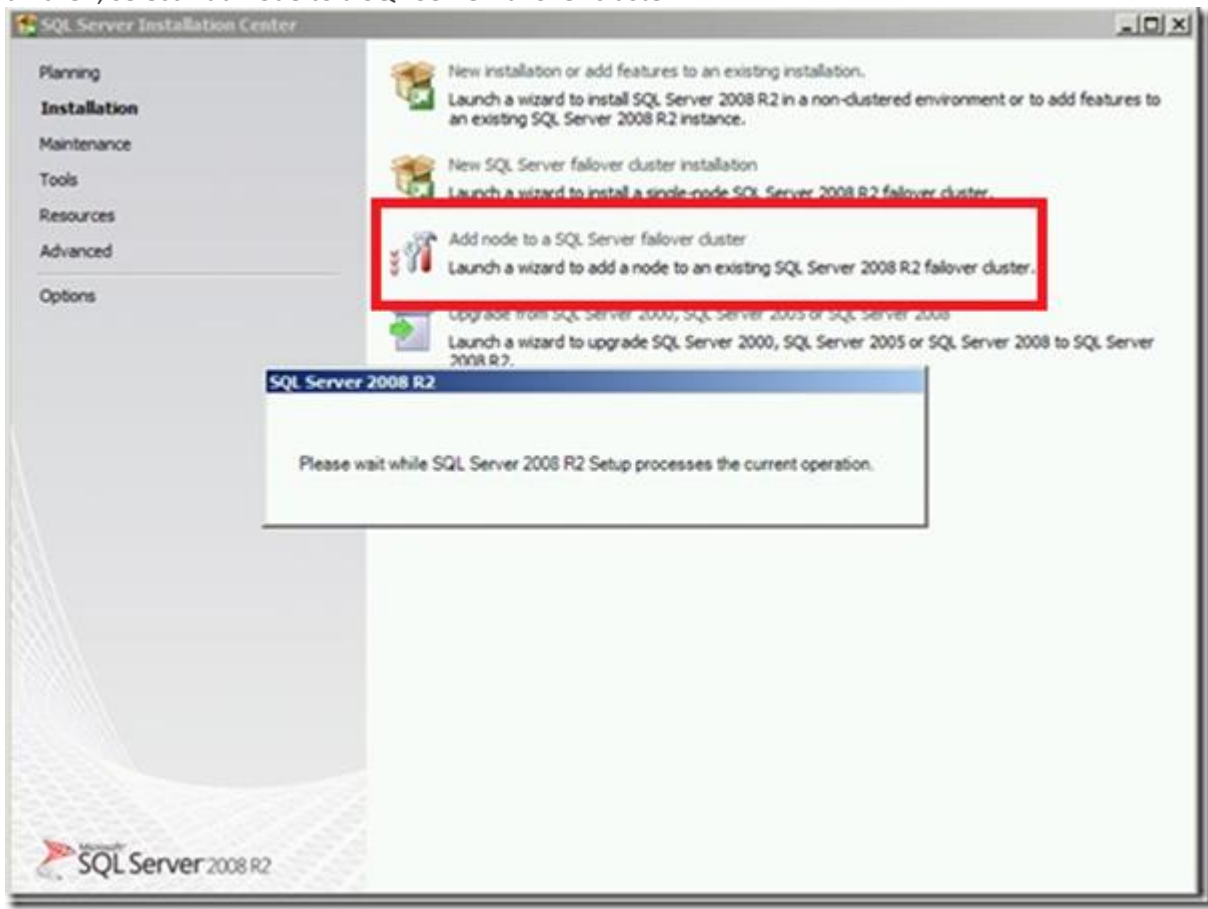
We can install the SQL server failover cluster in remaining nodes by selecting Add node to a SQL Server failover cluster in installation centre window. To add node to existing SQL Server failover cluster follow the steps in [“How to add node to SQL Server cluster”](#)

How to add node to SQL Server cluster

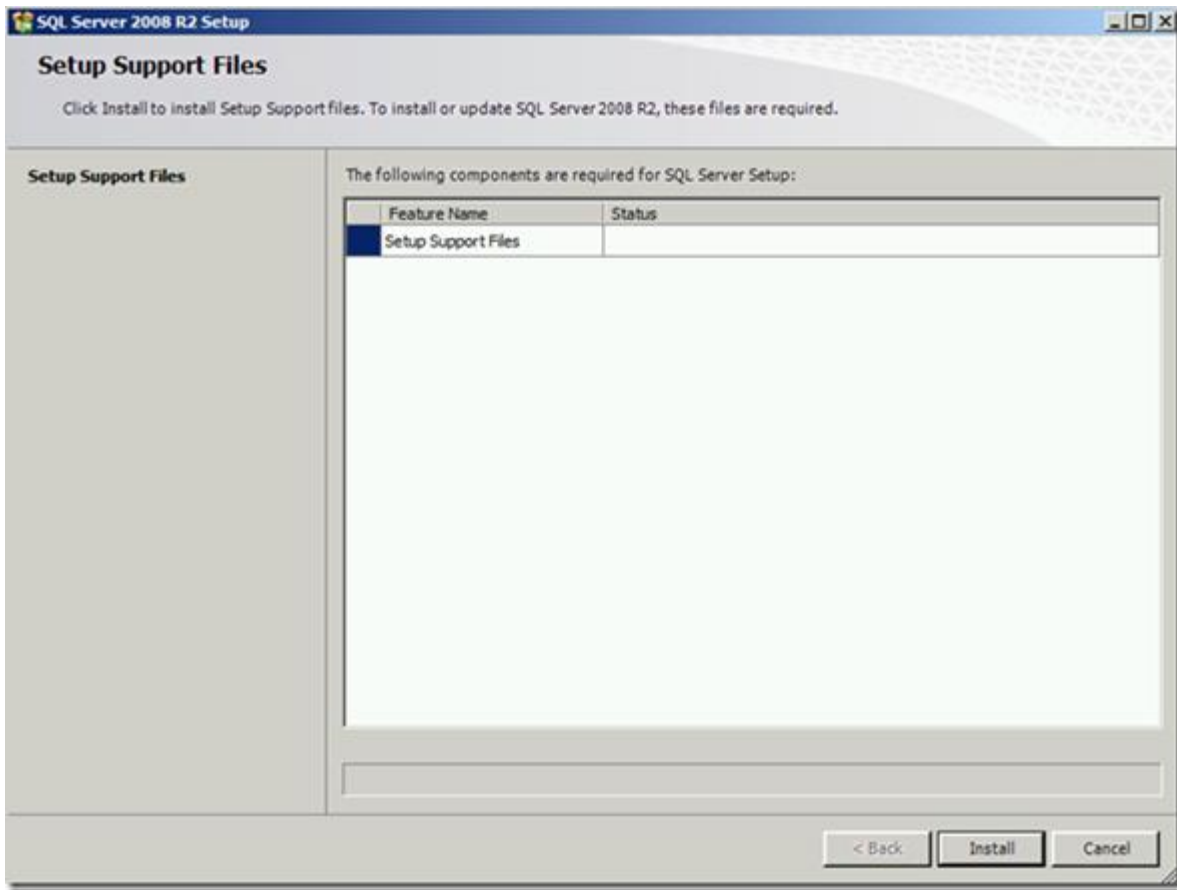
<https://sqlserverscribbles.com/2013/06/24/how-to-add-node-to-sql-server-cluster/>

We can add nodes to the existing SQL server failover cluster by following below steps. (Below method can be followed to add node in SQL Server 2008, SQL Server 2008r2 and SQL Server2012)

1. To add new node to existing SQL Server failover cluster, from new node invoke Setup.exe installation media .
- 2.The Installation Wizard will show the SQL Server Installation Center as shown in below image.Click Installation an then, select Add node to a SQL Server failover cluster.

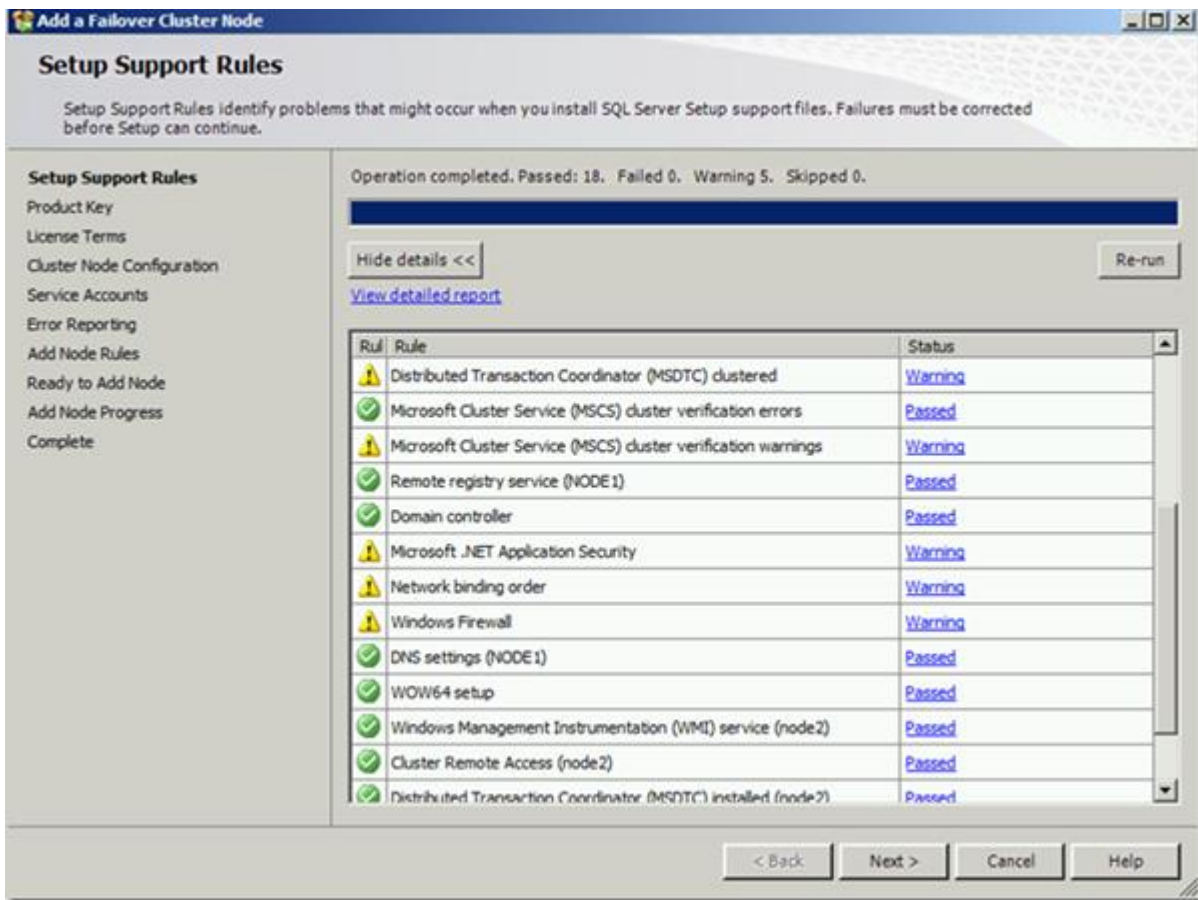


- 3.Install the setup support files.



REPORT THIS AD

4.After setup validation ,you will get details about operations completed,Click Next and continue.



5. On the Product key page, specify the Product ID key of the product. Click Next.

The screenshot shows the 'Add a Failover Cluster Node' wizard at the 'Product Key' step. The left sidebar lists the steps: Setup Support Rules, Product Key (selected), License Terms, Cluster Node Configuration, Service Accounts, Error Reporting, Add Node Rules, Ready to Add Node, Add Node Progress, and Complete. The main area contains instructions to validate the SQL Server 2008 R2 instance by entering a 25-character key. Two options are available: 'Specify a free edition' (with a dropdown menu showing 'Evaluation') and 'Enter the product key' (selected, with a text box containing 'MC46H-3QR3C-2JRH-YXRLKY-QWPM'). At the bottom are buttons for '< Back', 'Next >', 'Cancel', and 'Help'.

6. Then read and select the check box to accept terms and agreements. Click Next.

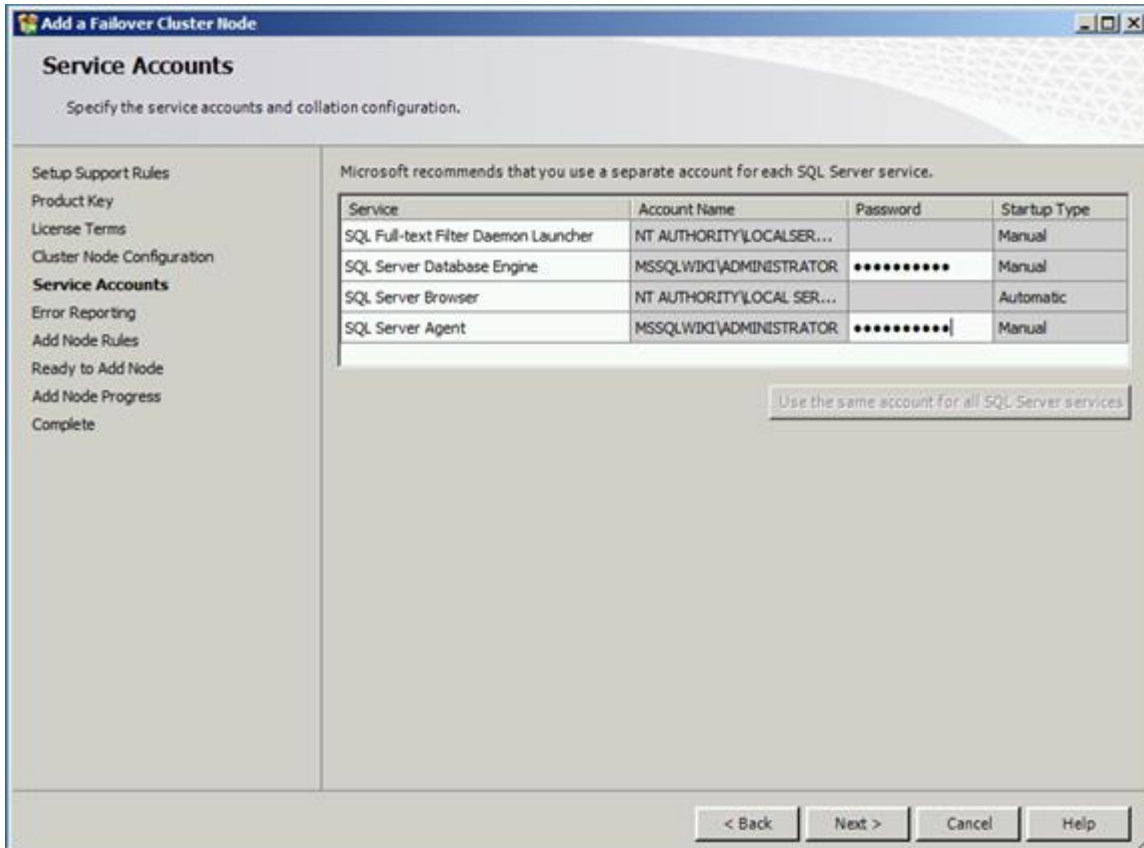
The screenshot shows the 'Add a Failover Cluster Node' wizard at the 'License Terms' step. The left sidebar lists the steps: Setup Support Rules, Product Key, License Terms (selected), Cluster Node Configuration, Service Accounts, Error Reporting, Add Node Rules, Ready to Add Node, Add Node Progress, and Complete. The main area displays the 'MICROSOFT SOFTWARE LICENSE TERMS' for 'MICROSOFT SQL SERVER 2008 R2 DEVELOPER'. It includes a paragraph of terms and a bulleted list: updates, supplements, Internet-based services, and support services. Below the list is a checkbox 'I accept the license terms.' which is checked. There is also an unchecked checkbox for 'Send feature usage data to Microsoft'. At the bottom are buttons for '< Back', 'Next >', 'Cancel', and 'Help'. There are also 'Copy' and 'Print' icons on the right side of the license terms area.

7. Then in cluster Node configuration choose the SQL Server **instance name** and Current **Node** which we are adding would automatically appear in “Name of this node” . Click Next and continue.

The screenshot shows the 'Add a Failover Cluster Node' wizard, specifically the 'Cluster Node Configuration' step. The window title is 'Add a Failover Cluster Node'. The main heading is 'Cluster Node Configuration' with the subtitle 'Add a node to an existing SQL Server failover cluster.' On the left, a navigation pane lists steps: Setup Support Rules, Product Key, License Terms, Cluster Node Configuration (selected), Service Accounts, Error Reporting, Add Node Rules, Ready to Add Node, Add Node Progress, and Complete. The main area contains two input fields: 'SQL Server instance name:' with a dropdown menu showing 'MSSQLWIKI', and 'Name of this node:' with a text box containing 'NODE1'. Below these is a table with four columns: Instance Name, Cluster Network Name, Features, and Nodes. The table has one row with the values 'MSSQLWIKI', 'NETWORKNAME', 'SQLEngine, SQL...', and 'NODE2'. At the bottom right, there are four buttons: '< Back', 'Next >', 'Cancel', and 'Help'.

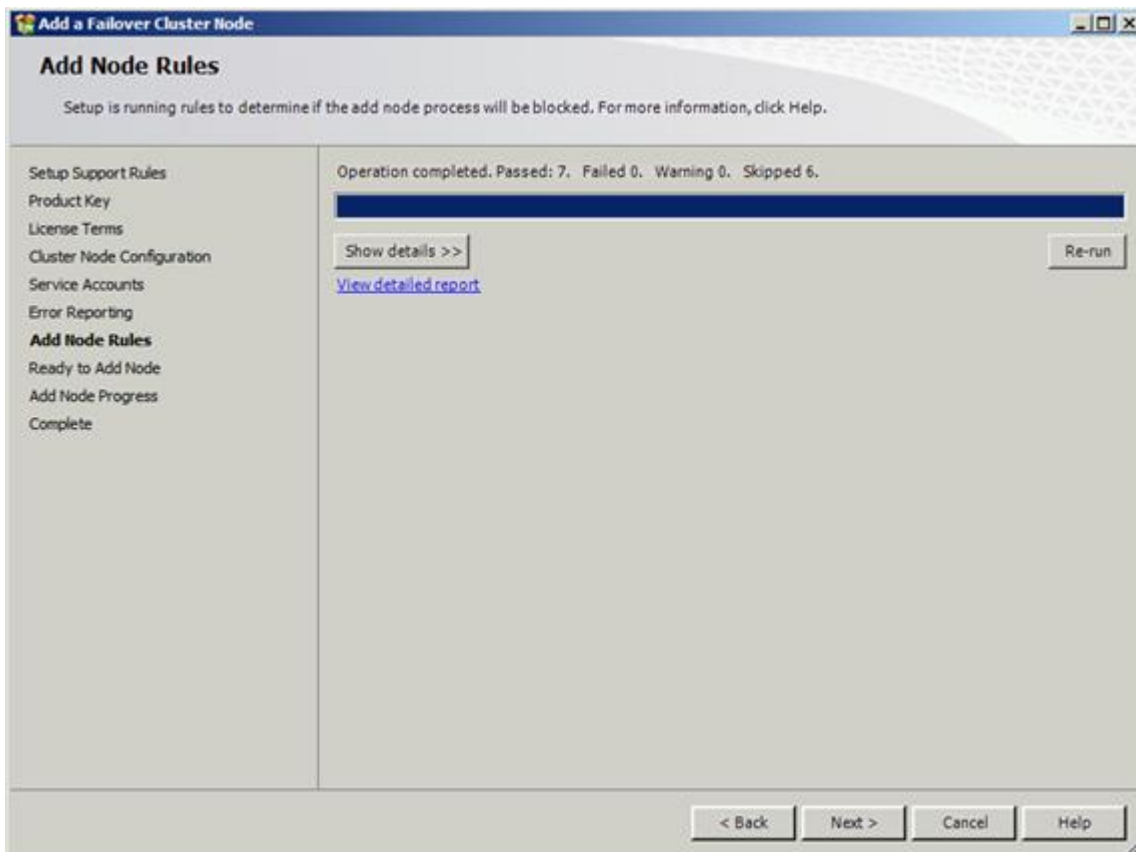
Instance Name	Cluster Network Name	Features	Nodes
MSSQLWIKI	NETWORKNAME	SQLEngine, SQL...	NODE2

8. On the Service Accounts page, specify login accounts for SQL Server services. Specify strong password for each account. Click Next after finishing.

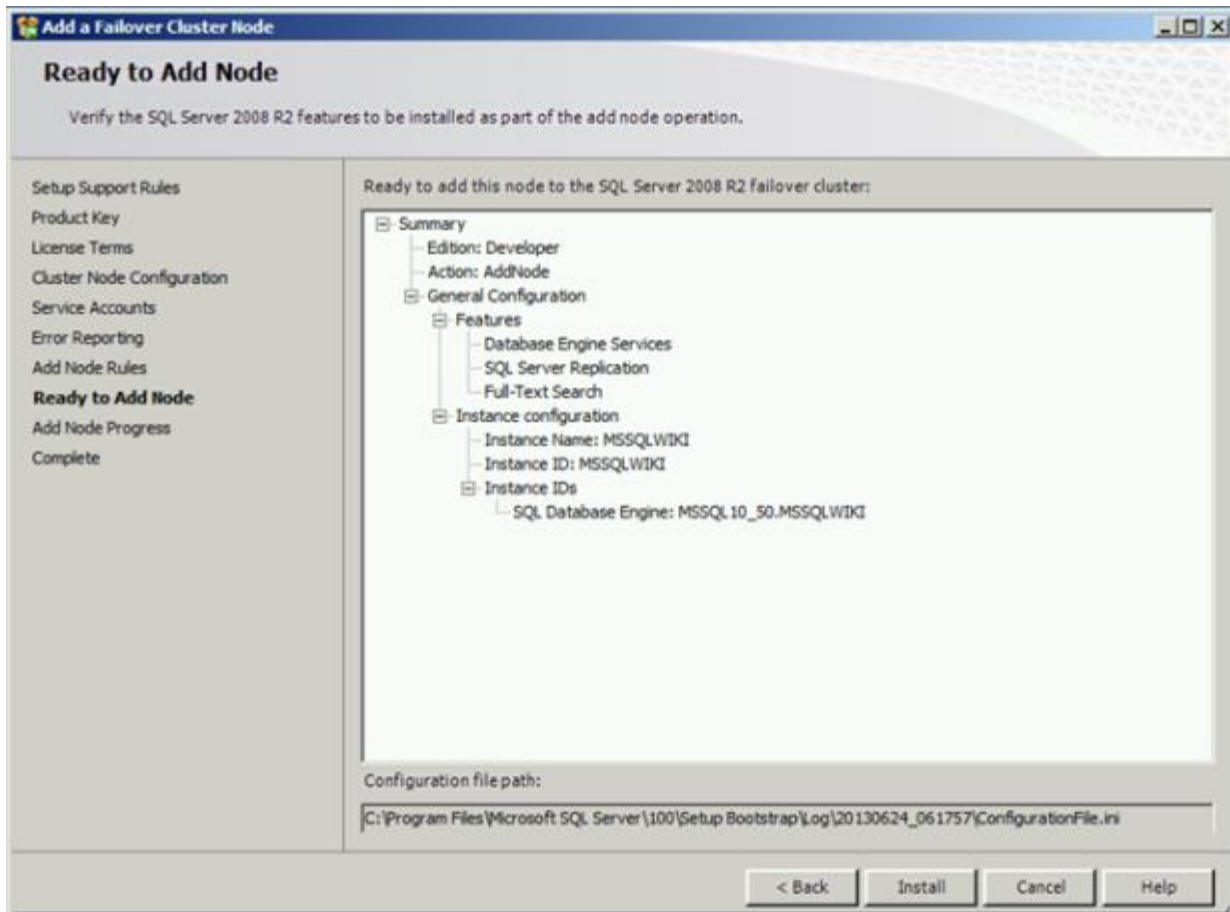


9. On the Reporting page by default the option for error reporting is enabled. If you don't want, disable it and continue.

10. Add Node rules page provides the information about operation completed.

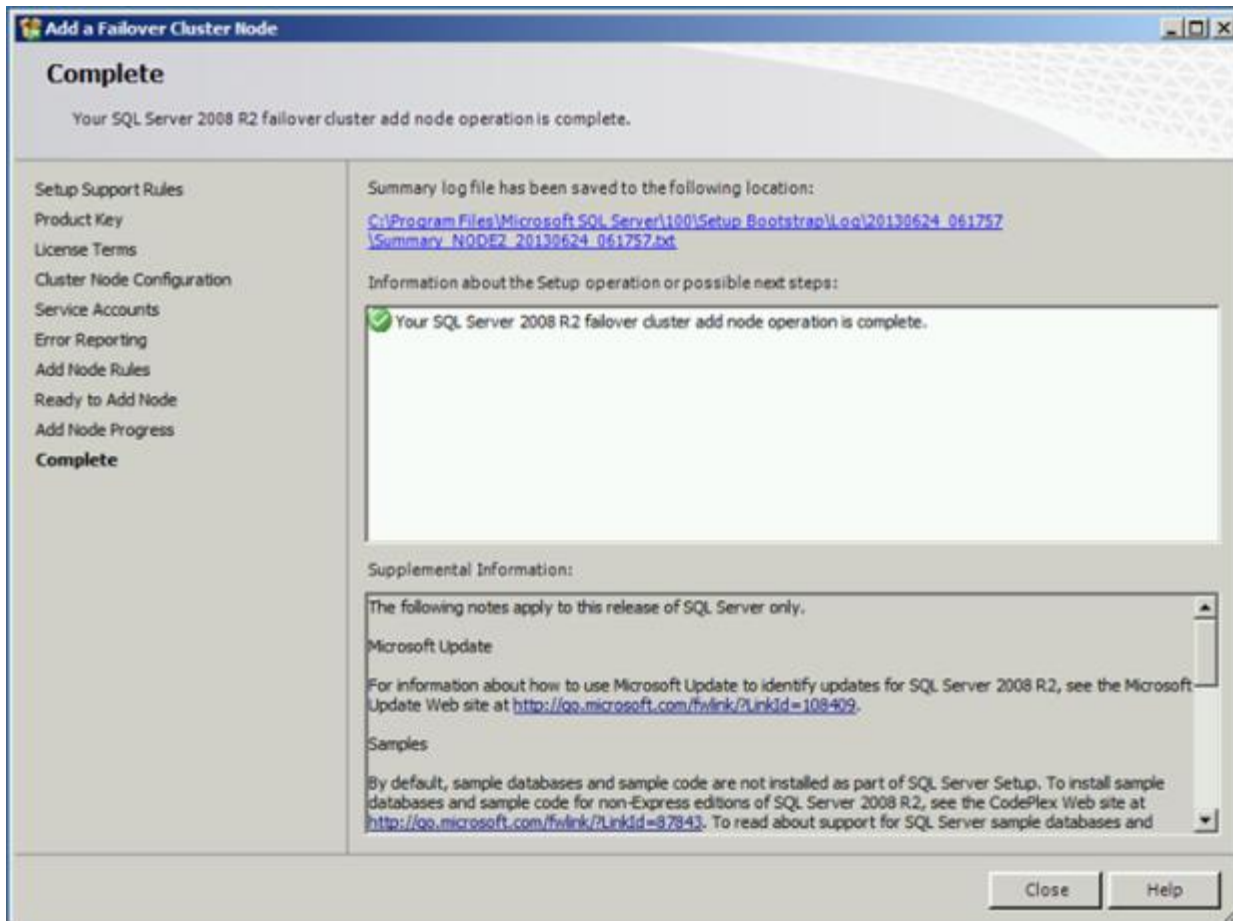


11. Ready to Add Node page displays the options that were specified during Setup. Click Install.



12. Add Node Progress page provides the status of the installation process.

13. After installation, the page provides the summary for the installation.

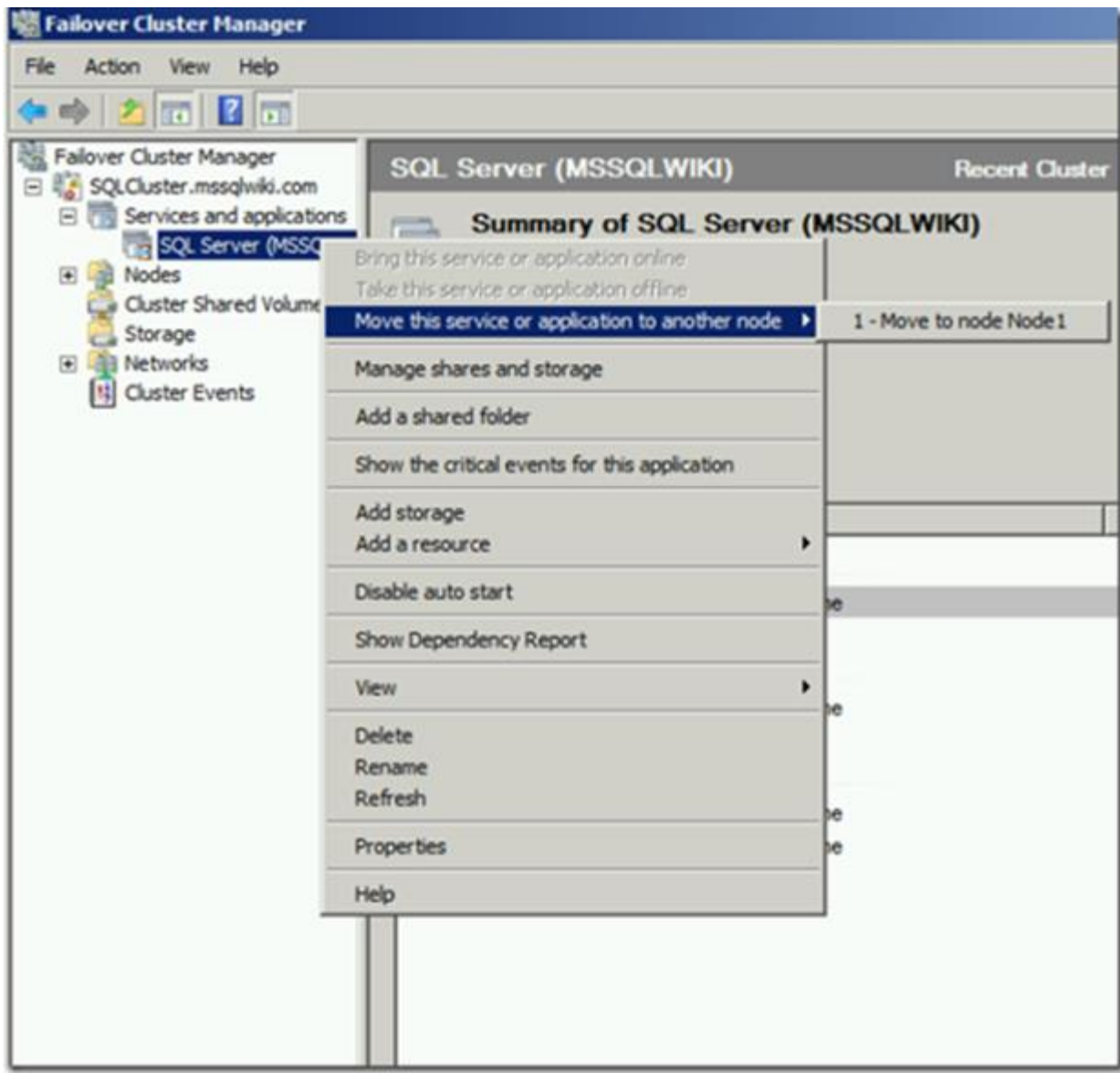


To complete the SQL Server installation process, click Close.

Move services or application between nodes to verify if SQL Server resources are able to failover successfully:

1.After installation you could move the service between all nodes.

2.Open failover cluster manager->expand SQL Cluster->expand services and applications->select the SQL Server group → Move this service or application to another node →Choose the newly added node.



3. After moving the resources to newly added Node check if all the resources are able to come online on newly added node.

